Advanced Standard Mode: Safe-to-Use Technique Combinations for Transferring Solver State Across Related Verifications

VHAGaR / neta-s-paper — Internal technical note

Abstract

Advanced standard (advstd) mode is an execution mode of the VHAGaR global-robustness verifier that computes the global robustness bound $\delta^*_{N_2}$ for a target network N_2 by *reusing solver state* from a prior solve of a structurally similar network N_1.

This note documents only the **safe-to-use** technique combinations — those empirically validated to produce average speedups with no severe slowdowns on any cell (geometric-mean time ratio $< 0.952\times$ and worst-cell time ratio $\leq 1.333\times$; i.e. $\text{perf\_tier} \in \{\text{dominant}, \text{avg-win}\}$). Out of the full combinatorial space, 25 combinations satisfy this criterion. They use four independent techniques (Sections 4.2–4.4), all of which preserve the optimal global-robustness bound δ_{exact}. Section 5 gives the complete algorithm for every safe combination.

1 Motivation and Setting

Let N_1 and N_2 be two neural networks with the same architecture — same layer sizes, same ReLU structure, same input domain — but different parameter values. A typical instance is $N_2 = N_1$ after one additional SGD epoch.

The global robustness verification problem, as formulated in VHAGaR, asks: for a given source class c_s, target class c_t, perturbation type P, and bound ε, what is the smallest δ such that no input classified with confidence at least δ as c_s can be perturbed within $\mathcal{I}_\varepsilon(\cdot)$ to be classified as c_t? This is encoded as a mixed-integer program (MIP) and solved with Gurobi.

When N_1 and N_2 are structurally identical but numerically close, the two MIPs have identical variable names, identical constraint structure (only coefficients differ), and similar solution geometry. Advanced standard mode exploits this similarity through a set of concrete, optional techniques.

2 Two-Phase Execution

Advanced standard mode runs in two phases, possibly in separate processes:

Phase 1 — *N1 solve*

For each (c_s, c_t) class pair, solve the standard-mode MIP for N_1 to either optimality or a user-specified timeout. Extract:

- The full primal solution $(\text{var}_i, \text{val}_i)_{i=1}^{|V|}$.
- The per-neuron interval bounds $(l_{m,k}, u_{m,k})$ on all split ReLUs (the `layers_info` dictionary).

Serialise to disk. The resulting `n1_state` directory is a self-contained snapshot of “everything we learned while solving N_1”.

Phase 2 — *N2 sweep*

For each (c_s, c_t) pair, build the standard-mode MIP for N_2, load the N_1 state, apply each enabled technique, and call `optimize!`. Phase 2 never re-solves N_1.

3 Background: ReLU Encoding in the MIP

The VHAGaR MIP encodes each ReLU neuron $z^+ = \max(0, \hat{z})$ using a *big-M formulation* with a binary variable $a \in \{0, 1\}$. Given pre-activation bounds $\hat{z} \in [\underline{z}, \overline{z}]$ (where $\underline{z} < 0 < \overline{z}$ for a "split" neuron), the encoding introduces:

$$z^+ \geq 0, \tag{1}$$
$$z^+ \geq \hat{z}, \tag{2}$$
$$z^+ \leq \hat{z} - \underline{z}\,(1 - a), \tag{3}$$
$$z^+ \leq \overline{z}\,a. \tag{4}$$

This is *exact*: when $a = 1$ (neuron active), constraints force $z^+ = \hat{z}$; when $a = 0$ (neuron inactive), they force $z^+ = 0$. The binary a is what makes the problem a *mixed-integer* program — solving over all possible $\{0, 1\}$ assignments is what makes verification NP-hard.

The LP relaxation of ReLU. If we *drop* the binary a and keep only the three constraints that do not involve a, we get the *triangle LP relaxation*:

$$z^+ \geq 0, \tag{5}$$
$$z^+ \geq \hat{z}, \tag{6}$$
$$z^+ \leq \frac{\overline{z}}{\overline{z} - \underline{z}}\,(\hat{z} - \underline{z}). \tag{7}$$

This is a *convex relaxation*: it allows z^+ to take any value inside a triangle in the $(\hat{z}, z^+)$ plane, not just the two exact ReLU branches. There is no binary variable, so the problem becomes a linear program (LP) — much faster to solve, but the feasible set is *larger* than the exact one. Any solution to the exact encoding also satisfies the LP relaxation, but not vice versa.

The LP relaxation is used in Technique 4: selected binary variables in the final N_2 MIP are replaced with triangle relaxations to reduce the integer search space.

4 Techniques Used in Safe Combinations

The safe combinations use exactly four techniques. Two techniques from the original advstd toolchain — MIP Start and LP Basis — are **not** used in any safe combination and are therefore omitted from this document.

Correctness invariant

Let $\mathcal{F}_2$ denote the feasible set of N_2's standard-mode MIP and let

$$\delta_{\text{exact}} \triangleq \delta^*_{N_2} = \max_{x \in \mathcal{F}_2} \text{obj}_{N_2}(x).$$

For every subset of techniques, the optimal value reported by the solver when it terminates at $\mathsf{OPTIMAL}$ is exactly δ_{exact}. The techniques only change the *path* Gurobi takes through the branch-and-bound tree, not the set of feasible solutions or the objective function.

4.1 Technique 1 — Branch Priorities (bp=rank, bp=decay)

Flag: `--adv_std_branch_priorities` $\in$ {off, rank, decay}. **Gurobi attribute:** `BranchPriority`. **Retired:** bounds (magic-number 10/5/1 step rule) and pseudocost (never used in practice). Both produce a migration error pointing at the new modes.

Idea. For each binary $a_{m,k}$ in N_2, look up N_1's final interval bounds $(l_{m,k}, u_{m,k})$ and compute the gap $g_{m,k} = |u_{m,k} - l_{m,k}|$. The two modes share this gap signal but turn it into a Gurobi priority differently. Both keep the existing direction — smaller gap (tighter, more contested) gets higher priority — and the only constant left is 100 (Gurobi's natural integer ceiling).

Mode rank (uniform spacing). Sort N_2's split-ReLU binaries by $g_{m,k}$ ascending. For neuron (m, k) at rank $r \in \{1, \ldots, N\}$ in this list:

$$\mathrm{pri}_{\mathrm{rank}}(m,k) \;=\; \max\bigl(1, \mathrm{round}\bigl(100 \cdot (1 - (r-1)/N)\bigr)\bigr).$$

Equivalently we compute $\mathrm{score}(m,k) = G_{\max} - g_{m,k} + \varepsilon$ (with $G_{\max} = \max_{(m,k)} g_{m,k}$ and ε a tiny positive constant to keep all scores strictly positive) and feed it through the generic `rank_to_priority` helper. Since the priority depends only on the position in the sorted list, this mode is invariant to the absolute gap values — a heavy-tailed distribution produces the same priorities as a uniformly distributed one with the same ordering.

Mode decay (magnitude-aware). Let $g_{\mathrm{med}} = \mathrm{median}\{g_{m,k}\}$ over the same split-ReLU set (with fallback $g_{\mathrm{med}} = 1$ if every gap is zero). Then

$$\mathrm{pri}_{\mathrm{decay}}(m,k) \;=\; \max\bigl(1, \mathrm{round}\bigl(100 \cdot \exp(-g_{m,k}/g_{\mathrm{med}})\bigr)\bigr).$$

The data-derived scale g_{med} replaces hand-picked thresholds. Magnitude is preserved: a neuron with $g = 0.01$ gets a markedly higher priority than one with $g = 0.5$, even if their ranks are adjacent. Heavy-tail outliers ($g \gg g_{\mathrm{med}}$) collapse to priority 1.

Tradeoff. rank is robust to outliers and uses the order signal only; decay responds to gap magnitudes and is sensitive to the shape of the distribution. Neither is provably faster a priori — the right choice depends on N_1's gap distribution for the problem at hand, which is what the sweep harness measures empirically.

Soundness. `BranchPriority` only changes the order in which the branch-and-bound tree is explored. No constraint is added or removed: $\mathcal{F}_2^{(\{2\})} = \mathcal{F}_2$ and δ_{exact} is preserved. This argument is independent of the formula choice and applies identically to rank and decay.

4.2 Technique 2 — Bound Tightening (boundTight)

Flag: `--adv_std_bound_tightening` (always true in safe combos). **Model modification:** tightens neuron bounds before the N_2 MIP is built.

This technique runs *before* the MIP is constructed. Its output — tighter per-neuron intervals — feeds into the MIP encoding itself (smaller big-M constants, fewer binary variables). The other techniques (branch priorities, variable hints, triangle relaxation) operate *on* the already-built MIP.

Technique 2 is independent of the *perturbed interval constraints* that are added to the MIP after construction (see the note at the end of this subsection).

Goal. For each ReLU neuron (m, k) in N_2, compute a sound interval $[d^{\downarrow}_{m,k}, d^{\uparrow}_{m,k}]$ on the *weight-drift difference* $\hat{z}^{N_2}_{m,k} - \hat{z}^{N_1}_{m,k}$ (how much the pre-activation changed due to N_2 having different weights than N_1). Once we have this interval, we combine it with N_1's known bounds from Phase 1 to get N_2 bounds:

$$l^{N_2}_{m,k} \;\geq\; l^{N_1}_{m,k} + d^{\downarrow}_{m,k}, \qquad u^{N_2}_{m,k} \;\leq\; u^{N_1}_{m,k} + d^{\uparrow}_{m,k}.$$

When a neuron's tightened interval lies strictly in $\mathbb{R}_{<0}$ or $\mathbb{R}_{\geq 0}$, the binary $a_{m,k}$ is fixed and removed from the MIP.

There are two methods to compute $[d^{\downarrow}_{m,k}, d^{\uparrow}_{m,k}]$. The safe combinations use one of these two, controlled by the `zono` flag.

4.2.1 Method 1: Interval arithmetic (baseline)

Used when: `zono`=false.

This is the default method. From Phase 1 we have N_1's per-neuron bounds $(l^{N_1}_{m,k}, u^{N_1}_{m,k})$, so the post-ReLU range is $z^{N_1}_{m,k} \in [\max(0, l^{N_1}_{m,k}),\ \max(0, u^{N_1}_{m,k})]$.

Weight-drift difference. Because N_1 and N_2 share the same architecture, every neuron (m, k) in N_2 has a corresponding neuron in N_1. The *difference* in their pre-activations, $z^{N_2}_{m,k} - z^{N_1}_{m,k}$, depends only on the weight difference $\Delta W = W_2 - W_1$ (the "weight drift" from training). Sound intervals on this difference are computed layer by layer using interval arithmetic.

Affine layers. Let $\Delta w_{m,k} = w^2_{m,k} - w^1_{m,k}$ and $\Delta b_{m,k} = b^2_{m,k} - b^1_{m,k}$ be the weight and bias differences at layer m. From Phase 1 we have N_1's per-neuron bounds $(l^{N_1}_{m,k}, u^{N_1}_{m,k})$ for every ReLU neuron. Since the post-ReLU output is $z_{m,k} = \max(0, \hat{z}_{m,k})$, the post-ReLU range for each neuron k in layer $m-1$ is $z^{N_1}_{m-1,k} \in [\max(0, l^{N_1}_{m-1,k}),\ \max(0, u^{N_1}_{m-1,k})]$.

The pre-activation difference at neuron k of layer m decomposes as:

$$\hat{z}^{N_2}_{m,k} - \hat{z}^{N_1}_{m,k} \;=\; \underbrace{\sum_{k'} w^2_{m,k,k'}(z^{N_2}_{m-1,k'} - z^{N_1}_{m-1,k'})}_{\text{propagated diff from previous layer}} \;+\; \underbrace{\sum_{k'} \Delta w_{m,k,k'} \cdot z^{N_1}_{m-1,k'} + \Delta b_{m,k}}_{\text{direct weight-drift contribution}}.$$

Interval arithmetic bounds each term per output neuron k:

$$[d^{\downarrow}_{m,k},\ d^{\uparrow}_{m,k}] \;=\; \sum_{k'} w^2_{m,k,k'} \cdot [d^{\downarrow}_{m-1,k'},\ d^{\uparrow}_{m-1,k'}] + \sum_{k'} \Delta w_{m,k,k'} \cdot [\max(0, l^{N_1}_{m-1,k'}),\ \max(0, u^{N_1}_{m-1,k'})] + \Delta b_{m,k},$$

ReLU layers. The ReLU difference satisfies $\sigma(z + d) - \sigma(z) \in [\min(0, d^{\downarrow}), \max(0, d^{\uparrow})]$, so the diff interval is clipped: $d^{\uparrow} \leftarrow \max(0, d^{\uparrow})$ and $d^{\downarrow} \leftarrow -\max(0, -d^{\downarrow})$.

The final output at each layer is a sound interval:

$$d^{\downarrow}_{m,k} \;\leq\; z^{N_2}_{m,k} - z^{N_1}_{m,k} \;\leq\; d^{\uparrow}_{m,k}.$$

These are then combined with N_1's known bounds to get N_2 bounds:

$$l^{N_2}_{m,k} \;\geq\; l^{N_1}_{m,k} + d^{\downarrow}_{m,k}, \qquad u^{N_2}_{m,k} \;\leq\; u^{N_1}_{m,k} + d^{\uparrow}_{m,k}.$$

This method is fast but loose: it treats every neuron as independent, losing all correlations between neurons in the same layer.

4.2.2 Method 2: Zonotope arithmetic (Source A + B)

Used when: `zono`=true. **Flag:** `--adv_std_zono_bounds`.

This method computes the same $[d^{\downarrow}_{m,k}, d^{\uparrow}_{m,k}]$ as Method 1, but using *zonotope arithmetic* instead of interval arithmetic. A zonotope tracks correlations between neurons by representing each neuron's value as a linear combination of shared noise variables, producing tighter bounds. Additionally, a second pass (Source B) independently bounds N_2's absolute activations and intersects the result with the diff-based bounds.

Source A — Zonotope diff bounds. Each neuron's difference is represented as a zonotope — a centre plus generators sharing noise variables $\epsilon_j \in [-1, 1]$ across all neurons in the layer. We store a centre vector $\mathbf{c}_m \in \mathbb{R}^K$ and a generator matrix $G_m \in \mathbb{R}^{K\times p}$ (rows = K neurons, columns = p generators, where p is the number of columns accumulated so far — initially $p = K_0$ at the input layer, and each affine layer adds K' columns while each ReLU adds one column per split neuron):

$$d_{m,k} = c_{m,k} + \sum_j g_{m,k,j}\,\epsilon_j, \qquad d^{\downarrow}_{m,k} = c_{m,k} - \textstyle\sum_j |g_{m,k,j}|, \qquad d^{\uparrow}_{m,k} = c_{m,k} + \textstyle\sum_j |g_{m,k,j}|.$$

Shared ϵ_j variables encode correlations: interval arithmetic is the special case where G_m is diagonal (no correlations).

Initialisation. At the input layer, N_1 and N_2 receive the same input, so the diff is zero: $\mathbf{c}_0 = \mathbf{0}$, $G_0 = \mathbf{0}_{K_0\times 0}$ (no generators yet, $p = 0$). The first affine update creates the initial generators via G'_{indep}.

Affine layer update. Let K' = neurons in layer $m-1$, K = neurons in layer m:

$$c_{m,k} = \sum_{k'} w^2_{m,k,k'}\, c_{m-1,k'} + \sum_{k'} \Delta w_{m,k,k'} \cdot \text{mid}_{k'} + \Delta b_{m,k} \tag{8}$$

$$\underbrace{G'_{\text{corr}}}_{K\times p} = \underbrace{W_2}_{K\times K'}\ \underbrace{G_{m-1}}_{K'\times p} \tag{9}$$

$$\underbrace{G'_{\text{indep}}}_{K\times K'} = \underbrace{\Delta W}_{K\times K'} \cdot \underbrace{\text{diag}(\text{rad}_1, \ldots, \text{rad}_{K'})}_{K'\times K'} \tag{10}$$

$$\underbrace{G_m}_{K\times(p+K')} = \left[\, G'_{\text{corr}} \;\middle|\; G'_{\text{indep}} \,\right] \tag{11}$$

where $\text{mid}_{k'}$ and $\text{rad}_{k'}$ are the midpoint and half-width of N_1's post-ReLU interval at neuron k': $\text{mid}_{k'} = \frac{1}{2}(\max(0, u^{N_1}_{m-1,k'}) + \max(0, l^{N_1}_{m-1,k'}))$, $\text{rad}_{k'} = \frac{1}{2}(\max(0, u^{N_1}_{m-1,k'}) - \max(0, l^{N_1}_{m-1,k'}))$.

After the affine update, the column count grows from p to $p + K'$.

ReLU layer update (DeepZ). Compute hull per neuron: $l_k = c_{m,k} - \sum_j |g_{m,k,j}|$, $u_k = c_{m,k} + \sum_j |g_{m,k,j}|$.

- $u_k \le 0$: $c_{m,k} \leftarrow 0$, $g_{m,k,\cdot} \leftarrow 0$.
- $l_k \ge 0$: unchanged.
- Split ($l_k < 0 < u_k$): $\lambda_k = \frac{u_k}{u_k - l_k}$, $\mu_k = -\frac{u_k \cdot l_k}{2(u_k - l_k)}$,

$$c_{m,k} \leftarrow \lambda_k\, c_{m,k} + \mu_k, \qquad g_{m,k,\cdot} \leftarrow \lambda_k\, g_{m,k,\cdot},$$

  append column to G_m: value μ_k at row k, zero elsewhere.

After the ReLU update, $p \leftarrow p + |\{k : l_k < 0 < u_k\}|$ (one new column per split neuron).

So for a network with L affine layers of widths $K_1, \ldots, K_L$ and s_m split neurons at ReLU layer m:

$$p_{\text{final}} = K_0 + \sum_{m=1}^{L} (K_{m-1} + s_m).$$

Source B — N1-tightened absolute N2 zonotope. A separate zonotope propagates N_2's *absolute* pre-activations (not the diff) through N_2 alone, using the same zonotope machinery as Source A but applied to N_2's own values. Initialise with a single zonotope whose hull covers both the clean input $v_{\text{in}} \in [0,1]^n$ *and* the perturbed input $v_{x_0} = v_{\text{in}} + v_e \in [-\varepsilon, 1+\varepsilon]^n$ simultaneously:

$$\mathbf{c}_0^{\text{B}} = \tfrac{1}{2}\mathbf{1}, \qquad G_0^{\text{B}} = \operatorname{diag}\bigl(\tfrac{1}{2} + \varepsilon, \ldots, \tfrac{1}{2} + \varepsilon\bigr) \in \mathbb{R}^{n \times n},$$

so the initial hull is $[-\varepsilon, 1+\varepsilon]^n$ rather than $[0,1]^n$. This widens the input box by ε on each side relative to the clean-only range, which is the price of using one zonotope to bound N_2's pre-activations under both the clean and perturbed inputs (rather than propagating two separate zonotopes and taking their union). Propagate through N_2's affine layers ($c \leftarrow W_2 c + b_2$, $G \leftarrow W_2 G$) and DeepZ ReLU relaxations exactly as in Source A. At each ReLU layer, the zonotope hull gives per-neuron bounds:

$$u_{m,k}^{\text{hull}} = c_{m,k}^{\text{B}} + \textstyle\sum_j |g_{m,k,j}^{\text{B}}|, \qquad l_{m,k}^{\text{hull}} = c_{m,k}^{\text{B}} - \textstyle\sum_j |g_{m,k,j}^{\text{B}}|.$$

These are then *intersected* with Source A's diff-based bounds:

$$u_{m,k}^{N_2} = \min\bigl(u_{m,k}^{\text{hull}},\ u_{m,k}^{N_1} + d_{m,k}^{\uparrow}\bigr), \qquad l_{m,k}^{N_2} = \max\bigl(l_{m,k}^{\text{hull}},\ l_{m,k}^{N_1} + d_{m,k}^{\downarrow}\bigr). \tag{12}$$

The first term in each min/max comes from Source B's own zonotope pass on N_2; the second term comes from Source A's diff bounds applied to N_1's known bounds (the same formula as in the Goal paragraph above). Taking the tighter of the two at each neuron is sound because both are valid over-approximations.

The DeepZ ReLU relaxation is then applied using these tightened bounds, so Source B's zonotope passes through a narrower "tube" at each layer.

4.2.3 Precision guarantees

The two methods are ordered in precision, and Source A and Source B within Method 2 are incomparable.

Theorem 1 (Zonotope diff dominates interval diff)**.** *Let $[d_{m,k}^{\downarrow,I}, d_{m,k}^{\uparrow,I}]$ denote the diff bound produced by Method 1 (interval arithmetic, Section 4.2.1) and $[d_{m,k}^{\downarrow,Z}, d_{m,k}^{\uparrow,Z}]$ the diff bound produced by Source A of Method 2 (zonotope, Section 4.2.2) when both start from the same N_1 state. Then for every ReLU neuron (m,k),*

$$d_{m,k}^{\downarrow,I} \;\le\; d_{m,k}^{\downarrow,Z} \;\le\; \hat{z}_{m,k}^{N_2} - \hat{z}_{m,k}^{N_1} \;\le\; d_{m,k}^{\uparrow,Z} \;\le\; d_{m,k}^{\uparrow,I}.$$

The inequality is strict whenever some intermediate affine layer combines two generators whose signs disagree after multiplication by the layer's weights.

Proof sketch. Interval arithmetic is the special case of zonotope arithmetic in which the generator matrix G_m is restricted to be diagonal (each neuron has its own independent noise variable). Zonotope preserves correlations: when a subsequent affine layer computes $\sum_{k'} w_{k'} z_{m-1,k'}$, the shared noise ϵ_j combines as $\left(\sum_{k'} w_{k'} g_{m-1,k',j}\right)\epsilon_j$, contributing $\left|\sum_{k'} w_{k'} g_{m-1,k',j}\right|$ to the half-width. Interval arithmetic discards the correlation and contributes $\sum_{k'} |w_{k'} g_{m-1,k',j}|$ instead. The triangle inequality gives the former $\leq$ the latter, with equality iff all $w_{k'} g_{m-1,k',j}$ share a sign. Inducting on depth and applying the sound DeepZ ReLU relaxation at each layer yields the claim. The ReLU relaxation $\lambda_k z + \mu_k$ is affine with $\lambda_k \in [0,1]$ and therefore preserves width ordering. □

Corollary 2 (Precision gap grows with depth and drift). *Let $w^{\bullet}_{m,k} \triangleq d^{\uparrow,\bullet}_{m,k} - d^{\downarrow,\bullet}_{m,k}$ denote the width of the diff bound at neuron (m,k) under Method $\bullet \in \{I, Z\}$. The width gap $w^{I}_{m,k} - w^{Z}_{m,k}$ is non-decreasing in m and grows monotonically with $\|\Delta W\|_F$.*

Theorem 3 (Source A and Source B are incomparable). *Let $[l^A_{m,k}, u^A_{m,k}] \triangleq [l^{N_1}_{m,k} + d^{\downarrow}_{m,k},\ u^{N_1}_{m,k} + d^{\uparrow}_{m,k}]$ be the Source-A N_2 bound and $[l^B_{m,k}, u^B_{m,k}] \triangleq [l^{\text{hull}}_{m,k}, u^{\text{hull}}_{m,k}]$ the Source-B bound.*

(a) There exist instances on which $u^A < u^B$ (equivalently $l^A > l^B$) strictly, and instances on which the opposite strict inequality holds.
(b) The intersection bound $[\max(l^A, l^B), \min(u^A, u^B)]$ (Eq. 12) has width strictly less than $\min\big(w^A, w^B\big)$ on any neuron where the two sources disagree in orientation (i.e. $u^A < u^B$ at the upper end and $l^A > l^B$ at the lower end, or vice-versa).

Proof sketch. (a) *A tighter than B.* Take $N_1 = N_2$ and a generic $\varepsilon > 0$. Then $\Delta W = 0$, so every Source-A generator column $G'_{\text{indep}} = 0$ and the diff zonotope remains zero, giving $[d^{\downarrow}, d^{\uparrow}] = 0$ and $l^A = l^{N_1}$, $u^A = u^{N_1}$ (i.e. N_1's MIP-level bounds). Source B propagates the wider hull $[-\varepsilon, 1+\varepsilon]^n$ through DeepZ relaxations, which discard ReLU correlations and accumulate error, so $l^B \leq l^A - c_\varepsilon$ and $u^B \geq u^A + c_\varepsilon$ for some $c_\varepsilon > 0$ on any neuron with a split ancestor.

B tighter than A. Take an N_1 whose Phase-1 solve timed out before closing the root relaxation, so its MIP bounds are vacuous: $l^{N_1}_{m,k} = -\infty$ (in practice, the big-M input bound). Source A inherits these, giving l^A, u^A effectively vacuous. Source B is independent of N_1's MIP state and produces finite bounds from N_2's weights alone.

(b) Immediate from the min/max definition: when $u^A < u^B$ and $l^A > l^B$ the intersection keeps u^A and l^A (width w^A); when the sources disagree in orientation, say $u^A < u^B$ and $l^B > l^A$, the intersection is $[l^B, u^A]$ with width strictly less than both w^A and w^B. □

Implication for the safe combinations. Theorem 1 explains the empirical pattern that zono=true never *loses* precision relative to zono=false but may cost more to compute. The incomparability in Theorem 3 explains why the Source A + Source B intersection is always at least as useful as either alone, and why on benchmarks where N_1 is solved to optimality the benefit of Source B over Source A alone is small, whereas on timed-out N_1 instances it is large.

4.2.4 Final bound integration

When N_2's MIP is built, each ReLU neuron's bounds are the tightest (intersection) of all available sources:

$$u^{N_2}_{m,k} \leftarrow \min\big(u^{N_1}_{m,k} + d^{\uparrow}_{m,k},\ u^{\text{B}}_{m,k}\big), \tag{13}$$

$$l^{N_2}_{m,k} \leftarrow \max\big(l^{N_1}_{m,k} + d^{\downarrow}_{m,k},\ l^{\text{B}}_{m,k}\big), \tag{14}$$

Terms from disabled methods are simply omitted from the min/max. When the final interval is single-signed ($u^{N_2}_{m,k} \leq 0$ or $l^{N_2}_{m,k} \geq 0$), the associated binary is eliminated from the MIP entirely.

Soundness. Both methods produce sound over-approximations:

- Method 1 (interval arithmetic): standard interval arithmetic rules over-approximate the true range at each layer.
- Method 2 (zonotope): zonotope arithmetic is a sound abstract interpretation; Source B's intersection of two sound over-approximations is itself sound.

Every resulting bound is satisfied by every $x \in \mathcal{F}_2$, so x^* remains feasible and δ_{exact} is preserved.

Summary of what operates where:

- **Technique 4** (bound tightening, Sources A/B): runs *before* MIP construction. Output: tighter per-neuron intervals → smaller big-M constants, fewer binaries.
- **Perturbed interval constraints**: runs *after* MIP construction. Output: additional linear constraints linking the org and perturbation copies of N_2.
- Both are always active in the safe combinations and complement each other.

4.3 Technique 3 — Variable Hints (varHint=prev/direct/direct_pgd/prev_pgd)

Flag: `--adv_std_var_hint` $\in$ {off, prev, direct, direct_pgd, prev_pgd} (five-valued; legacy true/false still accepted, mapped to prev/off with a deprecation warning). **Gurobi attributes:** `VarHintVal`, `VarHintPri` (prev/direct) or `Start` via `set_start_value` (prev_pgd/direct_pgd). **Filename tags:** `_varHintFixed` (prev), `_varHintDirect` (direct), `_varHintDirectPGD` (direct_pgd), `_varHintPrevPGD` (prev_pgd). **Retired:** the previously opt-in `--adv_std_var_hint_fix` flag has been merged into `--adv_std_var_hint`; the "fix" (filtering N_1 pseudocosts down to N_2-surviving binaries before ranking) is now the default and only behavior. Pre-merger result files tagged `_varHint` or `_varHint_varHintFix` are reported as `vh_legacy` by the post-processing extractor and excluded from the auto-tex tables (they are not directly comparable to the merged-behavior runs).

Background: how Gurobi solves a MIP. When Gurobi solves N_2's MIP, it uses *branch-and-bound*: it maintains a search tree where each node is a sub-problem with some binaries fixed and the rest relaxed to continuous $[0, 1]$. At each node, Gurobi solves an LP relaxation of that sub-problem. The LP solution gives fractional values for the unfixed binaries (e.g. $a_{m,k} = 0.37$). Gurobi then needs to decide:

1. *Which* fractional binary to branch on next.
2. *Which direction* to try first ($a_i = 0$ or $a_i = 1$).
3. Whether to run *heuristics* (RINS, diving, rounding) that try to round the fractional LP solution into a feasible integer solution (an "incumbent").

Variable hints influence all three decisions as soft suggestions.

What the two hint attributes do. For each binary a_i in N_2's MIP, we set:

- `VarHintVal`(a_i): "what value should a_i take?" — Gurobi's heuristics prefer rounding a_i toward this value.
- `VarHintPri`(a_i): "how confident are we?" — higher priority means Gurobi follows the hint more strongly. Priority 0 means "ignore unless nothing else is available."

Both are *soft*: they do not add constraints or change the MIP. If the LP relaxation at some node clearly disagrees with the hint, Gurobi overrides it. A wrong hint wastes some effort but cannot change the final answer.

Transfer-probability signal. The right question for the hint is not "how useful was this binary during N_1's branch-and-bound" — it is *"what is the chance that N_1's binary value for this neuron is still correct for N_2?"* We build a continuous signal $p_i \in [0,1]$ that answers this question directly, using only state that N_1's solve and Technique 2 already produce.

Flag values. The `--adv_std_var_hint` flag takes five values. off sets no hint on any binary. The remaining four are the Cartesian product of two independent choices — (i) how the transfer probability p_i is computed (*prev-rule* vs. *direct-rule*), and (ii) which Gurobi channel carries the result (*soft VarHint* vs. *PGD-consensus Start*):

- prev — p_i from prev-rule, emitted via `VarHintVal`/`VarHintPri`.
- direct — p_i from direct-rule, emitted via `VarHintVal`/`VarHintPri`.
- prev_pgd — p_i from prev-rule, emitted via `Start` with PGD-consensus filtering (see below). No `VarHintVal`/`VarHintPri` is written.
- direct_pgd — p_i from direct-rule, emitted via `Start` with PGD-consensus filtering. No `VarHintVal`/`VarHintPr` is written.

The two rules for p_i:

- *prev-rule*: build a heuristic interval $\tilde{I}_i$ by shifting N_1's pre-activation $\hat{z}_i^{N_1}$ by the diff bound $[d_i^{\downarrow}, d_i^{\uparrow}]$ and clipping to $[l_i^{N_2}, u_i^{N_2}]$, then derive p_i from interval-length ratios.
- *direct-rule*: derive p_i directly from Technique 2's tightened bounds $[l_i^{N_2}, u_i^{N_2}]$, without constructing $\tilde{I}_i$ or using $\hat{z}_i^{N_1}$.

All four non-off modes share the same hint-value formula (agree with $v_i^{N_1}$ when $p_i \geq 0.5$, flip otherwise); they differ only in how p_i is computed and how the resulting `hint_val` is delivered to Gurobi.

Ingredients. For each ReLU neuron $i = (m,k)$ that survives Technique 2's elimination (i.e., $l_{m,k}^{N_2} < 0 < u_{m,k}^{N_2}$):

- $v_i^{N_1} \in \{0,1\}$ — N_1's optimal binary (from N_1's primal). Used by both modes to select N1's supporting side of zero.
- $[l_i^{N_2}, u_i^{N_2}]$ — N_2's final tightened bounds after Technique 2's integration step (Section 4.2.4). Used by both modes.
- *(prev only)* $\hat{z}_i^{N_1}$ — N_1's pre-activation scalar. When $v_i^{N_1} = 1$ this equals N_1's post-ReLU value z_i^{+,N_1} and is read directly from the primal. When $v_i^{N_1} = 0$ the big-M encoding only constrains $\hat{z}_i^{N_1} \in [l_i^{N_1}, 0]$; the prev rule takes its midpoint $l_i^{N_1}/2$ as a proxy.
- *(prev only)* $[d_i^{\downarrow}, d_i^{\uparrow}]$ — the weight-drift diff bound from Technique 2.

Mode prev — previous rule. Shift N_1's achieved pre-activation by the diff bound, clip to N_2's sound bounds, and take p_i as the fraction of the clipped interval on N_1's supporting side of zero:

$$I_i \;=\; \big[\hat{z}_i^{N_1} + d_i^{\downarrow},\; \hat{z}_i^{N_1} + d_i^{\uparrow}\big], \qquad \tilde{I}_i \;=\; I_i \cap [l_i^{N_2},\; u_i^{N_2}],$$

$$p_i \;=\; \begin{cases} \dfrac{\operatorname{length}\big(\tilde{I}_i \cap (-\infty, 0]\big)}{\operatorname{length}(\tilde{I}_i)} & \text{if } v_i^{N_1} = 0, \\[2ex] \dfrac{\operatorname{length}\big(\tilde{I}_i \cap [0, +\infty)\big)}{\operatorname{length}(\tilde{I}_i)} & \text{if } v_i^{N_1} = 1. \end{cases}$$

$\tilde{I}_i$ is a heuristic prior on where $\hat{z}_i^{N_2}$ lands under the assumption that N_2's optimal input is close to N_1's. It is not a sound over-approximation of $\hat{z}_i^{N_2}$ at N_2's own optimum (which is allowed to use a different input), but for an advisory hint this is the right quantity.

Mode direct — new rule. Skip the $\tilde{I}_i$ construction entirely. p_i is the fraction of N_2's tightened pre-activation range on N_1's supporting side of zero:

$$p_i = \begin{cases} \dfrac{-\,l_i^{N_2}}{u_i^{N_2} - l_i^{N_2}} & \text{if } v_i^{N_1} = 0, \\ \dfrac{u_i^{N_2}}{u_i^{N_2} - l_i^{N_2}} & \text{if } v_i^{N_1} = 1. \end{cases}$$

Technique 2's integration step already folds N_1's ReLU bounds and the weight drift into $[l_i^{N_2}, u_i^{N_2}]$, so re-deriving an interval by shifting $\hat{z}_i^{N_1}$ only re-introduces the midpoint-proxy error (when $v_i^{N_1} = 0$) and the diff-width noise (when $[d_i^{\downarrow}, d_i^{\uparrow}]$ is wide). Using $[l_i^{N_2}, u_i^{N_2}]$ directly self-calibrates: a wider diff bound flows through Technique 2 to produce a wider $[l_i^{N_2}, u_i^{N_2}]$, which pulls p_i toward 0.5 and lowers the priority automatically.

Three regimes of p_i. Under either mode:

- $p_i = 1$: the source range lies entirely on N_1's side — N_1's choice almost certainly transfers.
- $0 < p_i < 1$: the source range straddles zero; p_i measures how much of its length still backs N_1.
- $p_i = 0$: the source range lies entirely on the opposite side — the drift has likely flipped the neuron; N_1's raw choice is wrong for N_2.

The "source range" is $\tilde{I}_i$ under prev and $[l_i^{N_2}, u_i^{N_2}]$ under direct.

Continuous hint rule (all four non-off modes). Agree with N_1 when $p_i \geq 0.5$, flip otherwise:

$$\texttt{hint\_val}(a_i) \leftarrow \begin{cases} v_i^{N_1} & \text{if } p_i \geq 0.5, \\ 1 - v_i^{N_1} & \text{if } p_i < 0.5. \end{cases}$$

Under prev/direct this is paired with a V-shaped priority, maximal at both extremes and minimal at the 50/50 line:

$$\texttt{VarHintVal}(a_i) \leftarrow \texttt{hint\_val}(a_i), \qquad \texttt{VarHintPri}(a_i) \leftarrow \max\!\big(1,\ \text{round}\big(100 \cdot |2p_i - 1|\big)\big).$$

Under prev_pgd/direct_pgd the same `hint_val` is instead routed through `Start` with a consensus filter against the PGD warm-start (see below); no `VarHintVal`/`VarHintPri` is written.

Anchor points, for intuition:

p_i	`VarHintVal`	`VarHintPri`	Interpretation
1.00	$v_i^{N_1}$	100	Source range fully on N_1's side
0.75	$v_i^{N_1}$	50	Majority still backs N_1
0.50	either	1	No usable signal
0.25	$1 - v_i^{N_1}$	50	Majority opposes N_1
0.00	$1 - v_i^{N_1}$	100	Source range fully opposite to N_1; drift flipped the neuron

The hint value flips exactly at the $p_i = 0.5$ crossing. The priority is continuous in p_i and never requires a hand-tuned threshold beyond the floor at 1 (priority 0 is reserved by Gurobi for "ignore this hint").

PGD-consensus `Start` channel (prev_pgd/direct_pgd). The "_pgd" modes use the same `hint_val`(a_i) as their non-pgd siblings but deliver it via `Start` (`set_start_value` in `JuMP`) instead of `VarHintVal`/`VarHintPri`. Before Technique 3 runs, PGD has already populated `Start` on some binaries via `hyper_attack_hints()`; for each surviving binary a_i we read `JuMP.start_value`(a_i) and apply a three-way consensus filter:

- **PGD silent** (`start_value` = `nothing`): N_1's transfer hint is the only signal available — call `set_start_value`(a_i, `hint_val`) ("*fill*").
- **PGD agrees** (`start_value` = `hint_val`): PGD and the N_1 transfer point at the same branch; leave PGD's `Start` in place, no update ("*leave*").
- **PGD disagrees** (`start_value` $\neq$ `hint_val`): the two signals contradict — call `set_start_value`(a_i, `nothing` to withdraw PGD's value, leaving the binary unstarted for Gurobi to decide ("*withdraw*").

The intent is to keep only the warmstart bits both sources agree on (or that only one source filled in), instead of fighting between two incomplete MIP starts. If PGD failed entirely (all starts are `nothing`), the modes degrade to "variable hints fill every gap via `Start`" — equivalent to unconditional N_1-transfer warmstart.

Relation to prev/direct. The p_i formulas, the flip rule, and the handling of eliminated binaries are unchanged; only the Gurobi channel differs. Priority information is discarded (Gurobi's `Start` API has no analogue of `VarHintPri`) — every `hint_val` that survives the consensus filter has equal weight.

Optimal-hint characterization of direct. The p_i formula for direct is not a heuristic — it is a posterior probability under a canonical prior derived from Technique 2's output. Specifically:

Theorem 4 (Bayes-optimal hint under max-entropy prior). *Fix a surviving binary a_i (so $l_i^{N_2} < 0 < u_i^{N_2}$). Let π_i be the maximum-entropy prior on $\hat{z}_i^{N_2}$ whose support is the Technique-2 bound, i.e. $\hat{z}_i^{N_2} \sim \mathrm{Unif}([l_i^{N_2}, u_i^{N_2}])$, and let $v_i^{N_2,*} \in \{0,1\}$ be the (unknown) value taken by a_i at N_2's optimum. Then:*

(a) The transfer probability of the direct rule equals the posterior probability of agreement with N_1:

$$\Pr_{\pi_i}\left[v_i^{N_2,*} = v_i^{N_1} \,\middle|\, l_i^{N_2}, u_i^{N_2}\right] \;=\; p_i^{\textsf{direct}}.$$

(b) The hint value `VarHintVal`$(a_i) = (p_i \geq 0.5)\,?\,v_i^{N_1} \;:\; 1 - v_i^{N_1}$ is the maximum a posteriori *estimate of $v_i^{N_2,*}$ under π_i.*

(c) The hint priority `VarHintPri`$(a_i) = \max(1, \mathrm{round}(100 \cdot |2p_i - 1|))$ is monotone in the Bayes-risk reduction $\mathbb{E}_{\pi_i}[\mathbf{1}\{v_i^{N_2,*} \neq \hat{v}\}] - \mathbb{E}_{\pi_i}[\mathbf{1}\{v_i^{N_2,*} \neq MAP\}]$ over the uniform random hint $\hat{v}$.

Proof. The big-M encoding ties $a_i = \mathbf{1}\{\hat{z}_i^{N_2} \geq 0\}$ (up to tie-breaking on the measure-zero set $\hat{z}_i^{N_2} = 0$). Under π_i,

$$\Pr_{\pi_i}\left[\hat{z}_i^{N_2} > 0\right] \;=\; \frac{u_i^{N_2}}{u_i^{N_2} - l_i^{N_2}}, \qquad \Pr_{\pi_i}\left[\hat{z}_i^{N_2} < 0\right] \;=\; \frac{-\,l_i^{N_2}}{u_i^{N_2} - l_i^{N_2}}.$$

Substituting into the direct case split gives (a). Part (b) follows because the MAP estimate under a Bernoulli posterior with parameter q is $\mathbf{1}\{q \geq 0.5\}$; the direct rule assigns $v_i^{N_1}$ when $p_i \geq 0.5$ and $1 - v_i^{N_1}$ otherwise, which is exactly the MAP value. Part (c) is immediate from $|2p_i - 1|$ being the total-variation distance between the posterior and the uniform distribution. $\square$

Why prev is biased. The prev rule computes p_i on the interval $\tilde{I}_i = [\hat{z}_i^{N_1}+d_i^{\downarrow}, \hat{z}_i^{N_1}+d_i^{\uparrow}]\cap[l_i^{N_2}, u_i^{N_2}]$, which is a strict subinterval of $[l_i^{N_2}, u_i^{N_2}]$ whenever the diff bound is narrower than the tightened bound. $\tilde{I}_i$ is not the maximum-entropy support of $\hat{z}_i^{N_2}$: it presumes that N_2's optimum will evaluate $\hat{z}_i^{N_2}$ near $\hat{z}_i^{N_1}$, a commitment that the big-M encoding does not license (the two MIPs can pick different inputs). Consequently prev is a biased estimator of $\Pr[v_i^{N_2,*} = v_i^{N_1}]$, with bias proportional to the diff-bound width $d_i^{\uparrow} - d_i^{\downarrow}$. This explains the empirical self-calibration observation already noted above: on instances where the diff bound is wide relative to $[l_i^{N_2}, u_i^{N_2}]$, direct's Bayes-optimal pull toward $p_i = 0.5$ correctly suppresses the priority, whereas prev emits a confidently-wrong hint.

Applicability. The rule runs only for binaries that survive Technique 2's elimination. When $l_i^{N_2} \geq 0$ or $u_i^{N_2} \leq 0$ the binary is already fixed and removed from the MIP, so there is nothing to hint.

Hints vs. candidate solution — and why we use both channels. Gurobi exposes two advisory channels for biasing the search. Soft *variable hints* (`VarHintVal`/`VarHintPri`) nudge each binary independently with a confidence weight; Gurobi is free to ignore any individual hint the LP relaxation disagrees with. A full *MIP Start* (`Start`) submits a candidate assignment that Gurobi's start heuristic tries to turn into a feasible incumbent. Start is a stronger commitment: if N_1's solution is infeasible for N_2, Gurobi spends effort repairing (or discarding) it. Blindly transferring the entire N_1 primal as a Start was harmful in all tested combinations ("MIP Start mode", all LOSER) — which motivated first the soft-hint modes (prev, direct), and then the PGD-consensus Start modes (prev_pgd, direct_pgd) that only emit a Start bit when the N_1-transfer hint and the PGD warmstart agree (or when only one of them has an opinion). The consensus filter replaces the "repair-everything" failure mode with a "warmstart only the bits both signals trust" policy.

Soundness: δ_{exact} is preserved under all five modes. We claim $\delta_{\text{varHint}} = \delta_{\text{exact}}$ for every mode $\in$ {off, prev, direct, prev_pgd, direct_pgd} whenever Gurobi terminates at OPTIMAL. The argument is uniform:

- **No constraint, coefficient, or bound is modified by any mode.** prev/direct only set the advisory Gurobi attributes `VarHintVal`/`VarHintPri`; prev_pgd/direct_pgd only call `set_start_value` (which maps to Gurobi's `Start` attribute) or withdraw it (`set_start_value`$(\cdot,$`nothing`$)$). None of `VarHintVal`, `VarHintPri`, `Start` participates in the constraint system; they are inputs to Gurobi's heuristics and branching priority, not to feasibility. In particular, `Start` does *not* bound the variable — the feasible set is whatever the big-M/triangle constraints define, independent of any Start value.
- **Feasible-set invariance.** Consequently:

$$\mathcal{F}_2^{(\mathsf{off})} = \mathcal{F}_2^{(\mathsf{prev})} = \mathcal{F}_2^{(\mathsf{direct})} = \mathcal{F}_2^{(\mathsf{prev\_pgd})} = \mathcal{F}_2^{(\mathsf{direct\_pgd})} = \mathcal{F}_2.$$

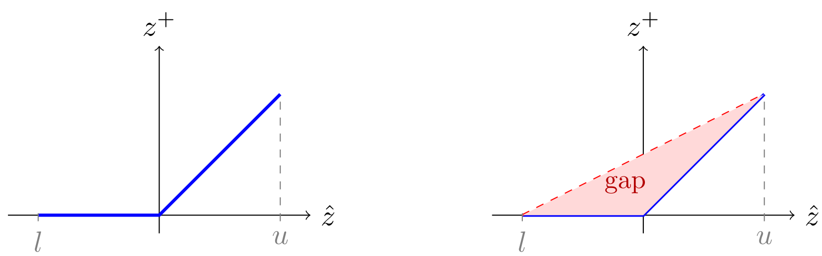
  Since $\delta_{\text{exact}} \triangleq \max_{x\in\mathcal{F}_2} \text{obj}_{N_2}(x)$ is defined purely in terms of $\mathcal{F}_2$ and the unchanged objective, δ_{exact} is identical across all five modes.
- **Consequence at termination.** At OPTIMAL, Gurobi returns $\max_{x\in\mathcal{F}_2^{(\mathsf{mode})}} \text{obj}_{N_2}(x) = \delta_{\text{exact}}$ in every mode. At TIME_LIMIT the reported incumbent is a sound lower bound on δ_{exact} whose numeric value may differ across modes (hints shift the branching path and which incumbents are found first), but the underlying δ_{exact} is fixed.

This covers both the soft-hint modes and the PGD-consensus Start modes in a single argument — neither channel writes a constraint, so the optimum is locked to δ_{exact} regardless of whether the hint is right, wrong, or contradictory. The only thing a bad hint costs is search time.

4.4 Technique 4 — N2 Triangle Relaxation (BoundTightPertRelax)

Flag: `--adv_std_n2_relax_threshold` $\in$ {off, 0.01, 0.1, 0.5, 1.0} (in safe combos; off means Technique 4 is disabled). **Model modification:** replaces selected N2 binary variables with LP relaxations. **Filename tag:** runs with $\tau \geq 0$ emit `_BoundTightPertRelax`$\{\tau\}$ in place of `_boundTight` (the tag implies bound-tightening is active). The retired `_relaxT`$\{\tau\}$ tag is not accepted by the post-processing pipeline.

Background: exact vs. relaxed ReLU encoding. Recall from Section 3 that a split ReLU neuron ($l < 0 < u$) can be encoded in two ways:



(a) Exact encoding
4 constraints + binary a
z^+ on the blue line only

(b) Triangle relaxation
3 constraints, no binary
z^+ anywhere in shaded region

- **Exact** (a): uses a binary variable $a \in \{0, 1\}$ and 4 constraints (Eqs. (1)–(4)). The post-ReLU value z^+ is forced onto the exact ReLU curve (blue line). This is precise but expensive: the binary makes the problem NP-hard.
- **Relaxed** (b): drops the binary and uses 3 constraints (Eqs. (5)–(7)). z^+ can be anywhere inside the triangle (shaded region). This is an over-approximation: the true ReLU value is always inside the triangle, but the triangle also contains points that are not on the ReLU curve. The shaded area between the blue line and the red dashed line is the *gap* — the approximation error introduced by the relaxation.

The idea. When the gap is very small (the triangle is very thin), the relaxation is almost as precise as the exact encoding, so there is little benefit in paying for a binary variable. Technique 4 identifies these "nearly stable" neurons using N_2's per-copy tightened bounds (from Technique 2), evaluated separately for the original copy $N_2(v_{\text{in}})$ and the perturbed copy $N_2(v_{x_0})$. Because the two copies can have substantially different tightened bounds under perturbation, a single neuron may be nearly stable in one copy and fully split in the other — Technique 4 exploits this asymmetry with a tiered rule (below).

Triangle gap area. For a ReLU neuron with pre-activation bounds $[l, u]$ from the relevant N2 copy, the gap area is:

$$\text{gap}(l, u) \;=\; \begin{cases} 0 & \text{if } l \geq 0 \text{ or } u \leq 0 \quad (\text{stable — no gap at all}), \\ \dfrac{u \cdot |l|}{2\,(u - l)} & \text{otherwise (split neuron).} \end{cases} \tag{15}$$

This is the area of the shaded triangle in diagram (b) above. When l is close to 0 (neuron almost always active) or u is close to 0 (neuron almost always inactive), the gap is small. When $|l| \approx u$ (neuron is maximally ambiguous), the gap is largest.

Decision rule (tiered, per-copy). For each ReLU neuron (m, k) during N_2's MIP construction, using the per-copy final tightened bounds from Technique 2's integration step (Sources A/B), compute the two per-copy gap areas:

$$g^{\mathrm{org}}_{m,k} = \mathrm{gap}\big(l^{N_2,\mathrm{org}}_{m,k},\, u^{N_2,\mathrm{org}}_{m,k}\big), \qquad g^{\mathrm{pert}}_{m,k} = \mathrm{gap}\big(l^{N_2,\mathrm{pert}}_{m,k},\, u^{N_2,\mathrm{pert}}_{m,k}\big).$$

Apply the **tiered rule**:

1. If $\max\big(g^{\mathrm{org}}_{m,k},\, g^{\mathrm{pert}}_{m,k}\big) \le \tau$: **relax both** copies — both are nearly stable, so pay for no binary variable at this neuron.
2. Else if $\min\big(g^{\mathrm{org}}_{m,k},\, g^{\mathrm{pert}}_{m,k}\big) \le \tau$: **relax only the smaller-gap copy**; the other copy keeps its exact big-M encoding. Tier 2 is what makes this BoundTightPertRelax rather than a symmetric τ-rule: it picks up asymmetric opportunities where one copy is nearly stable and the other is fully split (common under perturbation, where pert-bounds widen relative to org).
3. Else: **keep both exact** — both copies are genuinely contested, so the binary is worth its cost.

The perturbed-interval coupling constraints (always active in advstd safe combinations) link the relaxed copy's post-ReLU value to the exact one via $z^+_{\mathrm{pert}} - z^+_{\mathrm{org}} \in [\Delta^{\downarrow}, \Delta^{\uparrow}]$, so relaxing one copy does not let the two copies drift apart in the joint feasible set.

Why N_2's per-copy bounds (not N_1's)? Three reasons. (i) The triangle constraints we emit use these same per-copy bounds — by deciding on them we avoid any drift between "the interval we decided was thin" and "the interval of the triangle we wrote into the MIP"; see the consistency requirement below. (ii) Under non-trivial perturbations, N_2's org- and pert-copy tightened bounds can differ substantially (the pert copy sees wider input intervals), so a single-gap rule would miss the asymmetric opportunities Tier 2 exploits. (iii) The name BoundTightPertRelax reflects this dependency on Technique 2's per-copy bound tightening.

Consistency requirement. The decision-time gaps $g^{\mathrm{org}}_{m,k}$ and $g^{\mathrm{pert}}_{m,k}$ are computed from the *same* intersected bounds used by the emitted triangle constraints. Concretely, both sites route through a shared helper `intersect_per_copy_bounds` in `core_ops.jl`, so there is no drift between "bounds we decided on" and "bounds we wrote into the MIP." This is the load-bearing invariant of the soundness argument below.

Effect on the MIP. Each relaxed neuron removes one binary variable and one constraint from the MIP. For a network with hundreds of split neurons, relaxing those with small gap can remove dozens of binaries, making the branch-and-bound search significantly faster.

Effect on the result. The relaxation is an *over-approximation*: the triangle contains all points on the exact ReLU curve plus some extra points. This means:

$$\delta_{\mathrm{relaxed}} \;\ge\; \delta_{\mathrm{exact}}.$$

The reported δ may be slightly higher (looser) than the exact optimum, but it is always a valid upper bound. In practice the error grows with τ, but only neurons with gap area $\leq \tau$ are relaxed, so the encoding remains close to exact for the thin-triangle neurons that dominate the relaxation budget.

Threshold values. Technique 4 is swept over $\tau \in \{\mathsf{off}, 0.01, 0.1, 0.5, 1.0\}$:

- $\tau = \mathsf{off}$: Technique 4 disabled — every split neuron keeps both binaries. Useful as a baseline when the other techniques already deliver speedup.
- $\tau = 0.01$: very conservative — only neurons with a tiny per-copy gap are relaxed. Few binaries removed, but almost no looseness.
- $\tau = 0.1$: modestly aggressive — more neurons relaxed, more binaries removed, slightly looser bounds.
- $\tau = 0.5$ or 1.0: most aggressive — relaxes any split neuron whose triangle gap is up to 0.5 or 1.0 in area. The tiered rule mitigates the worst-case cost of aggressive thresholds (only the smaller-gap copy is relaxed when the wider copy would blow past τ), but per-instance slowdowns are still possible; empirical safe-set selection under BoundTightPertRelax is pending regeneration of the ranking.

Soundness. Let $R \subseteq \{(m, k, \text{copy})\}$ be the set of (neuron, copy) pairs selected for relaxation by the tiered rule, and let $\mathcal{F}_2^{(\{6\})}$ denote the feasible set of the resulting N2 MIP. We claim $\mathcal{F}_2 \subseteq \mathcal{F}_2^{(\{6\})}$, hence $\delta_{\text{BTPR}} \geq \delta_{\text{exact}}$.

- For $(m, k, \text{copy}) \notin R$ the exact big-M encoding is emitted, identical to the baseline; every $x \in \mathcal{F}_2$ trivially satisfies the same constraints.
- For $(m, k, \text{copy}) \in R$ the binary $a_{m,k,\text{copy}}$ and its four big-M constraints are dropped; the three triangle inequalities are emitted on the *same* per-copy bound $[l_{m,k}^{N_2,\text{copy}}, u_{m,k}^{N_2,\text{copy}}]$ that would have gated the exact encoding (the consistency requirement above). By the triangle-relaxation lemma (Eqs. (5)–(7)), every $(\hat{z}, z^+)$ satisfying the exact ReLU encoding on $[l_{m,k}^{N_2,\text{copy}}, u_{m,k}^{N_2,\text{copy}}]$ lies inside the triangle on the same interval. So every $x \in \mathcal{F}_2$ satisfies the relaxed constraints.

All other constraints (perturbed-interval coupling, VHAGaR dependencies, inter-layer linking, warmstart hints as non-constraining hints) are unchanged. Therefore $\mathcal{F}_2 \subseteq \mathcal{F}_2^{(\{6\})}$, and taking the max of the unchanged objective over a superset can only raise the value: $\delta_{\text{BTPR}} = \max_{x \in \mathcal{F}_2^{(\{6\})}} \text{obj}_{N_2}(x) \geq \max_{x \in \mathcal{F}_2} \text{obj}_{N_2}(x) = \delta_{\text{exact}}$.

5 Algorithms

All 25 safe combinations share the same two-phase structure (Algorithm 1 and Algorithm 2). They differ only in which optional sub-steps are enabled, as indicated by the mode flag $\textsc{bp} \in \{\text{off}, \text{rank}, \text{decay}\}$, the Boolean flag $\textsc{zono}$, the five-valued mode flag $\textsc{varHint} \in \{\text{off}, \text{prev}, \text{direct}, \text{prev\_pgd}, \text{direct\_pg}$
and the threshold τ.

Algorithm 1 Phase 1: Solve N_1 and extract state (shared by all safe combos)

Require: Network N_1, perturbation (P, ε), class pairs (c_s, c_t), timeout T
Ensure: State directory $\mathcal{S}$ on disk

for each class pair (c_s, c_t) **do**
  $x_{\text{pgd}} \leftarrow \text{HyperAttack}(N_1, c_s, c_t, P, \varepsilon)$ {PGD warm-start}
  Build MIP $\mathcal{M}_1$ for N_1 with standard encoding
  Apply PGD hints to $\mathcal{M}_1$
  Add VHAGaR dependency constraints to $\mathcal{M}_1$
  Add perturbed interval constraints to $\mathcal{M}_1$
  `optimize!`$(\mathcal{M}_1, T)$
  Extract primal solution $(\text{var}_i, \text{val}_i)_i$
  Extract neuron bounds $(l_{m,k}, u_{m,k})_{\forall m,k}$ from `layers_info`
  Serialise to $\mathcal{S}/(c_s, c_t)$
end for

Algorithm 2 Phase 2: Solve N_2 with advstd techniques (parameterised)

Require: Networks N_1, N_2, perturbation (P, ε), class pairs, timeout T, N1 state $\mathcal{S}$
Require: Flags: $\textsc{bp} \in \{\text{off}, \text{rank}, \text{decay}\}$, $\textsc{zono} \in \{\text{false}, \text{true}\}$, $\textsc{varHint} \in \{\text{off}, \text{prev}, \text{direct}, \text{prev\_pgd}, \text{direct\_pgd}\}$, $\tau \in \{\text{off}, 0.01, 0.1, 0.5, 1.0\}$
Ensure: Global robustness bound $\delta^*_{N_2}$ for each (c_s, c_t)

// **Technique 2: Precompute N2 bounds from N1 (once, before the loop)**
Load N_1 diff bounds from $\mathcal{S}$
if $\textsc{zono}$ **then**
  Compute zonotope diff bounds (Source A) between N_1 and N_2
  Compute N1-tightened absolute N2 zonotope (Source B), intersect with Source A
else
  Compute interval-arithmetic diff bounds between N_1 and N_2
end if
for each class pair (c_s, c_t) **do**
  // **Skip if result already exists for this pair**
  if result file contains (c_s, c_t) **then**
    continue
  end if
  Load N_1 state from $\mathcal{S}/(c_s, c_t)$: primal $(\text{var}_i, \text{val}_i)$, bounds $(l_{m,k}, u_{m,k})$
  // **Technique 2: Set N1 neuron bounds for N2 encoding**
  Inject $(l_{m,k}, u_{m,k})$ as initial N2 bounds {fixes stable ReLUs, tightens big-M}
  if $\tau \neq \text{off}$ **then**
    // **Technique 4 (BoundTightPertRelax): tiered per-copy N2 relaxation**
    Call `compute_n2_relax_decision!`(τ) {fills dict $(m, k) \to (\text{relax\_org}, \text{relax\_pert})$}
    for each ReLU neuron (m, k) in N2, separately for copy $\in \{\text{org}, \text{pert}\}$ **do**
      Compute $g^{\text{copy}}_{m,k} = \text{gap}\big(l^{N_2,\text{copy}}_{m,k}, u^{N_2,\text{copy}}_{m,k}\big)$ from per-copy final bounds
      Set $(\text{relax\_org}, \text{relax\_pert}) \leftarrow \begin{cases} (\text{true}, \text{true}) & \max(g^{\text{org}}, g^{\text{pert}}) \leq \tau \\ (\text{true}, \text{false}) & g^{\text{org}} \leq g^{\text{pert}},\ g^{\text{org}} \leq \tau \\ (\text{false}, \text{true}) & g^{\text{pert}} < g^{\text{org}},\ g^{\text{pert}} \leq \tau \\ (\text{false}, \text{false}) & \text{else} \end{cases}$
    end for
    During MIP build: for each (m, k, copy), if flag true emit triangle LP relaxation on $[l^{N_2,\text{copy}}_{m,k}, u^{N_2,\text{copy}}_{m,k}]$, else emit exact big-$M$
  end if
  Build MIP $\mathcal{M}_2$ for N_2 {with tightened bounds and relaxed neurons}
  $x_{\text{pgd}} \leftarrow \text{HyperAttack}(N_2, c_s, c_t, P, \varepsilon)$
  Apply PGD hints to $\mathcal{M}_2$
  if $\textsc{bp} \in \{\text{rank}, \text{decay}\}$ **then**
    // **Technique 1: Branch priorities from N1 bound gaps (Section 4.2)**
    Build $\{(m, k) \to g_{m,k} = |u_{m,k} - l_{m,k}|\}$ for every split-ReLU binary in $\mathcal{M}_2$
    if $\textsc{bp} = \text{rank}$ **then**
      Set `BranchPriority`$(a_{m,k}) \leftarrow \text{rank\_to\_priority}(G_{\max} - g_{m,k} + \varepsilon)$ $\{G_{\max} = \max_{m,k} g_{m,k}$, $\varepsilon \approx 10^{-9}$, image $\subseteq [1, 100]\}$
    else
      $g_{\text{med}} \leftarrow \text{median}\{g_{m,k}\}$ {fallback to 1 if all $g = 0$}
      Set `BranchPriority`$(a_{m,k}) \leftarrow \max\big(1, \text{round}\big(100 \cdot \exp(-g_{m,k}/g_{\text{med}})\big)\big)$
    end if
  end if
  if $\textsc{varHint} \neq \text{off}$ **then**
    // **Technique 3: Transfer-probability hints. Mode $\in$ {prev, direct, prev_pgd, direct_pgd}.**
    for each surviving binary a_i in $\mathcal{M}_2$ (with $l^{N_2}_i < 0 < u^{N_2}_i$) **do**

5.1 Instantiation for Each Safe Combination

[STALE] The 25 combinations below were ranked under the retired `_relaxT`-mode Technique 4. Under the new BoundTightPertRelax rule (Section 4.4) the ranking is being regenerated; until then, treat these rows as a historical reference and do not use them to pick safe combos for new experiments.

Table 1 maps each safe combination to its flag settings in Algorithm 2. To reproduce any combination, set the five flags accordingly; all other aspects of the algorithm are identical.

#	BP	VARHINT	ZONO	τ
1	off	true	true	0.5
2	bounds	true	true	1.0
3	off	true	false	0.5
4	bounds	true	false	1.0
5	bounds	false	false	0.5
6	off	false	false	0.5
7	off	false	true	1.0
8	bounds	false	false	1.0
9	off	false	true	0.1
10	off	true	true	off
11	bounds	false	true	0.5
12	off	true	true	1.0
13	off	true	false	1.0
14	off	false	true	0.5
15	bounds	false	true	1.0
16	bounds	true	true	off
17	off	true	false	off
18	off	true	false	0.1
19	bounds	true	false	0.01
20	bounds	true	false	0.5
21	off	false	false	1.0
22	bounds	false	true	0.1
23	bounds	false	false	0.01
24	bounds	true	false	off
25	off	true	false	0.01

Table 1: Flag instantiation for each safe combination from the full ranking (rows 1–19 are dominant, rows 20–25 are avg-win). boundTight is always true and is not shown. n1Probe is always off and is not shown. τ = off disables Technique 4 entirely.

6 Summary of Effects

Table 2 summarises which Gurobi attribute each safe technique uses and whether it modifies the mathematical problem.

7 Command-Line Interface

A minimal invocation using the top-ranked safe combination (#1) is:

```
julia run.jl --mode advanced_standard_n2 \
  --dataset mnist --model_name cnn1 \
```

Technique	Flag (short)	Gurobi API	Modifies MIP?	N_1 state consu
1. Branch Prio rank/decay	bp=rank or bp=decay	`BranchPriority`	no	final neuron bo
2. Bound Tightening	boundTight	(linear constr.)	yes†	final neuron bo
2+. Zonotope Bounds	zono	(linear constr.)	yes†	diff bounds
3. Variable Hints	varHint=prev/direct	`VarHintVal`+`VarHintPri`	no	prev: primal +
3+. VarHints via PGD-consensus Start	varHint=prev_pgd/direct_pgd	`Start` (`set_start_value`)	no	same p_i-inputs
4. N2 Relaxation	BTPRelax=τ	(replaces binaries)	yes‡	N2 per-copy bo

Table 2: Summary of the safe advstd techniques. †Adds sound constraints derived from over-approximations; the optimum is preserved. ‡Replaces binaries with LP relaxations; the reported δ is a sound upper bound ($\geq \delta_{\text{exact}}$).

```
  --model_path  /path/to/N1/model.p \
  --model_path2 /path/to/N2/model.p \
  --perturbation patch --perturbation_size "1,14,14,3" \
  --ctag 1 --ct "4,5,6" --timout 1800 \
  --n1_state_dir /path/to/n1_state \
  --adv_std_mip_start          false  \
  --adv_std_branch_priorities  off    \  # also: rank / decay
  --adv_std_lp_basis           false  \
  --adv_std_bound_tightening   true   \
  --adv_std_zono_bounds        true   \
  --adv_std_n2_relax_threshold 0.5    \  # tiered per-copy; output tag
     _BoundTightPertRelax0.5
  --adv_std_var_hint           prev      # also: off / direct (see
     Technique 3)
```

The Python sweep script can run all 25 safe combinations automatically:

```
python3 run_relaxation_sweep.py --timeout 1800 \
  --arch_models cnn1=/path/to/n1 3x10=/path/to/n1 \
  --advanced_standard \
  --sweep_adv_std_mip_start false \
  --sweep_adv_std_lp_basis false \
  --sweep_adv_std_bound_tightening true \
  --sweep_adv_std_zono_bounds false true \
  --sweep_adv_std_n1_probe off \
  --sweep_adv_std_n2_relax_threshold off 0.01 0.1 0.5 1.0 \
  --sweep_adv_std_branch_priorities off rank decay \
  --sweep_adv_std_var_hint off prev direct \
  --advstd_safe_combos_only \
      paper_experiments/mnist/advstd_combo_ranking_vs_withPerturbed.csv \
  --sweep_gurobi_seed 0 1 2 3 4 --ct 4,5,6
```

The `--advstd_safe_combos_only` flag filters the cross-product to only validated combinations (perf_tier $\in$ {dominant, avg-win}), blocking known loser/risky combos while allowing untested new combinations through. Note: with the recent BoundTightPertRelax (Technique 4) and the new rank/decay branch-priority modes (Technique 1), combos from the legacy ranking are treated as "untested" (the CSV either lacks a `relax_mode` column, or its `branch_priorities` value is now mapped to `bp_legacy`) and admitted permissively — regenerate the ranking once a new sweep completes.

8 Safe-to-Use Combinations

Tables below list *every* recorded advstd combination at all seeds, all τ values, evaluated per-architecture and broken out by perturbation type and size. Each row corresponds to a single $(c_{\text{src}}, c_{\text{tgt}})$ cell of one combo (rows for the same combo are grouped, with the flag columns shown only on the first cell of the combo). We report wall-clock t_std and t_adv per cell, the per-cell speedup sp= t_std/t_adv (higher is better; >1 means advstd finished faster than the with-perturbed-intervals baseline), $\overline{\text{sp}}_{c\_\text{src}}$ (mean sp over all tested c_{tgt} for the same c_{src} within a combo, shown once per c_{src} group), and n_relax (per-cell number of N2 binary variables removed by Technique 4). The bound_tight column labels which Technique 2 variant is used (`interval` / `zono` / `none`; n1Probe is always off), and the τ column is the Technique 4 threshold. Within each (perturbation, size) block combos are sorted by descending mean sp (fastest advstd first); cells within a combo are sorted by $(c_{\text{src}}, c_{\text{tgt}})$. mipStart and lpBasis are always off and omitted for brevity. Cells where both modes hit Gurobi's `TIME_LIMIT` are shown with their wall-clock times; for those cells, a separate per-arch gap-comparison table reports how often advstd reached a tighter $u-l$ bound on the objective than standard, alongside the mean gap of each method.

#	bound_tight	bp	varHint	τ	$\overline{\text{sp}}$	n_wins	n_cells	win_rate	⟨arch, pert, p_size⟩ with $\overline{\text{sp}} < 1$ on every c_{src}	⟨arch, c_{src}⟩ with $\overline{\text{sp}} > 1$ on every (pert, p_size)	missing ⟨arch, pert, p_size⟩ (not yet examined)
1	zono	off	direct_pgd	0.5	**1.297**	94	139	**67.63%**	`<3x10,translation,3,3>`; `<3x10,occ,5,5,5>`	*none*	*none*
2	zono	off	prev_pgd	0.5	**1.302**	91	137	**66.42%**	`<3x10,translation,3,3>`	`<cnn1,1>`	*none*
3	interval	off	prev_pgd	0.5	**1.262**	90	139	**64.75%**	`<3x10,translation,3,3>`; `<3x10,occ,5,5,5>`	*none*	*none*
4	zono	off	prev_pgd	0.0	**1.269**	88	136	**64.71%**	`<3x10,translation,3,3>`; `<3x10,occ,5,5,5>`	*none*	*none*
5	interval	off	no	0.5	**1.219**	87	135	**64.44%**	*none*	*none*	*none*
6	interval	off	prev	0.0	**1.230**	26	41	**63.41%**	`<cnn1,translation,3,1>`	`<cnn1,1>`*	`<3x10,occ,5,5,5>`; `<3x10,patch,1,14,14,3>`; `<3x10,translation,1,1>`; `<3x10,translation,1,3>`; `<3x10,translation,3,1>`; `<3x10,translation,3,3>`
7	interval	off	prev_pgd	0.0	**1.242**	87	139	**62.59%**	`<3x10,translation,3,3>`; `<3x10,occ,5,5,5>`	*none*	*none*
8	interval	off	direct_pgd	0.5	**1.268**	85	137	**62.04%**	`<3x10,translation,3,3>`; `<3x10,occ,5,5,5>`	`<cnn1,0>`	*none*
9	zono	off	no	0.5	**1.145**	84	139	**60.43%**	`<cnn1,translation,3,1>`	*none*	*none*

Table 3: Overall advstd combination ranking at all seeds, all τ values (part 1/3, combos ranked 1–9 of 26 by win_rate) — one row per flag combination, aggregated over every (arch, pert, size, c_{src}, c_{tgt}) cell in the per-arch tables below. $\overline{\text{sp}}$: mean per-cell sp = t_std/t_adv; n_wins counts cells with sp > 1; win_rate=n_wins/n_cells (bold when $> 50\%$). losers: ⟨arch, pert, p_size⟩ triples where every tested c_{src} averaged sp < 1. winners: ⟨arch, c_{src}⟩ pairs where every tested (pert, p_size) averaged sp > 1; * flags partial coverage. missing: ⟨arch, pert, p_size⟩ triples present somewhere in the dataset but not exercised by this combo (i.e., coverage gaps yet to be examined). Sorted by descending win_rate, then $\overline{\text{sp}}$; $\overline{\text{sp}} > 1$ in **bold**. Blue rows: bp=off.

#	bound_tight	bp	varHint	τ	$\overline{\text{sp}}$	n_wins	n_cells	win_rate	⟨arch, pert, p_size⟩ with $\overline{\text{sp}} < 1$ on every c_{src}	⟨arch, c_{src}⟩ with $\overline{\text{sp}} > 1$ on every (pert, p_size)	missing ⟨arch, pert, p_size⟩ (not yet examined)

10	interval	off	no	0.1	**1.119**	51	85	**60.00%**	`<3x10,translation,3,1>;`
									`<3x10,translation,1,3>;`
									`<cnn1,occ,5,5,5>;`
									`<3x10,patch,1,14,14,3>`
										`<cnn1,1>`*	*none*

11	interval	off	prev	0.5	**1.285**	22	37	**59.46%**	`<cnn1,patch,1,14,14,3>`	`<cnn1,1>`*	`<3x10,cnn1,occ,5,5,5>;`
											`<3x10,patch,1,14,14,3>;`
											`<3x10,translation,1,1>;`
											`<3x10,translation,1,3>;`
											`<3x10,translation,3,1>;`
											`<3x10,translation,3,3>`

12	interval	off	direct_pgd	0.0	**1.207**	79	133	**59.40%**	`<3x10,translation,3,3>`	*none*	*none*

13	zono	off	prev	0.0	**1.104**	19	32	**59.38%**	`<cnn1,translation,1,3>;`
									`<cnn1,translation,3,1>;`
									`<cnn1,patch,1,14,14,3>`
										none	`<3x10,cnn1,occ,5,5,5>;`
											`<3x10,patch,1,14,14,3>;`
											`<3x10,translation,1,1>;`
											`<3x10,translation,1,3>;`
											`<3x10,translation,3,1>;`
											`<3x10,translation,3,3>`

14	zono	off	no	0.0	**1.122**	81	137	**59.12%**	`<cnn1,translation,1,1>`	*none*	*none*
15	zono	off	direct_pgd	0.0	**1.213**	77	131	**58.78%**	`<3x10,translation,3,3>;` `<3x10,occ,5,5,5>`	*none*	*none*

16	zono	off	direct	0.5	**1.209**	48	85	**56.47%**	`<3x10,patch,1,14,14,3>;`
									`<3x10,translation,3,3>;`
									`<cnn1,occ,5,5,5>;`
									`<3x10,translation,1,3>;`
									`<3x10,translation,3,1>`
										`<cnn1,1>`*	*none*

17	interval	off	no	0.0	**1.127**	75	136	**55.15%**	`<3x10,translation,3,3>`	*none*	*none*

18	zono	off	no	0.01	**1.049**	44	81	**54.32%**	`<3x10,cnn1,`
									`translation,3,1>;`
									`<3x10,cnn1,`
									`translation,1,3>;`
									`<cnn1,occ,5,5,5>;`
									`<3x10,patch,1,14,14,3>`
										`<cnn1,1>`*	*none*

Table 4: Overall advstd combination ranking (part 2/3, combos ranked 10–18 of 26 by win_rate); continuation of Table 3. Column meanings: see Table 3.

#	bound_tight	bp	varHint	τ	$\overline{\mathrm{sp}}$	n_wins	n_cells	win_rate	$\langle$arch, pert, p_size$\rangle$ with $\overline{\mathrm{sp}} < 1$ on every c_{src}	$\langle$arch, $c_{\mathrm{src}}\rangle$ with $\overline{\mathrm{sp}} > 1$ on every (pert, p_size)	missing $\langle$arch, pert, p_size$\rangle$ (not yet examined)
19	interval	off	direct	0.5	**1.198**	46	85	**54.12%**	<3x10,patch,1,14,14,3>; <3x10,translation,1,3>; <3x10,translation,3,3>; <3x10,occ,5,5,5>	<cnn1,1>*	*none*
20	interval	off	no	0.05	**1.048**	44	84	**52.38%**	<3x10,cnn1, translation,3,1>; <3x10,translation,1,3>; <cnn1,occ,5,5,5>; <3x10,translation,3,3>	*none*	*none*
21	interval	off	no	0.01	**1.061**	41	81	**50.62%**	<3x10,cnn1, translation,3,1>; <3x10,cnn1, translation,1,3>; <3x10,translation,3,3>	<cnn1,1>*	*none*
22	zono	off	direct	0.0	**1.098**	42	83	**50.60%**	<3x10,patch,1,14,14,3>; <3x10,translation,3,3>; <3x10,cnn1, translation,3,1>; <3x10,occ,5,5,5>; <3x10,translation,1,3>	*none*	*none*
23	zono	off	no	0.1	**1.043**	39	82	47.56%	<3x10,cnn1, translation,3,1>; <3x10,cnn1, translation,1,3>; <cnn1,occ,5,5,5>; <3x10,translation,1,1>; <3x10,translation,3,3>	<cnn1,1>*	*none*
24	interval	off	direct	0.0	**1.098**	40	86	46.51%	<3x10,cnn1, translation,3,1>; <3x10,patch,1,14,14,3>; <3x10,occ,5,5,5>; <3x10,translation,3,3>	*none*	*none*
25	zono	off	no	0.05	**1.032**	36	80	45.00%	<3x10,cnn1, translation,3,1>; <3x10,cnn1, translation,1,3>; <cnn1,occ,5,5,5>; <3x10,patch,1,14,14,3>; <3x10,translation,1,1>	<cnn1,1>*	*none*
26	zono	off	prev	0.5	**1.004**	12	32	37.50%	<cnn1,translation,1,3>; <cnn1,translation,3,1>; <cnn1,patch,1,14,14,3>	<cnn1,1>*	<3x10,cnn1,occ,5,5,5>; <3x10,patch,1,14,14,3>; <3x10,translation,1,1>; <3x10,translation,1,3>; <3x10,translation,3,1>; <3x10,translation,3,3>

Table 5: Overall advstd combination ranking (part 3/3, combos ranked 19–26 of 26 by win_rate); continuation of Table 3. Column meanings: see Table 3.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}}-\delta_{\text{adv}}\rvert}_{c\_\text{src}}$
patch	1,14,14,3	interval	off	prev_pgd	0.5	0	3	0	14.75	12.51	**1.179**	*1.145*	*0.01*
						0	4	0	15.24	12.50	**1.219**		
						0	5	0	16.13	12.50	**1.290**		
						0	6	0	13.07	12.83	**1.019**		
						0	7	0	13.37	12.01	**1.113**		
						0	8	0	13.05	12.45	**1.048**		
		interval	off	no	0.5	0	3	0	14.75	12.87	**1.146**	*1.134*	*0.00*
						0	4	0	15.24	12.77	**1.193**		
						0	5	0	16.13	12.87	**1.253**		
						0	7	0	13.37	11.02	**1.213**		
						0	8	0	13.05	15.13	0.863		
		zono	off	no	0.5	0	3	0	14.75	12.02	**1.227**	*1.134*	*0.01*
						0	4	0	15.24	13.78	**1.106**		
						0	5	0	16.13	12.36	**1.305**		
						0	6	0	13.07	13.36	0.978		
						0	7	0	13.37	12.50	**1.070**		
						0	8	0	13.05	11.70	**1.115**		
		interval	off	direct_pgd	0.0	0	3	0	14.75	12.51	**1.179**	*1.122*	*0.01*
						0	4	0	15.24	12.85	**1.186**		
						0	5	0	16.13	13.00	**1.241**		
						0	6	0	13.07	13.27	0.985		
						0	8	0	13.05	12.80	**1.020**		
		interval	off	direct_pgd	0.5	0	3	0	14.75	12.90	**1.143**	*1.119*	*0.01*
						0	4	0	15.24	12.55	**1.214**		
						0	5	0	16.13	12.82	**1.258**		
						0	6	0	13.07	13.27	0.985		
						0	7	0	13.37	12.52	**1.068**		
						0	8	0	13.05	12.49	**1.045**		
		zono	off	direct_pgd	0.5	0	3	0	14.75	12.64	**1.167**	*1.098*	*0.01*
						0	4	0	15.24	12.68	**1.202**		
						0	5	0	16.13	12.63	**1.277**		
						0	6	0	13.07	13.68	0.955		
						0	7	0	13.37	14.30	0.935		
						0	8	0	13.05	12.40	**1.052**		

Table 6: Architecture **3x10** — every recorded advstd combination at all seeds, all τ values, perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean `sp`) (6 combos, 34 cell rows; see Tables labelled `tab:safe_3x10_*` for the other perturbations / c_{src} / τ-buckets of this arch). Rows shaded green satisfy `bp=off`; `sp`>1 (advstd faster) in **bold**. Combos sorted by descending mean `sp`; cells within a combo by $(c_{\text{src}}, c_{\text{tgt}})$. Total 1193 cell rows across this architecture.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
patch	`1,14,14,3`	zono	off	no	0.0	0	3	0	14.75	12.33	**1.196**	***1.085***	*0.00*
						0	4	0	15.24	14.17	**1.076**		
						0	5	0	16.13	13.72	**1.176**		
						0	6	0	13.07	12.88	**1.015**		
						0	7	0	13.37	13.88	0.963		
		zono	off	prev_pgd	0.0	0	3	0	14.75	13.67	**1.079**	***1.085***	*0.00*
						0	4	0	15.24	13.98	**1.090**		
						0	5	0	16.13	13.34	**1.209**		
						0	6	0	13.07	13.51	0.967		
						0	7	0	13.37	12.40	**1.078**		
		zono	off	prev_pgd	0.5	0	3	0	14.75	13.68	**1.078**	***1.069***	*0.01*
						0	4	0	15.24	14.87	**1.025**		
						0	5	0	16.13	13.04	**1.237**		
						0	6	0	13.07	12.64	**1.034**		
						0	7	0	13.37	12.20	**1.096**		
						0	8	0	13.05	13.82	0.944		
		zono	off	direct_pgd	0.0	0	3	0	14.75	14.53	**1.015**	***1.024***	*0.00*
						0	4	0	15.24	16.43	0.928		
						0	5	0	16.13	12.98	**1.243**		
						0	6	0	13.07	13.51	0.967		
						0	7	0	13.37	13.82	0.967		
		interval	off	no	0.0	0	3	0	14.75	14.00	**1.054**	***1.020***	*0.01*
						0	4	0	15.24	13.15	**1.159**		
						0	5	0	16.13	15.18	**1.063**		
						0	6	0	13.07	14.08	0.928		
						0	7	0	13.37	12.98	**1.030**		
						0	8	0	13.05	14.68	0.889		
		interval	off	prev_pgd	0.0	0	3	0	14.75	20.83	0.708	*0.981*	*0.01*
						0	4	0	15.24	17.97	0.848		
						0	5	0	16.13	12.34	**1.307**		
						0	6	0	13.07	12.80	**1.021**		
						0	7	0	13.37	11.89	**1.124**		
						0	8	0	13.05	14.86	0.878		

Table 7: Architecture **3x10** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean `sp`); continuation of Table 6 (6 combos, 33 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
patch	`1,14,14,3`	interval	off	direct	0.5	0	3	0	14.75	16.31	0.904	*0.832*	*0.01*
						0	4	0	15.24	17.46	0.873		
						0	5	0	16.13	16.60	0.972		
						0	6	0	13.07	16.82	0.777		
						0	7	0	13.37	17.78	0.752		
						0	8	0	13.05	18.35	0.711		
		interval	off	direct	0.0	0	3	0	14.75	19.38	0.761	*0.829*	*0.01*
						0	4	0	15.24	25.41	0.600		
						0	5	0	16.13	14.74	**1.094**		
						0	6	0	13.07	14.36	0.910		
						0	7	0	13.37	16.16	0.827		
						0	8	0	13.05	16.66	0.783		
		zono	off	direct	0.5	0	3	0	14.75	17.13	0.861	*0.778*	*0.01*
						0	4	0	15.24	21.55	0.707		
						0	5	0	16.13	17.06	0.945		
						0	6	0	13.07	19.12	0.684		
						0	7	0	13.37	16.72	0.800		
						0	8	0	13.05	19.49	0.670		
		zono	off	direct	0.0	0	3	0	14.75	16.84	0.876	*0.772*	*0.01*
						0	4	0	15.24	23.08	0.660		
						0	5	0	16.13	16.45	0.981		
						0	6	0	13.07	21.37	0.612		
						0	7	0	13.37	17.35	0.771		
						0	8	0	13.05	17.79	0.734		

Table 8: Architecture **3x10** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean `sp`); continuation of Table 6 (4 combos, 24 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
patch	`1,14,14,3`	interval	off	prev_pgd	0.0	1	3	0	22.55	15.39	**1.465**	***1.504***	*0.02*
						1	4	0	23.72	16.10	**1.473**		
						1	5	0	26.84	17.94	**1.496**		
						1	6	0	25.81	15.82	**1.631**		
						1	7	0	26.70	16.51	**1.617**		
						1	8	0	22.57	16.85	**1.339**		
		interval	off	prev_pgd	0.5	1	3	0	22.55	15.48	**1.457**	***1.499***	*0.02*
						1	4	0	23.72	16.64	**1.425**		
						1	5	0	26.84	17.57	**1.528**		
						1	6	0	25.81	16.05	**1.608**		
						1	7	0	26.70	16.37	**1.631**		
						1	8	0	22.57	16.80	**1.343**		
		zono	off	direct_pgd	0.0	1	3	0	22.55	15.40	**1.464**	***1.484***	*0.01*
						1	4	0	23.72	16.28	**1.457**		
						1	5	0	26.84	18.67	**1.438**		
						1	6	0	25.81	16.28	**1.585**		
						1	7	0	26.70	16.47	**1.621**		
						1	8	0	22.57	16.83	**1.341**		
		zono	off	prev_pgd	0.0	1	3	0	22.55	17.33	**1.301**	***1.476***	*0.02*
						1	4	0	23.72	16.52	**1.436**		
						1	5	0	26.84	17.18	**1.562**		
						1	6	0	25.81	16.15	**1.598**		
						1	7	0	26.70	16.30	**1.638**		
						1	8	0	22.57	17.12	**1.318**		
		interval	off	direct_pgd	0.0	1	3	0	22.55	16.11	**1.400**	***1.471***	*0.01*
						1	4	0	23.72	16.96	**1.399**		
						1	5	0	26.84	19.21	**1.397**		
						1	6	0	25.81	15.59	**1.656**		
						1	7	0	26.70	15.80	**1.690**		
						1	8	0	22.57	17.53	**1.288**		
		zono	off	prev_pgd	0.5	1	3	0	22.55	16.41	**1.374**	***1.464***	*0.02*
						1	4	0	23.72	16.22	**1.462**		
						1	5	0	26.84	18.40	**1.459**		
						1	6	0	25.81	16.24	**1.589**		
						1	7	0	26.70	16.28	**1.640**		
						1	8	0	22.57	17.92	**1.259**		

Table 9: Architecture **3x10** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
patch	`1,14,14,3`	zono	off	direct_pgd	0.5	1	3	0	22.55	15.98	**1.411**	***1.461***	*0.01*
						1	4	0	23.72	15.96	**1.486**		
						1	5	0	26.84	18.85	**1.424**		
						1	6	0	25.81	15.97	**1.616**		
						1	7	0	26.70	16.23	**1.645**		
						1	8	0	22.57	19.03	**1.186**		
		interval	off	direct_pgd	0.5	1	3	0	22.55	16.00	**1.409**	***1.447***	*0.01*
						1	4	0	23.72	16.80	**1.412**		
						1	5	0	26.84	18.52	**1.449**		
						1	6	0	25.81	16.12	**1.601**		
						1	7	0	26.70	17.76	**1.503**		
						1	8	0	22.57	17.27	**1.307**		
		interval	off	no	0.5	1	3	0	22.55	17.04	**1.323**	***1.430***	*0.02*
						1	4	0	23.72	17.04	**1.392**		
						1	5	0	26.84	19.35	**1.387**		
						1	6	0	25.81	16.77	**1.539**		
						1	7	0	26.70	15.49	**1.724**		
						1	8	0	22.57	18.57	**1.215**		
		interval	off	no	0.0	1	3	0	22.55	17.29	**1.304**	***1.412***	*0.02*
						1	4	0	23.72	16.20	**1.464**		
						1	5	0	26.84	19.44	**1.381**		
						1	6	0	25.81	16.93	**1.525**		
						1	7	0	26.70	17.17	**1.555**		
						1	8	0	22.57	18.12	**1.246**		
		zono	off	no	0.0	1	3	0	22.55	16.54	**1.363**	***1.411***	*0.02*
						1	4	0	23.72	16.41	**1.445**		
						1	5	0	26.84	19.87	**1.351**		
						1	6	0	25.81	16.61	**1.554**		
						1	7	0	26.70	17.87	**1.494**		
						1	8	0	22.57	17.93	**1.259**		
		zono	off	no	0.5	1	3	0	22.55	20.87	**1.080**	***1.360***	*0.02*
						1	4	0	23.72	16.98	**1.397**		
						1	5	0	26.84	18.94	**1.417**		
						1	6	0	25.81	18.42	**1.401**		
						1	7	0	26.70	15.91	**1.678**		
						1	8	0	22.57	19.03	**1.186**		

Table 10: Architecture **3x10** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
occ	5,5,5	zono	off	no	0.5	0	3	0	21.76	13.58	**1.602**	***1.353***	*0.01*
						0	4	0	15.18	11.43	**1.328**		
						0	5	0	15.59	11.55	**1.350**		
						0	6	0	14.17	11.33	**1.251**		
						0	7	0	17.19	11.29	**1.523**		
						0	8	0	14.13	13.29	**1.063**		
		interval	off	no	0.5	0	3	0	21.76	14.62	**1.488**	***1.174***	*0.01*
						0	4	0	15.18	14.46	**1.050**		
						0	5	0	15.59	16.53	0.943		
						0	7	0	17.19	14.17	**1.213**		
		zono	off	no	0.0	0	3	0	21.76	14.30	**1.522**	***1.169***	*0.01*
						0	4	0	15.18	14.98	**1.013**		
						0	5	0	15.59	13.52	**1.153**		
						0	6	0	14.17	12.85	**1.103**		
						0	7	0	17.19	13.16	**1.306**		
						0	8	0	14.13	15.39	0.918		
		interval	off	no	0.0	0	3	0	21.76	17.07	**1.275**	***1.106***	*0.01*
						0	4	0	15.18	13.35	**1.137**		
						0	5	0	15.59	14.81	**1.053**		
						0	6	0	14.17	13.87	**1.022**		
						0	7	0	17.19	14.10	**1.219**		
						0	8	0	14.13	15.15	0.933		
		zono	off	direct	0.5	0	3	0	21.76	14.89	**1.461**	***1.046***	*0.00*
						0	4	0	15.18	14.40	**1.054**		
						0	5	0	15.59	14.83	**1.051**		
						0	7	0	17.19	19.20	0.895		
						0	8	0	14.13	18.45	0.766		
		zono	off	direct_pgd	0.0	0	3	0	21.76	18.52	**1.175**	*0.986*	*0.01*
						0	4	0	15.18	14.80	**1.026**		
						0	6	0	14.17	14.84	0.955		
						0	7	0	17.19	15.37	**1.118**		
						0	8	0	14.13	21.58	0.655		

Table 11: Architecture **3x10** — perturbation **occ**(5,5,5), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 6 (6 combos, 32 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
occ	5,5,5	interval	off	direct_pgd	0.0	0	3	0	21.76	17.22	**1.264**	*0.980*	*0.01*
						0	4	0	15.18	16.92	0.897		
						0	5	0	15.59	15.95	0.977		
						0	6	0	14.17	15.86	0.893		
						0	7	0	17.19	16.75	**1.026**		
						0	8	0	14.13	17.22	0.821		
		zono	off	direct	0.0	0	3	0	21.76	18.29	**1.190**	*0.979*	*0.00*
						0	4	0	15.18	16.68	0.910		
						0	5	0	15.59	16.31	0.956		
						0	6	0	14.17	19.51	0.726		
						0	7	0	17.19	13.98	**1.230**		
						0	8	0	14.13	16.33	0.865		
		interval	off	direct	0.5	0	3	0	21.76	18.39	**1.183**	*0.978*	*0.00*
						0	4	0	15.18	16.97	0.895		
						0	6	0	14.17	16.33	0.868		
						0	7	0	17.19	15.71	**1.094**		
						0	8	0	14.13	16.61	0.851		
		interval	off	direct	0.0	0	3	0	21.76	18.62	**1.169**	*0.969*	*0.00*
						0	4	0	15.18	16.04	0.946		
						0	5	0	15.59	15.92	0.979		
						0	6	0	14.17	15.91	0.891		
						0	7	0	17.19	16.52	**1.041**		
						0	8	0	14.13	17.95	0.787		
		zono	off	direct_pgd	0.5	0	3	0	21.76	17.12	**1.271**	*0.967*	*0.01*
						0	4	0	15.18	15.65	0.970		
						0	5	0	15.59	17.05	0.914		
						0	6	0	14.17	16.81	0.843		
						0	7	0	17.19	17.00	**1.011**		
						0	8	0	14.13	17.89	0.790		
		zono	off	prev_pgd	0.5	0	3	0	21.76	19.24	**1.131**	*0.959*	*0.01*
						0	4	0	15.18	15.97	0.951		
						0	5	0	15.59	17.28	0.902		
						0	6	0	14.17	17.21	0.823		
						0	7	0	17.19	15.05	**1.142**		
						0	8	0	14.13	17.60	0.803		

Table 12: Architecture **3x10** — perturbation **occ**(5,5,5), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
occ	5,5,5	interval	off	prev_pgd	0.0	0	3	0	21.76	18.82	**1.156**	*0.957*	*0.01*
						0	4	0	15.18	16.69	0.910		
						0	5	0	15.59	16.90	0.922		
						0	6	0	14.17	16.08	0.881		
						0	7	0	17.19	15.51	**1.108**		
						0	8	0	14.13	18.50	0.764		
		interval	off	direct_pgd	0.5	0	3	0	21.76	18.26	**1.192**	*0.953*	*0.01*
						0	4	0	15.18	16.25	0.934		
						0	6	0	14.17	18.06	0.785		
						0	7	0	17.19	15.84	**1.085**		
						0	8	0	14.13	18.41	0.768		
		interval	off	prev_pgd	0.5	0	3	0	21.76	19.61	**1.110**	*0.949*	*0.01*
						0	4	0	15.18	15.77	0.963		
						0	6	0	14.17	16.19	0.875		
						0	7	0	17.19	16.38	**1.049**		
						0	8	0	14.13	18.92	0.747		
		zono	off	prev_pgd	0.0	0	3	0	21.76	20.00	**1.088**	*0.938*	*0.01*
						0	4	0	15.18	15.91	0.954		
						0	5	0	15.59	17.34	0.899		
						0	6	0	14.17	17.47	0.811		
						0	7	0	17.19	15.20	**1.131**		
						0	8	0	14.13	19.02	0.743		

Table 13: Architecture **3x10** — perturbation **occ**(5,5,5), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean sp); continuation of Table 6 (4 combos, 22 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}} - \delta_{\mathrm{adv}}\rvert}_{c\_src}$
occ	5,5,5	interval	off	direct_pgd	0.0	1	5	0	16.39	17.36	0.944	***1.043***	*0.03*
						1	6	0	17.03	14.97	**1.138**		
						1	7	0	16.60	14.52	**1.143**		
						1	8	0	15.69	16.55	0.948		
		zono	off	no	0.5	1	3	0	16.02	16.62	0.964	***1.022***	*0.02*
						1	4	0	14.21	14.00	**1.015**		
						1	5	0	16.39	16.13	**1.016**		
						1	6	0	17.03	14.91	**1.142**		
						1	7	0	16.60	15.32	**1.084**		
						1	8	0	15.69	17.22	0.911		
		zono	off	prev_pgd	0.5	1	3	0	16.02	17.47	0.917	***1.008***	*0.02*
						1	4	0	14.21	15.18	0.936		
						1	5	0	16.39	14.93	**1.098**		
						1	6	0	17.03	15.51	**1.098**		
						1	7	0	16.60	15.13	**1.097**		
						1	8	0	15.69	17.38	0.903		
		interval	off	prev_pgd	0.0	1	3	0	16.02	17.60	0.910	*0.997*	*0.02*
						1	4	0	14.21	14.45	0.983		
						1	5	0	16.39	14.92	**1.099**		
						1	6	0	17.03	15.46	**1.102**		
						1	7	0	16.60	15.19	**1.093**		
						1	8	0	15.69	19.73	0.795		
		zono	off	prev_pgd	0.0	1	3	0	16.02	17.07	0.938	*0.994*	*0.02*
						1	4	0	14.21	14.68	0.968		
						1	5	0	16.39	14.86	**1.103**		
						1	6	0	17.03	16.00	**1.064**		
						1	7	0	16.60	15.66	**1.060**		
						1	8	0	15.69	18.91	0.830		
		interval	off	prev_pgd	0.5	1	3	0	16.02	16.45	0.974	*0.981*	*0.02*
						1	4	0	14.21	14.78	0.961		
						1	5	0	16.39	18.19	0.901		
						1	6	0	17.03	15.26	**1.116**		
						1	7	0	16.60	15.25	**1.089**		
						1	8	0	15.69	18.62	0.843		

Table 14: Architecture **3x10** — perturbation **occ**(5,5,5), $c_{\mathrm{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean sp); continuation of Table 6 (6 combos, 34 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
occ	`5,5,5`	zono	off	no	0.0	1	3	0	16.02	17.94	0.893	*0.957*	*0.02*
						1	4	0	14.21	15.81	0.899		
						1	5	0	16.39	17.45	0.939		
						1	6	0	17.03	16.24	**1.049**		
						1	7	0	16.60	15.49	**1.072**		
						1	8	0	15.69	17.66	0.888		
		interval	off	direct_pgd	0.5	1	3	0	16.02	17.68	0.906	*0.952*	*0.02*
						1	4	0	14.21	15.18	0.936		
						1	5	0	16.39	18.06	0.908		
						1	7	0	16.60	15.04	**1.104**		
						1	8	0	15.69	17.32	0.906		
		zono	off	direct_pgd	0.0	1	3	0	16.02	19.43	0.824	*0.938*	*0.02*
						1	4	0	14.21	16.59	0.857		
						1	5	0	16.39	19.81	0.827		
						1	6	0	17.03	14.80	**1.151**		
						1	7	0	16.60	15.55	**1.068**		
						1	8	0	15.69	17.40	0.902		
		interval	off	no	0.0	1	3	0	16.02	17.38	0.922	*0.928*	*0.02*
						1	4	0	14.21	16.17	0.879		
						1	5	0	16.39	17.56	0.933		
						1	6	0	17.03	15.99	**1.065**		
						1	7	0	16.60	19.11	0.869		
						1	8	0	15.69	17.47	0.898		
		zono	off	direct_pgd	0.5	1	3	0	16.02	19.55	0.819	*0.915*	*0.02*
						1	4	0	14.21	15.98	0.889		
						1	5	0	16.39	18.93	0.866		
						1	6	0	17.03	17.46	0.975		
						1	7	0	16.60	15.51	**1.070**		
						1	8	0	15.69	18.08	0.868		
		interval	off	no	0.5	1	3	0	16.02	18.00	0.890	*0.910*	*0.02*
						1	4	0	14.21	14.99	0.948		
						1	5	0	16.39	17.98	0.912		
						1	6	0	17.03	17.40	0.979		
						1	7	0	16.60	17.60	0.943		
						1	8	0	15.69	19.83	0.791		

Table 15: Architecture **3x10** — perturbation **occ**(`5,5,5`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean `sp`); continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,1`	interval	off	direct_pgd	0.5	0	3	0	13.93	11.07	**1.258**	***1.483***	*0.03*
						0	4	0	15.18	11.23	**1.352**		
						0	5	0	15.40	11.05	**1.394**		
						0	6	0	18.67	10.79	**1.730**		
						0	7	0	20.37	11.21	**1.817**		
						0	8	0	16.12	11.95	**1.349**		
		zono	off	prev_pgd	0.0	0	3	0	13.93	11.17	**1.247**	***1.442***	*0.00*
						0	4	0	15.18	11.08	**1.370**		
						0	5	0	15.40	11.17	**1.379**		
						0	6	0	18.67	10.99	**1.699**		
						0	7	0	20.37	13.76	**1.480**		
						0	8	0	16.12	10.92	**1.476**		
		zono	off	direct_pgd	0.0	0	3	0	13.93	11.19	**1.245**	***1.434***	*0.03*
						0	4	0	15.18	12.59	**1.206**		
						0	5	0	15.40	11.46	**1.344**		
						0	6	0	18.67	11.72	**1.593**		
						0	7	0	20.37	11.48	**1.774**		
						0	8	0	16.12	11.16	**1.444**		
		interval	off	prev_pgd	0.5	0	3	0	13.93	11.95	**1.166**	***1.409***	*0.00*
						0	4	0	15.18	11.87	**1.279**		
						0	5	0	15.40	11.94	**1.290**		
						0	6	0	18.67	12.22	**1.528**		
						0	7	0	20.37	11.70	**1.741**		
						0	8	0	16.12	11.12	**1.450**		
		zono	off	direct_pgd	0.5	0	3	0	13.93	11.76	**1.185**	***1.362***	*0.03*
						0	4	0	15.18	12.23	**1.241**		
						0	5	0	15.40	11.69	**1.317**		
						0	6	0	18.67	11.87	**1.573**		
						0	7	0	20.37	13.77	**1.479**		
						0	8	0	16.12	11.73	**1.374**		
		zono	off	prev_pgd	0.5	0	4	0	15.18	13.20	**1.150**	***1.359***	*0.00*
						0	5	0	15.40	12.45	**1.237**		
						0	6	0	18.67	13.54	**1.379**		
						0	7	0	20.37	12.73	**1.600**		
						0	8	0	16.12	11.30	**1.427**		

Table 16: Architecture **3x10** — perturbation **translation**(`1,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean `sp`); continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	1,1	interval	off	direct_pgd	0.0	0	3	0	13.93	13.33	**1.045**	***1.333***	*0.03*
						0	4	0	15.18	12.11	**1.254**		
						0	5	0	15.40	11.69	**1.317**		
						0	6	0	18.67	12.62	**1.479**		
						0	7	0	20.37	12.18	**1.672**		
						0	8	0	16.12	13.08	**1.232**		
		interval	off	prev_pgd	0.0	0	3	0	13.93	12.84	**1.085**	***1.315***	*0.00*
						0	4	0	15.18	13.57	**1.119**		
						0	5	0	15.40	12.37	**1.245**		
						0	6	0	18.67	12.50	**1.494**		
						0	7	0	20.37	12.46	**1.635**		
						0	8	0	16.12	12.28	**1.313**		
		zono	off	direct	0.5	0	4	0	15.18	13.23	**1.147**	***1.255***	*0.00*
						0	5	0	15.40	14.01	**1.099**		
						0	6	0	18.67	14.90	**1.253**		
						0	7	0	20.37	13.73	**1.484**		
						0	8	0	16.12	12.47	**1.293**		
		interval	off	direct	0.5	0	3	0	13.93	13.01	**1.071**	***1.254***	*0.00*
						0	4	0	15.18	12.83	**1.183**		
						0	5	0	15.40	12.23	**1.259**		
						0	6	0	18.67	12.19	**1.532**		
						0	7	0	20.37	14.20	**1.435**		
						0	8	0	16.12	15.42	**1.045**		
		zono	off	direct	0.0	0	3	0	13.93	12.82	**1.087**	***1.232***	*0.00*
						0	4	0	15.18	14.23	**1.067**		
						0	5	0	15.40	13.85	**1.112**		
						0	6	0	18.67	13.83	**1.350**		
						0	7	0	20.37	13.38	**1.522**		
						0	8	0	16.12	12.87	**1.253**		
		zono	off	no	0.0	0	3	0	13.93	14.10	0.988	***1.176***	*0.00*
						0	4	0	15.18	13.74	**1.105**		
						0	5	0	15.40	14.85	**1.037**		
						0	6	0	18.67	13.58	**1.375**		
						0	7	0	20.37	14.49	**1.406**		
						0	8	0	16.12	14.08	**1.145**		

Table 17: Architecture **3x10** — perturbation **translation**(1,1), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_\mathrm{src}}$	$\overline{\lvert\delta_{\mathrm{std}} - \delta_{\mathrm{adv}}\rvert}_{c\_\mathrm{src}}$
translation	`1,1`	interval	off	no	0.5	0	3	0	13.93	13.64	**1.021**	***1.148***	*0.00*
						0	4	0	15.18	14.05	**1.080**		
						0	5	0	15.40	13.72	**1.122**		
						0	6	0	18.67	13.77	**1.356**		
						0	7	0	20.37	14.09	**1.446**		
						0	8	0	16.12	18.68	0.863		
		interval	off	no	0.0	0	3	0	13.93	14.13	0.986	***1.076***	*0.00*
						0	4	0	15.18	14.58	**1.041**		
						0	5	0	15.40	20.79	0.741		
						0	6	0	18.67	14.74	**1.267**		
						0	7	0	20.37	15.44	**1.319**		
						0	8	0	16.12	14.66	**1.100**		
		interval	off	direct	0.0	0	3	0	13.93	14.59	0.955	***1.062***	*0.00*
						0	4	0	15.18	15.01	**1.011**		
						0	5	0	15.40	15.79	0.975		
						0	6	0	18.67	15.36	**1.215**		
						0	7	0	20.37	16.11	**1.264**		
						0	8	0	16.12	16.91	0.953		
		zono	off	no	0.5	0	3	0	13.93	23.82	0.585	*0.731*	*0.00*
						0	4	0	15.18	27.68	0.548		
						0	5	0	15.40	15.65	0.984		
						0	6	0	18.67	23.79	0.785		
						0	7	0	20.37	33.73	0.604		
						0	8	0	16.12	18.28	0.882		

Table 18: Architecture **3x10** — perturbation **translation**(`1,1`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean `sp`); continuation of Table 6 (4 combos, 24 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	`1,1`	zono	off	prev_pgd	0.0	1	3	0	22.44	14.42	**1.556**	***1.580***	*0.01*
						1	4	0	22.97	14.79	**1.553**		
						1	5	0	21.27	13.40	**1.587**		
						1	6	0	21.10	13.65	**1.546**		
						1	7	0	22.94	13.68	**1.677**		
						1	8	0	22.62	14.49	**1.561**		
		zono	off	no	0.0	1	3	0	22.44	13.62	**1.648**	***1.579***	*0.00*
						1	4	0	22.97	14.62	**1.571**		
						1	5	0	21.27	14.13	**1.505**		
						1	6	0	21.10	13.54	**1.558**		
						1	7	0	22.94	13.95	**1.644**		
						1	8	0	22.62	14.60	**1.549**		
		interval	off	prev_pgd	0.0	1	3	0	22.44	14.60	**1.537**	***1.576***	*0.01*
						1	4	0	22.97	14.22	**1.615**		
						1	5	0	21.27	14.62	**1.455**		
						1	6	0	21.10	13.62	**1.549**		
						1	7	0	22.94	14.19	**1.617**		
						1	8	0	22.62	13.42	**1.686**		
		interval	off	direct_pgd	0.5	1	3	0	22.44	13.76	**1.631**	***1.575***	*0.01*
						1	4	0	22.97	13.93	**1.649**		
						1	5	0	21.27	14.81	**1.436**		
						1	6	0	21.10	14.10	**1.496**		
						1	7	0	22.94	14.01	**1.637**		
						1	8	0	22.62	14.15	**1.599**		
		interval	off	direct_pgd	0.0	1	3	0	22.44	13.35	**1.681**	***1.570***	*0.01*
						1	4	0	22.97	14.17	**1.621**		
						1	5	0	21.27	14.26	**1.492**		
						1	6	0	21.10	13.42	**1.572**		
						1	7	0	22.94	13.97	**1.642**		
						1	8	0	22.62	15.99	**1.415**		
		zono	off	prev_pgd	0.5	1	3	0	22.44	14.15	**1.586**	***1.569***	*0.01*
						1	4	0	22.97	14.13	**1.626**		
						1	5	0	21.27	13.71	**1.551**		
						1	6	0	21.10	15.23	**1.385**		
						1	7	0	22.94	14.56	**1.576**		
						1	8	0	22.62	13.40	**1.688**		

Table 19: Architecture **3x10** — perturbation **translation**(`1,1`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,1`	zono	off	direct_pgd	0.0	1	3	0	22.44	13.49	**1.663**	***1.566***	*0.01*
						1	4	0	22.97	14.08	**1.631**		
						1	5	0	21.27	14.67	**1.450**		
						1	6	0	21.10	13.56	**1.556**		
						1	7	0	22.94	13.54	**1.694**		
						1	8	0	22.62	16.12	**1.403**		
		interval	off	prev_pgd	0.5	1	3	0	22.44	13.53	**1.659**	***1.566***	*0.01*
						1	4	0	22.97	14.04	**1.636**		
						1	5	0	21.27	14.17	**1.501**		
						1	6	0	21.10	13.69	**1.541**		
						1	7	0	22.94	15.74	**1.457**		
						1	8	0	22.62	14.13	**1.601**		
		zono	off	no	0.5	1	3	0	22.44	13.80	**1.626**	***1.562***	*0.00*
						1	4	0	22.97	14.52	**1.582**		
						1	5	0	21.27	14.54	**1.463**		
						1	6	0	21.10	14.37	**1.468**		
						1	7	0	22.94	14.04	**1.634**		
						1	8	0	22.62	14.13	**1.601**		
		interval	off	no	0.0	1	3	0	22.44	13.97	**1.606**	***1.560***	*0.00*
						1	4	0	22.97	14.25	**1.612**		
						1	5	0	21.27	14.36	**1.481**		
						1	6	0	21.10	13.63	**1.548**		
						1	7	0	22.94	14.80	**1.550**		
						1	8	0	22.62	14.48	**1.562**		
		zono	off	direct_pgd	0.5	1	3	0	22.44	13.60	**1.650**	***1.556***	*0.01*
						1	4	0	22.97	14.39	**1.596**		
						1	5	0	21.27	15.79	**1.347**		
						1	6	0	21.10	13.77	**1.532**		
						1	7	0	22.94	14.35	**1.599**		
						1	8	0	22.62	14.04	**1.611**		
		interval	off	no	0.5	1	3	0	22.44	12.91	**1.738**	***1.542***	*0.00*
						1	4	0	22.97	15.83	**1.451**		
						1	5	0	21.27	15.43	**1.378**		
						1	6	0	21.10	14.78	**1.428**		
						1	7	0	22.94	14.15	**1.621**		
						1	8	0	22.62	13.83	**1.636**		

Table 20: Architecture **3x10** — perturbation **translation**(`1,1`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	1,3	interval	off	direct_pgd	0.0	0	3	0	12.26	12.49	0.982	***1.096***	*0.02*
						0	4	0	14.39	11.61	**1.239**		
						0	5	0	14.10	11.71	**1.204**		
						0	6	0	11.86	13.04	0.910		
						0	7	0	12.10	13.09	0.924		
						0	8	0	16.83	12.80	**1.315**		
		interval	off	prev_pgd	0.0	0	3	0	12.26	12.04	**1.018**	***1.051***	*0.02*
						0	4	0	14.39	13.40	**1.074**		
						0	5	0	14.10	15.15	0.931		
						0	6	0	11.86	12.82	0.925		
						0	7	0	12.10	13.92	0.869		
						0	8	0	16.83	11.31	**1.488**		
		zono	off	prev_pgd	0.5	0	3	0	12.26	13.69	0.896	***1.023***	*0.02*
						0	4	0	14.39	13.29	**1.083**		
						0	5	0	14.10	13.08	**1.078**		
						0	6	0	11.86	13.22	0.897		
						0	7	0	12.10	13.44	0.900		
						0	8	0	16.83	13.10	**1.285**		
		interval	off	direct	0.0	0	3	0	12.26	13.23	0.927	***1.018***	*0.02*
						0	4	0	14.39	14.21	**1.013**		
						0	5	0	14.10	13.28	**1.062**		
						0	6	0	11.86	13.56	0.875		
						0	7	0	12.10	13.88	0.872		
						0	8	0	16.83	12.39	**1.358**		
		interval	off	prev_pgd	0.5	0	3	0	12.26	13.37	0.917	***1.014***	*0.02*
						0	4	0	14.39	13.21	**1.089**		
						0	5	0	14.10	13.43	**1.050**		
						0	6	0	11.86	13.14	0.903		
						0	7	0	12.10	14.10	0.858		
						0	8	0	16.83	13.26	**1.269**		
		zono	off	prev_pgd	0.0	0	3	0	12.26	12.93	0.948	***1.010***	*0.02*
						0	4	0	14.39	12.78	**1.126**		
						0	5	0	14.10	13.87	**1.017**		
						0	6	0	11.86	13.62	0.871		
						0	7	0	12.10	13.59	0.890		
						0	8	0	16.83	13.96	**1.206**		

Table 21: Architecture **3x10** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	`1,3`	zono	off	direct_pgd	0.5	0	3	0	12.26	13.57	0.903	***1.001***	*0.02*
						0	4	0	14.39	13.64	**1.055**		
						0	5	0	14.10	13.30	**1.060**		
						0	6	0	11.86	13.04	0.910		
						0	7	0	12.10	14.52	0.833		
						0	8	0	16.83	13.53	**1.244**		
		zono	off	direct	0.0	0	3	0	12.26	13.51	0.907	*0.998*	*0.02*
						0	4	0	14.39	13.57	**1.060**		
						0	5	0	14.10	13.25	**1.064**		
						0	6	0	11.86	13.31	0.891		
						0	7	0	12.10	13.22	0.915		
						0	8	0	16.83	14.64	**1.150**		
		interval	off	direct_pgd	0.5	0	3	0	12.26	12.90	0.950	*0.974*	*0.02*
						0	4	0	14.39	13.57	**1.060**		
						0	5	0	14.10	13.80	**1.022**		
						0	6	0	11.86	13.14	0.903		
						0	7	0	12.10	14.15	0.855		
						0	8	0	16.83	15.98	**1.053**		
		zono	off	direct_pgd	0.0	0	3	0	12.26	13.63	0.899	*0.961*	*0.02*
						0	5	0	14.10	13.78	**1.023**		
						0	6	0	11.86	13.49	0.879		
						0	7	0	12.10	14.97	0.808		
						0	8	0	16.83	14.10	**1.194**		
		zono	off	direct	0.5	0	3	0	12.26	13.26	0.925	*0.953*	*0.02*
						0	4	0	14.39	14.17	**1.016**		
						0	5	0	14.10	14.83	0.951		
						0	6	0	11.86	15.44	0.768		
						0	7	0	12.10	14.53	0.833		
						0	8	0	16.83	13.75	**1.224**		
		interval	off	direct	0.5	0	3	0	12.26	15.63	0.784	*0.874*	*0.02*
						0	4	0	14.39	18.14	0.793		
						0	5	0	14.10	18.09	0.779		
						0	6	0	11.86	16.13	0.735		
						0	7	0	12.10	12.83	0.943		
						0	8	0	16.83	13.96	**1.206**		

Table 22: Architecture **3x10** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean `sp`); continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,3`	interval	off	no	0.0	0	3	0	12.26	36.57	0.335	*0.722*	*0.02*
						0	4	0	14.39	32.40	0.444		
						0	6	0	11.86	14.48	0.819		
						0	7	0	12.10	12.81	0.945		
						0	8	0	16.83	15.77	**1.067**		
		zono	off	no	0.5	0	3	0	12.26	34.90	0.351	*0.672*	*0.02*
						0	4	0	14.39	36.82	0.391		
						0	5	0	14.10	11.95	**1.180**		
						0	6	0	11.86	34.85	0.340		
						0	7	0	12.10	23.05	0.525		
						0	8	0	16.83	13.49	**1.248**		
		zono	off	no	0.0	0	3	0	12.26	39.92	0.307	*0.579*	*0.02*
						0	4	0	14.39	32.08	0.449		
						0	5	0	14.10	13.76	**1.025**		
						0	6	0	11.86	35.97	0.330		
						0	7	0	12.10	14.57	0.830		
						0	8	0	16.83	31.41	0.536		
		interval	off	no	0.5	0	3	0	12.26	15.44	0.794	*0.531*	*0.02*
						0	4	0	14.39	30.96	0.465		
						0	5	0	14.10	33.84	0.417		
						0	6	0	11.86	18.54	0.640		
						0	7	0	12.10	35.53	0.341		
						0	8	0	16.83	31.92	0.527		

Table 23: Architecture **3x10** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean `sp`); continuation of Table 6 (4 combos, 23 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	1,3	interval	off	prev_pgd	0.5	1	3	0	24.22	13.31	**1.820**	***1.658***	*0.00*
						1	4	0	19.98	13.48	**1.482**		
						1	5	0	23.18	14.36	**1.614**		
						1	6	0	20.60	13.59	**1.516**		
						1	7	0	25.10	13.68	**1.835**		
						1	8	0	22.29	13.25	**1.682**		
		interval	off	direct_pgd	0.5	1	3	0	24.22	14.15	**1.712**	***1.642***	*0.00*
						1	4	0	19.98	13.77	**1.451**		
						1	5	0	23.18	13.53	**1.713**		
						1	6	0	20.60	13.55	**1.520**		
						1	7	0	25.10	14.03	**1.789**		
						1	8	0	22.29	13.35	**1.670**		
		zono	off	prev_pgd	0.0	1	3	0	24.22	13.47	**1.798**	***1.627***	*0.00*
						1	4	0	19.98	14.70	**1.359**		
						1	5	0	23.18	13.96	**1.660**		
						1	6	0	20.60	13.29	**1.550**		
						1	7	0	25.10	13.57	**1.850**		
						1	8	0	22.29	14.42	**1.546**		
		zono	off	direct_pgd	0.5	1	3	0	24.22	12.93	**1.873**	***1.625***	*0.00*
						1	4	0	19.98	13.67	**1.462**		
						1	5	0	23.18	13.47	**1.721**		
						1	6	0	20.60	13.44	**1.533**		
						1	7	0	25.10	16.33	**1.537**		
						1	8	0	22.29	13.74	**1.622**		
		interval	off	no	0.5	1	3	0	24.22	13.77	**1.759**	***1.608***	*0.00*
						1	4	0	19.98	14.54	**1.374**		
						1	5	0	23.18	14.14	**1.639**		
						1	6	0	20.60	14.10	**1.461**		
						1	7	0	25.10	14.08	**1.783**		
						1	8	0	22.29	13.67	**1.631**		
		zono	off	prev_pgd	0.5	1	3	0	24.22	13.42	**1.805**	***1.590***	*0.00*
						1	4	0	19.98	15.16	**1.318**		
						1	5	0	23.18	15.33	**1.512**		
						1	6	0	20.60	14.65	**1.406**		
						1	7	0	25.10	13.45	**1.866**		
						1	8	0	22.29	13.65	**1.633**		

Table 24: Architecture **3x10** — perturbation **translation**(1,3), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean sp); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_\mathrm{src}}$	$\overline{\lvert\delta_{\mathrm{std}} - \delta_{\mathrm{adv}}\rvert}_{c\_\mathrm{src}}$
translation	`1,3`	interval	off	direct_pgd	0.0	1	3	0	24.22	14.02	**1.728**	***1.561***	*0.00*
						1	4	0	19.98	14.61	**1.368**		
						1	5	0	23.18	14.18	**1.635**		
						1	6	0	20.60	14.47	**1.424**		
						1	7	0	25.10	14.60	**1.719**		
						1	8	0	22.29	14.94	**1.492**		
		zono	off	no	0.0	1	3	0	24.22	14.24	**1.701**	***1.557***	*0.00*
						1	4	0	19.98	15.40	**1.297**		
						1	5	0	23.18	15.30	**1.515**		
						1	6	0	20.60	13.74	**1.499**		
						1	7	0	25.10	14.10	**1.780**		
						1	8	0	22.29	14.40	**1.548**		
		zono	off	direct_pgd	0.0	1	3	0	24.22	13.31	**1.820**	***1.536***	*0.00*
						1	4	0	19.98	17.84	**1.120**		
						1	5	0	23.18	16.49	**1.406**		
						1	6	0	20.60	13.51	**1.525**		
						1	7	0	25.10	14.25	**1.761**		
						1	8	0	22.29	14.07	**1.584**		
		interval	off	prev_pgd	0.0	1	3	0	24.22	13.71	**1.767**	***1.536***	*0.00*
						1	4	0	19.98	14.00	**1.427**		
						1	5	0	23.18	14.76	**1.570**		
						1	6	0	20.60	14.84	**1.388**		
						1	7	0	25.10	15.38	**1.632**		
						1	8	0	22.29	15.58	**1.431**		
		interval	off	no	0.0	1	3	0	24.22	15.25	**1.588**	***1.534***	*0.00*
						1	4	0	19.98	16.31	**1.225**		
						1	5	0	23.18	14.86	**1.560**		
						1	6	0	20.60	13.72	**1.501**		
						1	7	0	25.10	13.96	**1.798**		
						1	8	0	22.29	14.54	**1.533**		
		zono	off	no	0.5	1	3	0	24.22	17.31	**1.399**	***1.496***	*0.00*
						1	4	0	19.98	16.59	**1.204**		
						1	5	0	23.18	14.80	**1.566**		
						1	6	0	20.60	15.14	**1.361**		
						1	7	0	25.10	13.77	**1.823**		
						1	8	0	22.29	13.72	**1.625**		

Table 25: Architecture **3x10** — perturbation **translation**(`1,3`), $c_{\mathrm{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_\mathrm{src}}$	$\overline{\lvert\delta_{\mathrm{std}}-\delta_{\mathrm{adv}}\rvert}_{c\_\mathrm{src}}$
translation	`3,1`	interval	off	direct	0.5	0	3	0	13.40	12.97	**1.033**	***1.006***	*0.01*
						0	4	0	14.38	14.75	0.975		
						0	5	0	13.93	11.16	**1.248**		
						0	6	0	11.80	12.49	0.945		
						0	7	0	11.82	12.89	0.917		
						0	8	0	12.12	13.16	0.921		
		zono	off	direct	0.0	0	3	0	13.40	12.84	**1.044**	*0.996*	*0.01*
						0	4	0	14.38	12.20	**1.179**		
						0	5	0	13.93	13.43	**1.037**		
						0	6	0	11.80	13.98	0.844		
						0	7	0	11.82	12.89	0.917		
						0	8	0	12.12	12.69	0.955		
		zono	off	direct	0.5	0	3	0	13.40	13.83	0.969	*0.981*	*0.01*
						0	4	0	14.38	13.54	**1.062**		
						0	5	0	13.93	13.17	**1.058**		
						0	6	0	11.80	12.43	0.949		
						0	7	0	11.82	13.01	0.909		
						0	8	0	12.12	12.92	0.938		
		zono	off	prev_pgd	0.5	0	3	0	13.40	13.21	**1.014**	*0.972*	*0.02*
						0	4	0	14.38	13.06	**1.101**		
						0	5	0	13.93	12.93	**1.077**		
						0	6	0	11.80	13.07	0.903		
						0	7	0	11.82	14.03	0.842		
						0	8	0	12.12	13.59	0.892		
		zono	off	direct_pgd	0.0	0	3	0	13.40	13.14	**1.020**	*0.965*	*0.04*
						0	4	0	14.38	13.30	**1.081**		
						0	5	0	13.93	12.99	**1.072**		
						0	6	0	11.80	13.60	0.868		
						0	7	0	11.82	13.91	0.850		
						0	8	0	12.12	13.51	0.897		
		zono	off	direct_pgd	0.5	0	3	0	13.40	13.55	0.989	*0.953*	*0.04*
						0	4	0	14.38	13.57	**1.060**		
						0	5	0	13.93	13.22	**1.054**		
						0	6	0	11.80	14.15	0.834		
						0	7	0	11.82	13.54	0.873		
						0	8	0	12.12	13.32	0.910		

Table 26: Architecture **3x10** — perturbation **translation**(`3,1`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`3,1`	interval	off	direct_pgd	0.0	0	3	0	13.40	13.04	**1.028**	*0.941*	*0.04*
						0	4	0	14.38	13.43	**1.071**		
						0	5	0	13.93	13.21	**1.055**		
						0	6	0	11.80	13.58	0.869		
						0	7	0	11.82	16.05	0.736		
						0	8	0	12.12	13.63	0.889		
		interval	off	prev_pgd	0.5	0	3	0	13.40	13.08	**1.024**	*0.925*	*0.02*
						0	4	0	14.38	13.33	**1.079**		
						0	5	0	13.93	14.92	0.934		
						0	6	0	11.80	13.07	0.903		
						0	7	0	11.82	14.08	0.839		
						0	8	0	12.12	15.67	0.773		
		zono	off	prev_pgd	0.0	0	3	0	13.40	14.11	0.950	*0.914*	*0.02*
						0	4	0	14.38	13.60	**1.057**		
						0	5	0	13.93	14.55	0.957		
						0	6	0	11.80	13.51	0.873		
						0	7	0	11.82	13.90	0.850		
						0	8	0	12.12	15.26	0.794		
		interval	off	prev_pgd	0.0	0	3	0	13.40	15.33	0.874	*0.901*	*0.02*
						0	4	0	14.38	13.35	**1.077**		
						0	5	0	13.93	14.79	0.942		
						0	6	0	11.80	14.34	0.823		
						0	7	0	11.82	14.08	0.839		
						0	8	0	12.12	14.27	0.849		
		interval	off	direct_pgd	0.5	0	3	0	13.40	14.81	0.905	*0.887*	*0.03*
						0	4	0	14.38	14.57	0.987		
						0	6	0	11.80	13.73	0.859		
						0	7	0	11.82	13.96	0.847		
						0	8	0	12.12	14.47	0.838		
		interval	off	direct	0.0	0	3	0	13.40	16.01	0.837	*0.758*	*0.01*
						0	4	0	14.38	16.22	0.887		
						0	5	0	13.93	17.31	0.805		
						0	6	0	11.80	16.62	0.710		
						0	7	0	11.82	19.26	0.614		
						0	8	0	12.12	17.34	0.699		

Table 27: Architecture **3x10** — perturbation **translation**(`3,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean `sp`); continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	3,1	zono	off	no	0.5	0	3	0	13.40	35.84	0.374	*0.594*	*0.02*
						0	4	0	14.38	30.59	0.470		
						0	5	0	13.93	14.22	0.980		
						0	6	0	11.80	16.32	0.723		
						0	7	0	11.82	16.88	0.700		
						0	8	0	12.12	38.26	0.317		
		zono	off	no	0.0	0	3	0	13.40	36.02	0.372	*0.500*	*0.02*
						0	4	0	14.38	38.00	0.378		
						0	5	0	13.93	36.07	0.386		
						0	6	0	11.80	19.66	0.600		
						0	7	0	11.82	13.66	0.865		
						0	8	0	12.12	30.43	0.398		
		interval	off	no	0.5	0	3	0	13.40	14.20	0.944	*0.469*	*0.02*
						0	4	0	14.38	36.57	0.393		
						0	5	0	13.93	38.16	0.365		
						0	6	0	11.80	36.55	0.323		
						0	7	0	11.82	26.00	0.455		
						0	8	0	12.12	36.04	0.336		
		interval	off	no	0.0	0	3	0	13.40	35.54	0.377	*0.404*	*0.02*
						0	4	0	14.38	23.88	0.602		
						0	5	0	13.93	33.35	0.418		
						0	6	0	11.80	38.52	0.306		
						0	7	0	11.82	29.95	0.395		
						0	8	0	12.12	37.10	0.327		

Table 28: Architecture **3x10** — perturbation **translation**(3,1), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean sp); continuation of Table 6 (4 combos, 24 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	3,1	interval	off	no	0.5	1	3	0	23.91	13.71	**1.744**	***1.567***	*0.03*
						1	4	0	22.60	13.55	**1.668**		
						1	5	0	15.45	13.55	**1.140**		
						1	6	0	23.56	13.89	**1.696**		
						1	7	0	24.44	14.03	**1.742**		
						1	8	0	19.60	13.86	**1.414**		
		interval	off	direct_pgd	0.5	1	3	0	23.91	13.47	**1.775**	***1.554***	*0.04*
						1	4	0	22.60	14.26	**1.585**		
						1	5	0	15.45	13.57	**1.139**		
						1	6	0	23.56	13.69	**1.721**		
						1	7	0	24.44	13.90	**1.758**		
						1	8	0	19.60	14.59	**1.343**		
		zono	off	direct_pgd	0.0	1	3	0	23.91	13.48	**1.774**	***1.550***	*0.04*
						1	4	0	22.60	13.84	**1.633**		
						1	5	0	15.45	13.93	**1.109**		
						1	6	0	23.56	14.03	**1.679**		
						1	7	0	24.44	13.63	**1.793**		
						1	8	0	19.60	14.97	**1.309**		
		zono	off	prev_pgd	0.0	1	3	0	23.91	14.01	**1.707**	***1.527***	*0.04*
						1	4	0	22.60	13.58	**1.664**		
						1	5	0	15.45	13.50	**1.144**		
						1	6	0	23.56	13.72	**1.717**		
						1	7	0	24.44	15.31	**1.596**		
						1	8	0	19.60	14.67	**1.336**		
		interval	off	prev_pgd	0.0	1	3	0	23.91	14.07	**1.699**	***1.524***	*0.04*
						1	4	0	22.60	13.66	**1.654**		
						1	5	0	15.45	13.76	**1.123**		
						1	6	0	23.56	13.58	**1.735**		
						1	7	0	24.44	14.38	**1.700**		
						1	8	0	19.60	15.87	**1.235**		
		zono	off	direct_pgd	0.5	1	3	0	23.91	14.09	**1.697**	***1.519***	*0.04*
						1	4	0	22.60	14.43	**1.566**		
						1	5	0	15.45	13.12	**1.178**		
						1	6	0	23.56	14.46	**1.629**		
						1	7	0	24.44	14.76	**1.656**		
						1	8	0	19.60	14.14	**1.386**		

Table 29: Architecture **3x10** — perturbation **translation**(3,1), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean sp); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	`3,1`	interval	off	prev_pgd	0.5	1	3	0	23.91	13.58	**1.761**	***1.506***	*0.04*
						1	4	0	22.60	13.81	**1.636**		
						1	5	0	15.45	13.87	**1.114**		
						1	6	0	23.56	15.36	**1.534**		
						1	7	0	24.44	14.46	**1.690**		
						1	8	0	19.60	15.09	**1.299**		
		zono	off	prev_pgd	0.5	1	3	0	23.91	15.70	**1.523**	***1.500***	*0.04*
						1	4	0	22.60	13.75	**1.644**		
						1	5	0	15.45	14.66	**1.054**		
						1	6	0	23.56	14.66	**1.607**		
						1	7	0	24.44	14.04	**1.741**		
						1	8	0	19.60	13.68	**1.433**		
		interval	off	direct_pgd	0.0	1	3	0	23.91	14.19	**1.685**	***1.495***	*0.04*
						1	4	0	22.60	13.77	**1.641**		
						1	5	0	15.45	13.57	**1.139**		
						1	6	0	23.56	13.88	**1.697**		
						1	7	0	24.44	15.10	**1.619**		
						1	8	0	19.60	16.51	**1.187**		
		interval	off	no	0.0	1	3	0	23.91	13.76	**1.738**	***1.492***	*0.03*
						1	4	0	22.60	15.60	**1.449**		
						1	5	0	15.45	14.25	**1.084**		
						1	6	0	23.56	14.22	**1.657**		
						1	7	0	24.44	14.93	**1.637**		
						1	8	0	19.60	14.13	**1.387**		
		zono	off	no	0.5	1	3	0	23.91	15.48	**1.545**	***1.423***	*0.03*
						1	4	0	22.60	14.83	**1.524**		
						1	5	0	15.45	14.82	**1.043**		
						1	6	0	23.56	14.63	**1.610**		
						1	7	0	24.44	15.20	**1.608**		
						1	8	0	19.60	16.22	**1.208**		
		zono	off	no	0.0	1	3	0	23.91	14.50	**1.649**	***1.419***	*0.03*
						1	4	0	22.60	15.32	**1.475**		
						1	5	0	15.45	17.55	0.880		
						1	6	0	23.56	14.32	**1.645**		
						1	7	0	24.44	15.16	**1.612**		
						1	8	0	19.60	15.66	**1.252**		

Table 30: Architecture **3x10** — perturbation **translation**(`3,1`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean `sp`); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	3,3	zono	off	no	0.0	0	3	0	12.78	10.75	**1.189**	***1.130***	*0.03*
						0	4	0	14.57	11.46	**1.271**		
						0	5	0	14.10	11.20	**1.259**		
						0	6	0	10.86	11.53	0.942		
						0	7	0	11.99	12.22	0.981		
						0	8	0	12.37	10.88	**1.137**		
		interval	off	no	0.5	0	3	0	12.78	10.70	**1.194**	***1.129***	*0.03*
						0	4	0	14.57	11.48	**1.269**		
						0	5	0	14.10	10.66	**1.323**		
						0	6	0	10.86	11.24	0.966		
						0	7	0	11.99	13.40	0.895		
						0	8	0	12.37	10.99	**1.126**		
		zono	off	no	0.5	0	3	0	12.78	11.18	**1.143**	***1.102***	*0.03*
						0	4	0	14.57	11.47	**1.270**		
						0	5	0	14.10	12.17	**1.159**		
						0	6	0	10.86	10.40	**1.044**		
						0	7	0	11.99	11.33	**1.058**		
						0	8	0	12.37	13.17	0.939		
		interval	off	direct	0.0	0	3	0	12.78	13.05	0.979	*0.977*	*0.03*
						0	4	0	14.57	14.11	**1.033**		
						0	5	0	14.10	13.27	**1.063**		
						0	6	0	10.86	12.99	0.836		
						0	7	0	11.99	12.23	0.980		
						0	8	0	12.37	12.71	0.973		
		zono	off	prev_pgd	0.5	0	3	0	12.78	13.13	0.973	*0.962*	*0.03*
						0	4	0	14.57	13.28	**1.097**		
						0	5	0	14.10	13.30	**1.060**		
						0	6	0	10.86	13.16	0.825		
						0	7	0	11.99	13.40	0.895		
						0	8	0	12.37	13.39	0.924		
		interval	off	direct	0.5	0	3	0	12.78	12.02	**1.063**	*0.953*	*0.03*
						0	4	0	14.57	13.35	**1.091**		
						0	5	0	14.10	14.29	0.987		
						0	6	0	10.86	14.01	0.775		
						0	7	0	11.99	14.16	0.847		
						0	8	0	12.37	12.92	0.957		

Table 31: Architecture **3x10** — perturbation **translation**(3,3), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean sp); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	3,3	interval	off	prev_pgd	0.0	0	3	0	12.78	13.25	0.965	*0.953*	*0.03*
						0	4	0	14.57	13.40	**1.087**		
						0	5	0	14.10	13.28	**1.062**		
						0	6	0	10.86	13.13	0.827		
						0	7	0	11.99	13.22	0.907		
						0	8	0	12.37	14.26	0.867		
		interval	off	direct_pgd	0.5	0	3	0	12.78	13.00	0.983	*0.952*	*0.05*
						0	4	0	14.57	13.67	**1.066**		
						0	5	0	14.10	13.38	**1.054**		
						0	6	0	10.86	13.22	0.821		
						0	7	0	11.99	14.31	0.838		
						0	8	0	12.37	13.01	0.951		
		zono	off	direct_pgd	0.5	0	3	0	12.78	13.00	0.983	*0.936*	*0.05*
						0	4	0	14.57	13.55	**1.075**		
						0	5	0	14.10	13.17	**1.071**		
						0	6	0	10.86	13.26	0.819		
						0	7	0	11.99	15.40	0.779		
						0	8	0	12.37	13.91	0.889		
		zono	off	direct_pgd	0.0	0	3	0	12.78	13.12	0.974	*0.935*	*0.05*
						0	4	0	14.57	13.69	**1.064**		
						0	5	0	14.10	13.58	**1.038**		
						0	6	0	10.86	13.30	0.817		
						0	7	0	11.99	14.53	0.825		
						0	8	0	12.37	13.89	0.891		
		interval	off	prev_pgd	0.5	0	3	0	12.78	13.37	0.956	*0.934*	*0.03*
						0	4	0	14.57	14.16	**1.029**		
						0	5	0	14.10	13.35	**1.056**		
						0	6	0	10.86	14.22	0.764		
						0	7	0	11.99	12.44	0.964		
						0	8	0	12.37	14.84	0.834		
		zono	off	prev_pgd	0.0	0	3	0	12.78	14.89	0.858	*0.923*	*0.03*
						0	4	0	14.57	13.62	**1.070**		
						0	5	0	14.10	13.36	**1.055**		
						0	6	0	10.86	13.69	0.793		
						0	7	0	11.99	13.68	0.876		
						0	8	0	12.37	13.98	0.885		

Table 32: Architecture **3x10** — perturbation **translation**(3,3), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean sp); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	3,3	interval	off	direct_pgd	0.0	0	3	0	12.78	13.64	0.937	*0.888*	*0.03*
						0	5	0	14.10	13.14	**1.073**		
						0	6	0	10.86	13.53	0.803		
						0	7	0	11.99	16.07	0.746		
						0	8	0	12.37	14.00	0.884		
		zono	off	direct	0.5	0	3	0	12.78	14.63	0.874	*0.839*	*0.04*
						0	4	0	14.57	16.14	0.903		
						0	6	0	10.86	13.66	0.795		
						0	7	0	11.99	14.77	0.812		
						0	8	0	12.37	15.27	0.810		
		zono	off	direct	0.0	0	3	0	12.78	13.16	0.971	*0.825*	*0.04*
						0	4	0	14.57	14.92	0.977		
						0	6	0	10.86	15.15	0.717		
						0	7	0	11.99	15.92	0.753		
						0	8	0	12.37	17.46	0.708		
		interval	off	no	0.0	0	3	0	12.78	35.34	0.362	*0.670*	*0.03*
						0	4	0	14.57	20.81	0.700		
						0	5	0	14.10	12.69	**1.111**		
						0	6	0	10.86	15.43	0.704		
						0	7	0	11.99	15.06	0.796		
						0	8	0	12.37	35.55	0.348		

Table 33: Architecture **3x10** — perturbation **translation**(3,3), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean sp); continuation of Table 6 (4 combos, 21 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`3,3`	interval	off	prev_pgd	0.0	1	3	0	14.78	13.77	**1.073**	*0.982*	*0.04*
						1	4	0	13.53	14.15	0.956		
						1	6	0	12.78	14.29	0.894		
						1	7	0	14.15	13.74	**1.030**		
						1	8	0	13.03	13.61	0.957		
		interval	off	prev_pgd	0.5	1	3	0	14.78	13.66	**1.082**	*0.956*	*0.04*
						1	4	0	13.53	14.16	0.956		
						1	6	0	12.78	16.02	0.798		
						1	7	0	14.15	14.40	0.983		
						1	8	0	13.03	13.54	0.962		
		interval	off	no	0.5	1	3	0	14.78	15.25	0.969	*0.953*	*0.03*
						1	4	0	13.53	13.32	**1.016**		
						1	5	0	13.74	14.23	0.966		
						1	6	0	12.78	13.87	0.921		
						1	7	0	14.15	13.77	**1.028**		
						1	8	0	13.03	15.87	0.821		
		interval	off	direct_pgd	0.0	1	3	0	14.78	13.14	**1.125**	*0.942*	*0.05*
						1	4	0	13.53	13.71	0.987		
						1	6	0	12.78	15.33	0.834		
						1	8	0	13.03	15.81	0.824		
		zono	off	direct_pgd	0.0	1	3	0	14.78	13.81	**1.070**	*0.937*	*0.04*
						1	4	0	13.53	13.76	0.983		
						1	6	0	12.78	14.24	0.897		
						1	7	0	14.15	16.97	0.834		
						1	8	0	13.03	14.47	0.900		
		interval	off	no	0.0	1	3	0	14.78	14.28	**1.035**	*0.931*	*0.03*
						1	4	0	13.53	13.84	0.978		
						1	5	0	13.74	15.91	0.864		
						1	6	0	12.78	14.22	0.899		
						1	7	0	14.15	15.15	0.934		
						1	8	0	13.03	14.82	0.879		

Table 34: Architecture **3x10** — perturbation **translation**(`3,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean `sp`); continuation of Table 6 (6 combos, 31 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	3,3	zono	off	prev_pgd	0.0	1	3	0	14.78	14.18	**1.042**	*0.930*	*0.03*
						1	4	0	13.53	14.24	0.950		
						1	5	0	13.74	14.06	0.977		
						1	6	0	12.78	14.13	0.904		
						1	7	0	14.15	14.99	0.944		
						1	8	0	13.03	17.04	0.765		
		interval	off	direct_pgd	0.5	1	3	0	14.78	13.64	**1.084**	*0.928*	*0.03*
						1	4	0	13.53	13.75	0.984		
						1	5	0	13.74	14.25	0.964		
						1	6	0	12.78	14.33	0.892		
						1	7	0	14.15	17.38	0.814		
						1	8	0	13.03	15.66	0.832		
		zono	off	no	0.0	1	3	0	14.78	14.21	**1.040**	*0.921*	*0.03*
						1	4	0	13.53	15.10	0.896		
						1	5	0	13.74	16.91	0.813		
						1	6	0	12.78	14.84	0.861		
						1	7	0	14.15	13.96	**1.014**		
						1	8	0	13.03	14.45	0.902		
		zono	off	direct_pgd	0.5	1	3	0	14.78	15.24	0.970	*0.885*	*0.03*
						1	4	0	13.53	16.23	0.834		
						1	5	0	13.74	16.17	0.850		
						1	6	0	12.78	16.16	0.791		
						1	7	0	14.15	14.58	0.971		
						1	8	0	13.03	14.59	0.893		
		zono	off	prev_pgd	0.5	1	3	0	14.78	15.72	0.940	*0.879*	*0.03*
						1	4	0	13.53	15.91	0.850		
						1	5	0	13.74	16.46	0.835		
						1	6	0	12.78	17.32	0.738		
						1	7	0	14.15	14.49	0.977		
						1	8	0	13.03	13.91	0.937		
		zono	off	no	0.5	1	3	0	14.78	16.24	0.910	*0.845*	*0.03*
						1	4	0	13.53	18.49	0.732		
						1	5	0	13.74	16.85	0.815		
						1	6	0	12.78	14.31	0.893		
						1	7	0	14.15	16.64	0.850		
						1	8	0	13.03	14.99	0.869		

Table 35: Architecture **3x10** — perturbation **translation**(3,3), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean sp); continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
patch	`1,14,14,3`	zono	off	no	0.5	0	3	164	96.17	51.01	**1.885**	*1.524*	*0.04*
						0	4	164	328.21	101.25	**3.242**		
						0	5	164	62.33	49.39	**1.262**		
						0	6	164	34.38	42.42	0.810		
						0	7	164	71.72	97.50	0.736		
						0	8	164	55.30	45.77	**1.208**		
		interval	off	direct	0.5	0	3	164	96.17	122.92	0.782	*1.273*	*0.02*
						0	4	164	328.21	143.09	**2.294**		
						0	5	164	62.33	49.22	**1.266**		
						0	6	164	34.38	32.24	**1.066**		
						0	7	164	71.72	92.93	0.772		
						0	8	164	55.30	37.96	**1.457**		
		zono	off	prev_pgd	0.0	0	3	0	96.17	41.72	**2.305**	*1.240*	*0.03*
						0	4	0	328.21	276.30	**1.188**		
						0	5	0	62.33	47.36	**1.316**		
						0	6	0	34.38	45.06	0.763		
						0	7	0	71.72	77.54	0.925		
						0	8	0	55.30	58.76	0.941		
		interval	off	no	0.5	0	3	164	96.17	51.79	**1.857**	*1.215*	*0.04*
						0	4	164	328.21	161.03	**2.038**		
						0	5	164	62.33	60.41	**1.032**		
						0	6	164	34.38	52.81	0.651		
						0	7	164	71.72	124.15	0.578		
						0	8	164	55.30	48.84	**1.132**		
		zono	off	direct	0.5	0	3	164	96.17	121.05	0.794	*1.202*	*0.02*
						0	4	164	328.21	136.20	**2.410**		
						0	5	164	62.33	54.66	**1.140**		
						0	6	164	34.38	30.83	**1.115**		
						0	7	164	71.72	240.36	0.298		
						0	8	164	55.30	37.98	**1.456**		
		interval	off	prev	0.0	0	3	0	96.17	39.36	**2.443**	*1.193*	*0.01*
						0	4	0	328.21	275.28	**1.192**		
						0	5	0	62.33	45.85	**1.359**		
						0	6	0	34.38	87.08	0.395		
						0	7	0	71.72	104.67	0.685		
						0	8	0	55.30	51.03	**1.084**		
		zono	off	direct	0.0	0	3	0	96.17	47.19	**2.038**	*1.146*	*0.01*
						0	4	0	328.21	417.67	0.786		
						0	5	0	62.33	43.85	**1.421**		
						0	6	0	34.38	70.26	0.489		
						0	7	0	71.72	56.85	**1.262**		
						0	8	0	55.30	62.96	0.878		

Table 36: Architecture **cnn1** — every recorded advstd combination at all seeds, all τ values, perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–7 of 20 by mean `sp`) (7 combos, 42 cell rows; see Tables labelled `tab:safe_cnn1_*` for the other perturbations / c_{src} / τ-buckets of this arch). Rows shaded green satisfy `bp`=`off`; `sp`>1 (advstd faster) in **bold**. Combos sorted by descending mean `sp`; cells within a combo by $(c_{\text{src}}, c_{\text{tgt}})$. Total 1419 cell rows across this architecture. Timeout-cell gaps: Table 69.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}}-\delta_{\mathrm{adv}}\rvert}_{c\_src}$
patch	`1,14,14,3`	interval	off	direct	0.0	0	3	0	96.17	45.45	**2.116**	***1.114***	*0.02*
						0	4	0	328.21	414.52	0.792		
						0	5	0	62.33	52.38	**1.190**		
						0	6	0	34.38	69.70	0.493		
						0	7	0	71.72	52.48	**1.367**		
						0	8	0	55.30	75.87	0.729		
		interval	off	prev_pgd	0.5	0	3	164	96.17	76.72	**1.254**	***1.114***	*0.02*
						0	4	164	328.21	205.69	**1.596**		
						0	5	164	62.33	40.95	**1.522**		
						0	6	164	34.38	27.32	**1.258**		
						0	7	164	71.72	116.56	0.615		
						0	8	164	55.30	126.11	0.439		
		zono	off	prev_pgd	0.5	0	3	164	96.17	84.10	**1.144**	***1.105***	*0.02*
						0	4	164	328.21	366.58	0.895		
						0	5	164	62.33	45.80	**1.361**		
						0	6	164	34.38	26.99	**1.274**		
						0	7	164	71.72	109.11	0.657		
						0	8	164	55.30	42.52	**1.301**		
		interval	off	direct_pgd	0.5	0	3	164	96.17	66.52	**1.446**	***1.044***	*0.03*
						0	4	164	328.21	301.39	**1.089**		
						0	5	164	62.33	49.17	**1.268**		
						0	6	164	34.38	43.55	0.789		
						0	7	164	71.72	87.56	0.819		
						0	8	164	55.30	64.92	0.852		
		zono	off	prev	0.0	0	3	0	96.17	63.30	**1.519**	*0.994*	*0.02*
						0	4	0	328.21	290.62	**1.129**		
						0	5	0	62.33	44.32	**1.406**		
						0	6	0	34.38	101.62	0.338		
						0	7	0	71.72	124.55	0.576		
		zono	off	prev	0.5	0	3	164	96.17	1813.55	0.053	*0.975*	*0.81*
						0	4	164	328.21	207.45	**1.582**		
						0	5	164	62.33	90.33	0.690		
						0	6	164	34.38	59.43	0.578		
						0	7	164	71.72	96.81	0.741		
						0	8	164	55.30	25.09	**2.204**		
		interval	off	prev	0.5	0	3	164	96.17	1813.19	0.053	*0.936*	*0.81*
						0	4	164	328.21	213.59	**1.537**		
						0	5	164	62.33	109.38	0.570		
						0	6	164	34.38	67.41	0.510		
						0	7	164	71.72	97.36	0.737		
						0	8	164	55.30	25.01	**2.211**		

Table 37: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 8–14 of 20 by mean `sp`); continuation of Table 36 (7 combos, 41 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
patch	`1,14,14,3`	interval	off	no	0.0	0	3	0	96.17	53.51	**1.797**	*0.918*	*0.02*
						0	4	0	328.21	482.18	0.681		
						0	5	0	62.33	603.28	0.103		
						0	6	0	34.38	56.96	0.604		
						0	7	0	71.72	47.44	**1.512**		
						0	8	0	55.30	67.96	0.814		
		interval	off	prev_pgd	0.0	0	3	0	96.17	56.12	**1.714**	*0.897*	*0.03*
						0	4	0	328.21	428.98	0.765		
						0	5	0	62.33	57.21	**1.089**		
						0	6	0	34.38	89.39	0.385		
						0	7	0	71.72	115.10	0.623		
						0	8	0	55.30	68.82	0.804		
		zono	off	direct_pgd	0.5	0	3	164	96.17	67.40	**1.427**	*0.811*	*0.03*
						0	4	164	328.21	374.86	0.876		
						0	5	164	62.33	59.24	**1.052**		
						0	6	164	34.38	54.43	0.632		
						0	7	164	71.72	205.70	0.349		
						0	8	164	55.30	103.94	0.532		
		zono	off	no	0.0	0	3	0	96.17	76.55	**1.256**	*0.692*	*0.02*
						0	4	0	328.21	525.50	0.625		
						0	5	0	62.33	744.70	0.084		
						0	6	0	34.38	68.74	0.500		
						0	7	0	71.72	94.25	0.761		
						0	8	0	55.30	59.63	0.927		
		zono	off	direct_pgd	0.0	0	3	0	96.17	1816.27	0.053	*0.564*	*0.51*
						0	4	0	328.21	579.79	0.566		
						0	5	0	62.33	54.46	**1.145**		
						0	6	0	34.38	84.39	0.407		
						0	7	0	71.72	98.45	0.728		
						0	8	0	55.30	113.47	0.487		
		interval	off	direct_pgd	0.0	0	3	0	96.17	1813.17	0.053	*0.533*	*0.51*
						0	4	0	328.21	636.96	0.515		
						0	5	0	62.33	74.07	0.842		
						0	6	0	34.38	62.84	0.547		
						0	7	0	71.72	88.39	0.811		
						0	8	0	55.30	129.35	0.428		

Table 38: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 15–20 of 20 by mean `sp`); continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
patch	`1,14,14,3`	interval	off	prev_pgd	0.5	1	3	164	94.60	22.74	**4.160**	***2.280***	*0.02*
						1	4	164	113.19	73.66	**1.537**		
						1	5	164	138.25	70.21	**1.969**		
						1	6	164	206.61	133.40	**1.549**		
						1	7	164	83.24	37.29	**2.232**		
						1	8	164	67.32	30.14	**2.234**		
		zono	off	prev_pgd	0.5	1	3	164	94.60	24.94	**3.793**	***1.993***	*0.03*
						1	4	164	113.19	66.56	**1.701**		
						1	6	164	206.61	197.30	**1.047**		
						1	7	164	83.24	46.86	**1.776**		
						1	8	164	67.32	40.90	**1.646**		
		interval	off	no	0.5	1	3	164	94.60	30.30	**3.122**	***1.911***	*0.01*
						1	4	164	113.19	84.52	**1.339**		
						1	5	164	138.25	76.75	**1.801**		
						1	6	164	206.61	115.30	**1.792**		
						1	7	164	83.24	49.21	**1.692**		
						1	8	164	67.32	39.18	**1.718**		
		zono	off	direct	0.5	1	3	164	94.60	26.91	**3.515**	***1.853***	*0.01*
						1	4	164	113.19	73.12	**1.548**		
						1	5	164	138.25	97.23	**1.422**		
						1	6	164	206.61	151.18	**1.367**		
						1	7	164	83.24	45.49	**1.830**		
						1	8	164	67.32	46.80	**1.438**		
		interval	off	prev_pgd	0.0	1	3	0	94.60	60.23	**1.571**	***1.711***	*0.02*
						1	4	0	113.19	67.97	**1.665**		
						1	5	0	138.25	61.73	**2.240**		
						1	6	0	206.61	115.20	**1.793**		
						1	7	0	83.24	44.04	**1.890**		
						1	8	0	67.32	60.71	**1.109**		
		zono	off	prev_pgd	0.0	1	3	0	94.60	54.97	**1.721**	***1.709***	*0.03*
						1	4	0	113.19	55.31	**2.046**		
						1	5	0	138.25	39.19	**3.528**		
						1	6	0	206.61	203.15	**1.017**		
						1	7	0	83.24	128.46	0.648		
						1	8	0	67.32	52.12	**1.292**		

Table 39: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 16 by mean `sp`); continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
patch	`1,14,14,3`	zono	off	no	0.5	1	3	164	94.60	51.75	**1.828**	***1.646***	*0.02*
						1	4	164	113.19	69.92	**1.619**		
						1	5	164	138.25	68.04	**2.032**		
						1	6	164	206.61	149.88	**1.379**		
						1	7	164	83.24	52.87	**1.574**		
						1	8	164	67.32	46.60	**1.445**		
		interval	off	no	0.0	1	3	0	94.60	71.27	**1.327**	***1.612***	*0.03*
						1	4	0	113.19	56.19	**2.014**		
						1	5	0	138.25	78.56	**1.760**		
						1	6	0	206.61	105.37	**1.961**		
						1	7	0	83.24	66.38	**1.254**		
						1	8	0	67.32	49.69	**1.355**		
		zono	off	direct	0.0	1	3	0	94.60	51.97	**1.820**	***1.578***	*0.01*
						1	4	0	113.19	156.66	0.723		
						1	5	0	138.25	68.32	**2.024**		
						1	6	0	206.61	81.57	**2.533**		
						1	7	0	83.24	64.22	**1.296**		
						1	8	0	67.32	62.89	**1.070**		
		interval	off	direct_pgd	0.5	1	3	164	94.60	33.34	**2.837**	***1.542***	*0.54*
						1	4	164	113.19	1813.29	0.062		
						1	5	164	138.25	78.59	**1.759**		
						1	6	164	206.61	141.01	**1.465**		
						1	7	164	83.24	50.22	**1.658**		
						1	8	164	67.32	45.81	**1.470**		
		zono	off	direct_pgd	0.5	1	3	164	94.60	38.58	**2.452**	***1.517***	*0.52*
						1	4	164	113.19	1815.95	0.062		
						1	5	164	138.25	75.23	**1.838**		
						1	6	164	206.61	137.68	**1.501**		
						1	7	164	83.24	47.68	**1.746**		
						1	8	164	67.32	44.70	**1.506**		
		zono	off	no	0.0	1	3	0	94.60	60.39	**1.566**	***1.471***	*0.01*
						1	4	0	113.19	169.45	0.668		
						1	5	0	138.25	104.89	**1.318**		
						1	6	0	206.61	116.67	**1.771**		
						1	7	0	83.24	59.83	**1.391**		
						1	8	0	67.32	31.85	**2.114**		

Table 40: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 16 by mean `sp`); continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
patch	`1,14,14,3`	interval	off	direct	0.0	1	3	0	94.60	40.92	**2.312**	***1.442***	*0.02*
						1	4	0	113.19	188.34	0.601		
						1	5	0	138.25	92.32	**1.498**		
						1	6	0	206.61	102.15	**2.023**		
						1	7	0	83.24	59.92	**1.389**		
						1	8	0	67.32	80.83	0.833		
		interval	off	direct_pgd	0.0	1	3	0	94.60	50.45	**1.875**	***1.377***	*0.01*
						1	4	0	113.19	73.22	**1.546**		
						1	5	0	138.25	74.33	**1.860**		
						1	6	0	206.61	145.70	**1.418**		
						1	7	0	83.24	125.07	0.666		
						1	8	0	67.32	74.84	0.900		
		zono	off	direct_pgd	0.0	1	3	0	94.60	46.31	**2.043**	***1.344***	*0.01*
						1	4	0	113.19	83.31	**1.359**		
						1	5	0	138.25	77.27	**1.789**		
						1	6	0	206.61	154.97	**1.333**		
						1	7	0	83.24	199.19	0.418		
						1	8	0	67.32	60.07	**1.121**		
		interval	off	direct	0.5	1	3	164	94.60	47.13	**2.007**	***1.335***	*0.38*
						1	4	164	113.19	86.36	**1.311**		
						1	5	164	138.25	100.78	**1.372**		
						1	6	164	206.61	1815.70	0.114		
						1	7	164	83.24	50.31	**1.655**		
						1	8	164	67.32	43.33	**1.554**		

Table 41: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–16 of 16 by mean `sp`); continuation of Table 36 (4 combos, 24 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
occ	5,5,5	interval	off	direct_pgd	0.5	0	3	173	42.18	34.82	**1.211**	***1.399***	*0.02*
						0	4	173	536.01	335.80	**1.596**		
						0	5	173	110.91	48.63	**2.281**		
						0	6	173	33.37	41.37	0.807		
						0	7	173	56.20	48.60	**1.156**		
						0	8	173	57.73	42.98	**1.343**		
		zono	off	direct_pgd	0.5	0	3	173	42.18	34.73	**1.215**	***1.381***	*0.02*
						0	4	173	536.01	375.96	**1.426**		
						0	5	173	110.91	46.90	**2.365**		
						0	6	173	33.37	40.99	0.814		
						0	7	173	56.20	49.72	**1.130**		
						0	8	173	57.73	43.15	**1.338**		
		zono	off	prev_pgd	0.0	0	3	0	42.18	53.85	0.783	***1.355***	*0.03*
						0	4	0	536.01	325.55	**1.646**		
						0	5	0	110.91	57.06	**1.944**		
						0	6	0	33.37	34.23	0.975		
						0	7	0	56.20	44.09	**1.275**		
						0	8	0	57.73	38.37	**1.505**		
		interval	off	prev_pgd	0.0	0	3	0	42.18	75.61	0.558	***1.348***	*0.03*
						0	4	0	536.01	323.39	**1.657**		
						0	5	0	110.91	56.61	**1.959**		
						0	6	0	33.37	35.94	0.928		
						0	7	0	56.20	36.94	**1.521**		
						0	8	0	57.73	39.51	**1.461**		
		zono	off	direct_pgd	0.0	0	3	0	42.18	41.00	**1.029**	***1.324***	*0.01*
						0	4	0	536.01	332.14	**1.614**		
						0	5	0	110.91	51.62	**2.149**		
						0	6	0	33.37	51.67	0.646		
						0	7	0	56.20	58.31	0.964		
						0	8	0	57.73	37.46	**1.541**		
		interval	off	prev	0.0	0	3	0	42.18	39.88	**1.058**	***1.314***	*0.03*
						0	5	0	110.91	65.35	**1.697**		
						0	6	0	33.37	36.06	0.925		
						0	7	0	56.20	41.80	**1.344**		
						0	8	0	57.73	37.39	**1.544**		

Table 42: Architecture **cnn1** — perturbation **occ**(5,5,5), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 17 by mean sp); continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
occ	`5,5,5`	interval	off	prev_pgd	0.5	0	3	173	42.18	28.19	**1.496**	***1.296***	*0.03*
						0	4	173	536.01	460.90	**1.163**		
						0	5	173	110.91	72.10	**1.538**		
						0	6	173	33.37	37.94	0.880		
						0	7	173	56.20	30.40	**1.849**		
						0	8	173	57.73	67.72	0.852		
		interval	off	direct	0.0	0	3	0	42.18	43.78	0.963	***1.286***	*0.02*
						0	4	0	536.01	313.26	**1.711**		
						0	5	0	110.91	71.10	**1.560**		
						0	6	0	33.37	48.51	0.688		
						0	7	0	56.20	57.00	0.986		
						0	8	0	57.73	31.89	**1.810**		
		zono	off	direct	0.0	0	3	0	42.18	43.88	0.961	***1.263***	*0.02*
						0	4	0	536.01	292.89	**1.830**		
						0	5	0	110.91	82.30	**1.348**		
						0	6	0	33.37	49.85	0.669		
						0	7	0	56.20	52.58	**1.069**		
						0	8	0	57.73	33.94	**1.701**		
		zono	off	prev_pgd	0.5	0	3	173	42.18	31.56	**1.337**	***1.258***	*0.03*
						0	4	173	536.01	461.06	**1.163**		
						0	5	173	110.91	70.63	**1.570**		
						0	6	173	33.37	38.78	0.860		
						0	7	173	56.20	33.16	**1.695**		
						0	8	173	57.73	62.44	0.925		
		interval	off	no	0.5	0	3	173	42.18	38.63	**1.092**	***1.188***	*0.03*
						0	4	173	536.01	315.44	**1.699**		
						0	5	173	110.91	53.53	**2.072**		
						0	6	173	33.37	65.73	0.508		
						0	7	173	56.20	92.77	0.606		
						0	8	173	57.73	50.07	**1.153**		
		interval	off	no	0.0	0	3	0	42.18	65.05	0.648	***1.178***	*0.02*
						0	4	0	536.01	348.88	**1.536**		
						0	5	0	110.91	47.94	**2.314**		
						0	6	0	33.37	44.16	0.756		
						0	7	0	56.20	62.06	0.906		
						0	8	0	57.73	63.37	0.911		

Table 43: Architecture **cnn1** — perturbation **occ**(`5,5,5`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 17 by mean `sp`); continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
occ	`5,5,5`	zono	off	no	0.5	0	3	173	42.18	39.09	**1.079**	***1.145***	*0.03*
						0	4	173	536.01	341.61	**1.569**		
						0	5	173	110.91	52.15	**2.127**		
						0	6	173	33.37	67.59	0.494		
						0	7	173	56.20	100.10	0.561		
						0	8	173	57.73	55.67	**1.037**		
		interval	off	direct_pgd	0.0	0	3	0	42.18	64.93	0.650	***1.142***	*0.01*
						0	4	0	536.01	338.13	**1.585**		
						0	5	0	110.91	50.78	**2.184**		
						0	6	0	33.37	63.26	0.528		
						0	7	0	56.20	64.03	0.878		
						0	8	0	57.73	56.18	**1.028**		
		interval	off	direct	0.5	0	3	173	42.18	47.45	0.889	***1.019***	*0.02*
						0	4	173	536.01	549.47	0.976		
						0	5	173	110.91	51.95	**2.135**		
						0	6	173	33.37	61.14	0.546		
						0	7	173	56.20	151.12	0.372		
						0	8	173	57.73	48.24	**1.197**		
		zono	off	direct	0.5	0	3	173	42.18	52.34	0.806	*0.948*	*0.02*
						0	4	173	536.01	582.56	0.920		
						0	5	173	110.91	61.46	**1.805**		
						0	6	173	33.37	46.82	0.713		
						0	7	173	56.20	147.12	0.382		
						0	8	173	57.73	54.44	**1.060**		
		zono	off	no	0.0	0	3	0	42.18	122.06	0.346	*0.926*	*0.02*
						0	4	0	536.01	444.89	**1.205**		
						0	5	0	110.91	64.69	**1.714**		
						0	6	0	33.37	54.35	0.614		
						0	7	0	56.20	67.59	0.831		
						0	8	0	57.73	68.23	0.846		

Table 44: Architecture **cnn1** — perturbation **occ**(`5,5,5`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–17 of 17 by mean `sp`); continuation of Table 36 (5 combos, 30 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
occ	5,5,5	interval	off	prev_pgd	0.0	1	3	0	73.18	24.81	**2.950**	***1.348***	*0.03*
						1	4	0	35.85	39.50	0.908		
						1	5	0	35.61	50.26	0.709		
						1	6	0	41.49	38.01	**1.092**		
						1	7	0	38.38	40.74	0.942		
						1	8	0	40.33	27.05	**1.491**		
		interval	off	prev_pgd	0.5	1	3	173	73.18	32.71	**2.237**	***1.218***	*0.04*
						1	4	173	35.85	40.76	0.880		
						1	5	173	35.61	27.73	**1.284**		
						1	6	173	41.49	39.45	**1.052**		
						1	7	173	38.38	55.80	0.688		
						1	8	173	40.33	34.62	**1.165**		
		zono	off	direct_pgd	0.5	1	3	173	73.18	32.79	**2.232**	***1.193***	*0.03*
						1	4	173	35.85	42.36	0.846		
						1	5	173	35.61	31.22	**1.141**		
						1	6	173	41.49	43.46	0.955		
						1	7	173	38.38	34.96	**1.098**		
						1	8	173	40.33	45.53	0.886		
		zono	off	prev_pgd	0.0	1	3	0	73.18	43.33	**1.689**	***1.123***	*0.04*
						1	4	0	35.85	55.05	0.651		
						1	6	0	41.49	40.13	**1.034**		
						1	8	0	40.33	36.13	**1.116**		
		interval	off	direct_pgd	0.5	1	3	173	73.18	33.69	**2.172**	***1.118***	*0.02*
						1	4	173	35.85	43.22	0.829		
						1	5	173	35.61	40.50	0.879		
						1	6	173	41.49	55.21	0.751		
						1	7	173	38.38	31.14	**1.232**		
						1	8	173	40.33	47.95	0.841		
		zono	off	no	0.0	1	3	0	73.18	40.60	**1.802**	***1.106***	*0.01*
						1	4	0	35.85	49.65	0.722		
						1	5	0	35.61	37.87	0.940		
						1	6	0	41.49	40.64	**1.021**		
						1	7	0	38.38	39.79	0.965		
						1	8	0	40.33	34.03	**1.185**		

Table 45: Architecture **cnn1** — perturbation **occ**(5,5,5), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean sp); continuation of Table 36 (6 combos, 34 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
occ	5,5,5	zono	off	prev_pgd	0.5	1	3	173	73.18	59.29	**1.234**	***1.079***	*0.05*
						1	4	173	35.85	49.17	0.729		
						1	5	173	35.61	26.54	**1.342**		
						1	6	173	41.49	43.22	0.960		
						1	7	173	38.38	33.92	**1.131**		
		interval	off	no	0.5	1	3	173	73.18	49.58	**1.476**	*0.990*	*0.01*
						1	4	173	35.85	37.47	0.957		
						1	5	173	35.61	34.20	**1.041**		
						1	6	173	41.49	58.00	0.715		
						1	7	173	38.38	41.41	0.927		
						1	8	173	40.33	48.83	0.826		
		zono	off	direct_pgd	0.0	1	3	0	73.18	44.11	**1.659**	*0.990*	*0.01*
						1	4	0	35.85	41.46	0.865		
						1	5	0	35.61	40.23	0.885		
						1	6	0	41.49	54.65	0.759		
						1	7	0	38.38	49.07	0.782		
		zono	off	no	0.5	1	3	173	73.18	49.39	**1.482**	*0.988*	*0.01*
						1	4	173	35.85	60.99	0.588		
						1	5	173	35.61	34.24	**1.040**		
						1	6	173	41.49	41.05	**1.011**		
						1	7	173	38.38	41.95	0.915		
						1	8	173	40.33	45.21	0.892		
		interval	off	no	0.0	1	3	0	73.18	43.24	**1.692**	*0.986*	*0.02*
						1	4	0	35.85	43.11	0.832		
						1	5	0	35.61	48.13	0.740		
						1	7	0	38.38	48.75	0.787		
						1	8	0	40.33	45.90	0.879		
		interval	off	direct_pgd	0.0	1	3	0	73.18	51.82	**1.412**	*0.733*	*0.02*
						1	4	0	35.85	60.74	0.590		
						1	5	0	35.61	924.61	0.039		
						1	6	0	41.49	42.70	0.972		
						1	7	0	38.38	48.51	0.791		
						1	8	0	40.33	68.17	0.592		

Table 46: Architecture **cnn1** — perturbation **occ**(5,5,5), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean sp); continuation of Table 36 (6 combos, 33 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,1`	zono	off	direct_pgd	0.0	0	3	0	344.88	288.50	**1.195**	***2.171***	*0.03*
						0	4	0	442.26	259.88	**1.702**		
						0	5	0	393.83	96.14	**4.096**		
						0	6	0	526.31	217.26	**2.422**		
						0	7	0	568.70	263.45	**2.159**		
						0	8	0	1392.84	959.93	**1.451**		
		interval	off	direct_pgd	0.0	0	3	0	344.88	253.52	**1.360**	***2.021***	*0.03*
						0	4	0	442.26	344.95	**1.282**		
						0	5	0	393.83	113.37	**3.474**		
						0	6	0	526.31	198.91	**2.646**		
						0	7	0	568.70	278.84	**2.040**		
						0	8	0	1392.84	1049.74	**1.327**		
		zono	off	prev_pgd	0.5	0	3	172	344.88	131.45	**2.624**	***2.016***	*0.38*
						0	4	172	442.26	291.64	**1.516**		
						0	5	172	393.83	398.06	0.989		
						0	6	172	526.31	1812.37	0.290		
						0	7	172	568.70	151.25	**3.760**		
						0	8	172	1392.84	477.64	**2.916**		
		zono	off	direct_pgd	0.5	0	3	172	344.88	207.34	**1.663**	***1.976***	*0.07*
						0	4	172	442.26	234.89	**1.883**		
						0	5	172	393.83	209.41	**1.881**		
						0	6	172	526.31	200.65	**2.623**		
						0	7	172	568.70	263.08	**2.162**		
						0	8	172	1392.84	847.80	**1.643**		
		zono	off	prev_pgd	0.0	0	3	0	344.88	174.09	**1.981**	***1.793***	*0.02*
						0	4	0	442.26	300.31	**1.473**		
						0	5	0	393.83	285.28	**1.381**		
						0	6	0	526.31	171.34	**3.072**		
						0	7	0	568.70	355.62	**1.599**		
						0	8	0	1392.84	1110.92	**1.254**		
		interval	off	prev_pgd	0.0	0	3	0	344.88	213.91	**1.612**	***1.719***	*0.02*
						0	4	0	442.26	323.31	**1.368**		
						0	5	0	393.83	272.72	**1.444**		
						0	6	0	526.31	171.29	**3.073**		
						0	7	0	568.70	357.55	**1.591**		
						0	8	0	1392.84	1137.33	**1.225**		
		interval	off	direct	0.5	0	3	171	344.88	228.41	**1.510**	***1.566***	*0.02*
						0	4	171	442.26	382.36	**1.157**		
						0	5	171	393.83	241.57	**1.630**		
						0	6	171	526.31	281.62	**1.869**		
						0	7	171	568.70	269.51	**2.110**		
						0	8	171	1392.84	1242.66	**1.121**		

Table 47: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–7 of 20 by mean `sp`); continuation of Table 36 (7 combos, 42 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,1`	zono	off	no	0.5	0	3	172	344.88	296.98	**1.161**	***1.345***	*0.03*
						0	4	172	442.26	484.69	0.912		
						0	5	172	393.83	311.73	**1.263**		
						0	6	172	526.31	228.73	**2.301**		
						0	7	172	568.70	629.88	0.903		
						0	8	172	1392.84	910.59	**1.530**		
		interval	off	no	0.5	0	3	171	344.88	260.07	**1.326**	***1.334***	*0.02*
						0	4	171	442.26	336.89	**1.313**		
						0	5	171	393.83	326.17	**1.207**		
						0	6	171	526.31	313.88	**1.677**		
						0	7	171	568.70	336.40	**1.691**		
						0	8	171	1392.84	1757.53	0.792		
		zono	off	direct	0.5	0	3	172	344.88	322.98	**1.068**	***1.327***	*0.02*
						0	4	172	442.26	538.49	0.821		
						0	5	172	393.83	351.91	**1.119**		
						0	6	172	526.31	278.93	**1.887**		
						0	7	172	568.70	429.81	**1.323**		
						0	8	172	1392.84	799.06	**1.743**		
		interval	off	direct_pgd	0.5	0	3	171	344.88	302.42	**1.140**	***1.268***	*0.72*
						0	4	171	442.26	321.04	**1.378**		
						0	5	171	393.83	1811.96	0.217		
						0	6	171	526.31	282.73	**1.862**		
						0	7	171	568.70	253.32	**2.245**		
						0	8	171	1392.84	1822.33	0.764		
		interval	off	prev	0.0	0	3	0	344.88	283.93	**1.215**	***1.230***	*0.02*
						0	4	0	442.26	517.61	0.854		
						0	5	0	393.83	293.98	**1.340**		
						0	6	0	526.31	485.77	**1.083**		
						0	7	0	568.70	519.23	**1.095**		
						0	8	0	1392.84	776.08	**1.795**		
		zono	off	prev	0.0	0	3	0	344.88	353.67	0.975	***1.213***	*0.02*
						0	4	0	442.26	549.44	0.805		
						0	5	0	393.83	201.60	**1.954**		
						0	6	0	526.31	423.96	**1.241**		
						0	7	0	568.70	520.69	**1.092**		
						0	8	0	1392.84	1152.86	**1.208**		
		interval	off	prev_pgd	0.5	0	3	171	344.88	436.69	0.790	***1.155***	*0.02*
						0	4	171	442.26	310.01	**1.427**		
						0	5	171	393.83	463.68	0.849		
						0	6	171	526.31	432.89	**1.216**		
						0	7	171	568.70	325.99	**1.745**		
						0	8	171	1392.84	1545.65	0.901		

Table 48: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 8–14 of 20 by mean `sp`); continuation of Table 36 (7 combos, 42 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,1`	zono	off	direct	0.0	0	3	0	344.88	365.77	0.943	***1.125***	*0.02*
						0	4	0	442.26	359.49	**1.230**		
						0	5	0	393.83	354.14	**1.112**		
						0	6	0	526.31	569.71	0.924		
						0	7	0	568.70	416.37	**1.366**		
						0	8	0	1392.84	1185.70	**1.175**		
		interval	off	direct	0.0	0	3	0	344.88	379.13	0.910	***1.078***	*0.02*
						0	4	0	442.26	350.83	**1.261**		
						0	5	0	393.83	377.60	**1.043**		
						0	6	0	526.31	596.72	0.882		
						0	7	0	568.70	464.24	**1.225**		
						0	8	0	1392.84	1212.14	**1.149**		
		interval	off	prev	0.5	0	3	171	344.88	501.80	0.687	***1.068***	*0.01*
						0	4	171	442.26	852.19	0.519		
						0	5	171	393.83	268.31	**1.468**		
						0	6	171	526.31	393.13	**1.339**		
						0	7	171	568.70	447.58	**1.271**		
						0	8	171	1392.84	1235.53	**1.127**		
		interval	off	no	0.0	0	3	0	344.88	502.15	0.687	*0.970*	*0.29*
						0	4	0	442.26	348.09	**1.271**		
						0	5	0	393.83	413.77	0.952		
						0	6	0	526.31	1812.54	0.290		
						0	7	0	568.70	419.22	**1.357**		
						0	8	0	1392.84	1104.32	**1.261**		
		zono	off	no	0.0	0	3	0	344.88	554.54	0.622	*0.895*	*0.29*
						0	4	0	442.26	366.02	**1.208**		
						0	5	0	393.83	426.76	0.923		
						0	6	0	526.31	1811.14	0.291		
						0	7	0	568.70	391.80	**1.452**		
						0	8	0	1392.84	1590.32	0.876		
		zono	off	prev	0.5	0	3	172	344.88	504.35	0.684	*0.841*	*0.02*
						0	4	172	442.26	686.62	0.644		
						0	5	172	393.83	472.35	0.834		
						0	7	172	568.70	528.36	**1.076**		
						0	8	172	1392.84	1438.65	0.968		

Table 49: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 15–20 of 20 by mean `sp`); continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,1`	interval	off	prev	0.5	1	3	171	298.04	250.83	**1.188**	***1.400***	*0.02*
						1	4	171	932.29	602.10	**1.548**		
						1	5	171	226.98	279.55	0.812		
						1	6	171	336.06	221.33	**1.518**		
						1	7	171	776.98	333.88	**2.327**		
						1	8	171	1820.97	1812.42	**1.005**		
		interval	off	no	0.5	1	3	171	298.04	173.55	**1.717**	***1.380***	*1.17*
						1	4	171	932.29	614.35	**1.518**		
						1	5	171	226.98	444.97	0.510		
						1	6	171	336.06	108.23	**3.105**		
						1	7	171	776.98	1835.29	0.423		
						1	8	171	1820.97	1812.79	**1.005**		
		interval	off	prev	0.0	1	3	0	298.04	379.32	0.786	***1.347***	*0.01*
						1	5	0	226.98	139.97	**1.622**		
						1	6	0	336.06	142.61	**2.356**		
						1	7	0	776.98	804.83	0.965		
						1	8	0	1820.97	1811.96	**1.005**		
		interval	off	direct	0.5	1	3	171	298.04	267.61	**1.114**	***1.268***	*0.06*
						1	5	171	226.98	323.99	0.701		
						1	6	171	336.06	278.47	**1.207**		
						1	7	171	776.98	335.00	**2.319**		
						1	8	171	1820.97	1821.66	1.000		
		zono	off	direct	0.5	1	3	172	298.04	396.57	0.752	***1.205***	*0.07*
						1	4	172	932.29	535.04	**1.742**		
						1	5	172	226.98	359.46	0.631		
						1	6	172	336.06	355.71	0.945		
						1	7	172	776.98	360.82	**2.153**		
						1	8	172	1820.97	1813.83	**1.004**		
		interval	off	prev_pgd	0.5	1	3	171	298.04	417.83	0.713	***1.198***	*0.01*
						1	4	171	932.29	832.20	**1.120**		
						1	5	171	226.98	309.06	0.734		
						1	6	171	336.06	329.16	**1.021**		
						1	7	171	776.98	299.67	**2.593**		
						1	8	171	1820.97	1814.79	**1.003**		
		interval	off	no	0.0	1	3	0	298.04	381.75	0.781	***1.182***	*0.02*
						1	4	0	932.29	1169.82	0.797		
						1	5	0	226.98	165.19	**1.374**		
						1	6	0	336.06	204.49	**1.643**		
						1	7	0	776.98	519.38	**1.496**		
						1	8	0	1820.97	1820.06	**1.000**		

Table 50: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–7 of 20 by mean `sp`); continuation of Table 36 (7 combos, 40 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,1`	zono	off	direct_pgd	0.5	1	3	172	298.04	364.56	0.818	***1.160***	*0.04*
						1	4	172	932.29	557.51	**1.672**		
						1	5	172	226.98	264.20	0.859		
						1	6	172	336.06	251.07	**1.339**		
						1	7	172	776.98	611.92	**1.270**		
						1	8	172	1820.97	1815.10	**1.003**		
		zono	off	prev_pgd	0.5	1	3	172	298.04	349.65	0.852	***1.155***	*0.01*
						1	4	172	932.29	846.05	**1.102**		
						1	5	172	226.98	159.90	**1.420**		
						1	6	172	336.06	353.84	0.950		
						1	7	172	776.98	483.64	**1.607**		
						1	8	172	1820.97	1816.26	**1.003**		
		interval	off	prev_pgd	0.0	1	3	0	298.04	409.60	0.728	***1.081***	*0.02*
						1	4	0	932.29	1269.03	0.735		
						1	5	0	226.98	154.32	**1.471**		
						1	6	0	336.06	361.04	0.931		
						1	7	0	776.98	479.96	**1.619**		
						1	8	0	1820.97	1814.30	**1.004**		
		zono	off	prev	0.5	1	3	172	298.04	292.73	**1.018**	***1.076***	*0.05*
						1	4	172	932.29	777.36	**1.199**		
						1	5	172	226.98	388.02	0.585		
						1	6	172	336.06	285.39	**1.178**		
						1	7	172	776.98	554.68	**1.401**		
		interval	off	direct_pgd	0.5	1	3	171	298.04	488.40	0.610	***1.030***	*0.05*
						1	4	171	932.29	831.07	**1.122**		
						1	5	171	226.98	275.05	0.825		
						1	6	171	336.06	529.76	0.634		
						1	7	171	776.98	391.35	**1.985**		
						1	8	171	1820.97	1813.87	**1.004**		
		zono	off	prev	0.0	1	3	0	298.04	265.88	**1.121**	*0.995*	*0.03*
						1	5	0	226.98	259.33	0.875		
						1	6	0	336.06	361.93	0.929		
						1	7	0	776.98	744.43	**1.044**		
						1	8	0	1820.97	1812.51	**1.005**		
		interval	off	direct	0.0	1	3	0	298.04	328.19	0.908	*0.956*	*0.07*
						1	4	0	932.29	1655.04	0.563		
						1	5	0	226.98	274.76	0.826		
						1	6	0	336.06	298.91	**1.124**		
						1	7	0	776.98	590.87	**1.315**		
						1	8	0	1820.97	1819.53	**1.001**		

Table 51: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 8–14 of 20 by mean `sp`); continuation of Table 36 (7 combos, 40 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}}-\delta_{\mathrm{adv}}\rvert}_{c\_src}$
translation	`1,1`	zono	off	direct	0.0	1	3	0	298.04	271.96	**1.096**	*0.952*	*0.02*
						1	4	0	932.29	1427.24	0.653		
						1	5	0	226.98	332.76	0.682		
						1	6	0	336.06	320.67	**1.048**		
						1	7	0	776.98	631.45	**1.230**		
						1	8	0	1820.97	1813.96	**1.004**		
		zono	off	no	0.0	1	3	0	298.04	459.45	0.649	*0.945*	*0.04*
						1	4	0	932.29	870.50	**1.071**		
						1	5	0	226.98	230.31	0.986		
						1	6	0	336.06	442.83	0.759		
						1	7	0	776.98	644.45	**1.206**		
						1	8	0	1820.97	1817.80	**1.002**		
		zono	off	no	0.5	1	3	172	298.04	303.71	0.981	*0.916*	*0.04*
						1	4	172	932.29	1029.63	0.905		
						1	6	172	336.06	392.08	0.857		
						1	7	172	776.98	1416.25	0.549		
						1	8	172	1820.97	1411.65	**1.290**		
		zono	off	prev_pgd	0.0	1	3	0	298.04	292.76	**1.018**	*0.915*	*0.02*
						1	4	0	932.29	1135.71	0.821		
						1	5	0	226.98	270.13	0.840		
						1	6	0	336.06	321.11	**1.047**		
						1	7	0	776.98	1013.88	0.766		
						1	8	0	1820.97	1819.69	**1.001**		
		zono	off	direct_pgd	0.0	1	3	0	298.04	379.46	0.785	*0.802*	*0.03*
						1	4	0	932.29	1816.50	0.513		
						1	5	0	226.98	274.36	0.827		
						1	6	0	336.06	347.39	0.967		
						1	7	0	776.98	1089.86	0.713		
						1	8	0	1820.97	1814.85	**1.003**		
		interval	off	direct_pgd	0.0	1	3	0	298.04	670.24	0.445	*0.717*	*0.06*
						1	4	0	932.29	1805.24	0.516		
						1	5	0	226.98	293.86	0.772		
						1	6	0	336.06	378.70	0.887		
						1	7	0	776.98	1150.00	0.676		
						1	8	0	1820.97	1814.94	**1.003**		

Table 52: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\mathrm{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 15–20 of 20 by mean `sp`); continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}}-\delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	interval	off	direct_pgd	0.0	0	3	0	251.33	96.13	**2.614**	***1.235***	*0.02*
						0	5	0	270.72	321.11	0.843		
						0	6	0	999.68	1405.63	0.711		
						0	7	0	246.58	292.34	0.843		
						0	8	0	276.94	237.75	**1.165**		
		zono	off	direct_pgd	0.5	0	3	172	251.33	211.85	**1.186**	***1.224***	*0.03*
						0	5	172	270.72	264.62	**1.023**		
						0	6	172	999.68	602.44	**1.659**		
						0	7	172	246.58	207.30	**1.189**		
						0	8	172	276.94	261.07	**1.061**		
		interval	off	direct_pgd	0.5	0	3	171	251.33	136.31	**1.844**	***1.157***	*0.02*
						0	5	171	270.72	243.55	**1.112**		
						0	6	171	999.68	973.18	**1.027**		
						0	7	171	246.58	366.99	0.672		
						0	8	171	276.94	245.42	**1.128**		
		zono	off	no	0.0	0	3	0	251.33	174.86	**1.437**	***1.147***	*0.02*
						0	5	0	270.72	164.80	**1.643**		
						0	6	0	999.68	1115.07	0.897		
						0	7	0	246.58	284.29	0.867		
						0	8	0	276.94	311.46	0.889		
		zono	off	direct_pgd	0.0	0	3	0	251.33	113.41	**2.216**	***1.141***	*0.02*
						0	5	0	270.72	296.14	0.914		
						0	6	0	999.68	986.27	**1.014**		
						0	7	0	246.58	287.67	0.857		
						0	8	0	276.94	393.53	0.704		
		interval	off	no	0.5	0	3	171	251.33	226.36	**1.110**	***1.141***	*0.07*
						0	5	171	270.72	238.27	**1.136**		
						0	6	171	999.68	870.33	**1.149**		
						0	7	171	246.58	161.62	**1.526**		
						0	8	171	276.94	353.63	0.783		
		interval	off	prev_pgd	0.5	0	3	171	251.33	377.48	0.666	***1.112***	*0.01*
						0	5	171	270.72	321.88	0.841		
						0	6	171	999.68	789.28	**1.267**		
						0	7	171	246.58	409.77	0.602		
						0	8	171	276.94	126.67	**2.186**		

Table 53: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–7 of 20 by mean `sp`); continuation of Table 36 (7 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	interval	off	direct	0.0	0	3	0	251.33	202.77	**1.239**	***1.101***	*0.05*
						0	5	0	270.72	190.65	**1.420**		
						0	6	0	999.68	870.78	**1.148**		
						0	7	0	246.58	293.43	0.840		
						0	8	0	276.94	323.04	0.857		
		interval	off	no	0.0	0	3	0	251.33	227.13	**1.107**	***1.017***	*0.02*
						0	5	0	270.72	175.66	**1.541**		
						0	6	0	999.68	1432.94	0.698		
						0	7	0	246.58	278.73	0.885		
						0	8	0	276.94	323.31	0.857		
		zono	off	direct	0.0	0	3	0	251.33	212.44	**1.183**	***1.013***	*0.05*
						0	5	0	270.72	224.05	**1.208**		
						0	6	0	999.68	1107.76	0.902		
						0	7	0	246.58	314.49	0.784		
						0	8	0	276.94	281.02	0.985		
		zono	off	prev_pgd	0.5	0	3	172	251.33	479.48	0.524	*0.927*	*0.45*
						0	5	172	270.72	134.37	**2.015**		
						0	6	172	999.68	1811.81	0.552		
						0	7	172	246.58	327.27	0.753		
						0	8	172	276.94	350.71	0.790		
		zono	off	no	0.5	0	3	172	251.33	1812.66	0.139	*0.861*	*0.40*
						0	5	172	270.72	299.19	0.905		
						0	6	172	999.68	753.45	**1.327**		
						0	7	172	246.58	268.30	0.919		
						0	8	172	276.94	273.12	**1.014**		
		interval	off	direct	0.5	0	3	171	251.33	246.48	**1.020**	*0.849*	*0.05*
						0	5	171	270.72	312.52	0.866		
						0	6	171	999.68	1191.02	0.839		
						0	7	171	246.58	327.23	0.754		
						0	8	171	276.94	361.74	0.766		
		zono	off	direct	0.5	0	3	172	251.33	1811.80	0.139	*0.804*	*0.31*
						0	5	172	270.72	232.42	**1.165**		
						0	6	172	999.68	891.21	**1.122**		
						0	7	172	246.58	323.61	0.762		
						0	8	172	276.94	332.84	0.832		

Table 54: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 8–14 of 20 by mean `sp`); continuation of Table 36 (7 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	zono	off	prev	0.5	0	3	172	251.33	538.88	0.466	*0.737*	*0.02*
						0	5	172	270.72	435.27	0.622		
						0	6	172	999.68	955.54	**1.046**		
						0	7	172	246.58	275.86	0.894		
						0	8	172	276.94	421.57	0.657		
		zono	off	prev	0.0	0	3	0	251.33	233.85	**1.075**	*0.712*	*0.02*
						0	5	0	270.72	443.81	0.610		
						0	6	0	999.68	1813.16	0.551		
						0	7	0	246.58	366.90	0.672		
						0	8	0	276.94	425.91	0.650		
		interval	off	prev	0.0	0	3	0	251.33	261.17	0.962	*0.703*	*0.02*
						0	5	0	270.72	422.10	0.641		
						0	6	0	999.68	1703.34	0.587		
						0	7	0	246.58	365.15	0.675		
						0	8	0	276.94	425.98	0.650		
		interval	off	prev_pgd	0.0	0	3	0	251.33	335.46	0.749	*0.703*	*0.02*
						0	5	0	270.72	415.67	0.651		
						0	6	0	999.68	1105.52	0.904		
						0	7	0	246.58	475.05	0.519		
						0	8	0	276.94	401.12	0.690		
		zono	off	prev_pgd	0.0	0	3	0	251.33	418.20	0.601	*0.644*	*0.02*
						0	5	0	270.72	506.48	0.535		
						0	6	0	999.68	1100.21	0.909		
						0	7	0	246.58	479.44	0.514		
						0	8	0	276.94	418.26	0.662		
		interval	off	prev	0.5	0	3	171	251.33	471.92	0.533	*0.614*	*0.02*
						0	5	171	270.72	551.67	0.491		
						0	6	171	999.68	1265.48	0.790		
						0	7	171	246.58	350.14	0.704		
						0	8	171	276.94	501.43	0.552		

Table 55: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 15–20 of 20 by mean `sp`); continuation of Table 36 (6 combos, 30 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	interval	off	prev	0.5	1	4	171	384.34	281.40	**1.366**	***1.765***	*0.04*
						1	5	171	1814.60	1112.21	**1.632**		
						1	7	171	309.44	134.76	**2.296**		
		interval	off	direct	0.5	1	4	171	384.34	355.37	**1.082**	***1.652***	*0.06*
						1	5	171	1814.60	1111.17	**1.633**		
						1	6	171	1816.47	1821.83	0.997		
						1	7	171	309.44	87.22	**3.548**		
						1	8	171	1819.70	1817.26	**1.001**		
		zono	off	prev_pgd	0.5	1	4	172	384.34	254.21	**1.512**	***1.526***	*0.04*
						1	5	172	1814.60	1240.44	**1.463**		
						1	6	172	1816.47	1814.05	**1.001**		
						1	7	172	309.44	116.54	**2.655**		
						1	8	172	1819.70	1819.14	**1.000**		
		zono	off	direct	0.5	1	4	172	384.34	273.33	**1.406**	***1.504***	*0.05*
						1	5	172	1814.60	715.15	**2.537**		
						1	6	172	1816.47	1812.39	**1.002**		
						1	7	172	309.44	248.97	**1.243**		
						1	8	172	1819.70	1364.42	**1.334**		
		interval	off	prev	0.0	1	4	0	384.34	338.22	**1.136**	***1.481***	*0.08*
						1	6	0	1816.47	1817.79	0.999		
						1	7	0	309.44	134.06	**2.308**		
		interval	off	direct	0.0	1	4	0	384.34	343.49	**1.119**	***1.427***	*0.08*
						1	6	0	1816.47	1811.71	**1.003**		
						1	7	0	309.44	119.80	**2.583**		
						1	8	0	1819.70	1814.06	**1.003**		

Table 56: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–6 of 18 by mean `sp`); continuation of Table 36 (6 combos, 25 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,3`	zono	off	prev_pgd	0.0	1	4	0	384.34	260.32	**1.476**	***1.348***	*0.07*
						1	5	0	1814.60	1735.76	**1.045**		
						1	6	0	1816.47	1814.95	**1.001**		
						1	7	0	309.44	139.63	**2.216**		
						1	8	0	1819.70	1814.17	**1.003**		
		interval	off	prev_pgd	0.0	1	4	0	384.34	399.66	0.962	***1.293***	*0.03*
						1	5	0	1814.60	1677.99	**1.081**		
						1	6	0	1816.47	1814.30	**1.001**		
						1	7	0	309.44	127.95	**2.418**		
						1	8	0	1819.70	1814.48	**1.003**		
		zono	off	no	0.5	1	4	172	384.34	348.97	**1.101**	***1.246***	*0.06*
						1	5	172	1814.60	1305.05	**1.390**		
						1	6	172	1816.47	1817.46	0.999		
						1	7	172	309.44	207.33	**1.492**		
		zono	off	no	0.0	1	4	0	384.34	556.92	0.690	***1.232***	*0.04*
						1	5	0	1814.60	1438.19	**1.262**		
						1	6	0	1816.47	1814.84	**1.001**		
						1	7	0	309.44	140.61	**2.201**		
						1	8	0	1819.70	1811.90	**1.004**		
		interval	off	no	0.0	1	5	0	1814.60	1839.72	0.986	***1.215***	*0.06*
						1	6	0	1816.47	1812.28	**1.002**		
						1	7	0	309.44	165.57	**1.869**		
						1	8	0	1819.70	1811.77	**1.004**		
		interval	off	direct_pgd	0.5	1	4	171	384.34	314.59	**1.222**	***1.206***	*0.05*
						1	5	171	1814.60	778.84	**2.330**		
						1	6	171	1816.47	1815.03	**1.001**		
						1	7	171	309.44	647.34	0.478		
						1	8	171	1819.70	1820.99	0.999		

Table 57: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 7–12 of 18 by mean `sp`); continuation of Table 36 (6 combos, 28 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`1,3`	interval	off	no	0.5	1	4	171	384.34	420.42	0.914	***1.114***	*0.06*
						1	5	171	1814.60	1268.31	**1.431**		
						1	6	171	1816.47	1819.67	0.998		
		zono	off	direct	0.0	1	4	0	384.34	329.17	**1.168**	***1.083***	*0.04*
						1	8	0	1819.70	1822.94	0.998		
		interval	off	direct_pgd	0.0	1	4	0	384.34	324.55	**1.184**	***1.071***	*0.04*
						1	6	0	1816.47	1816.78	1.000		
						1	7	0	309.44	282.40	**1.096**		
						1	8	0	1819.70	1814.39	**1.003**		
		interval	off	prev_pgd	0.5	1	4	171	384.34	335.20	**1.147**	***1.033***	*0.06*
						1	5	171	1814.60	1367.16	**1.327**		
						1	6	171	1816.47	1816.02	**1.000**		
						1	7	171	309.44	449.13	0.689		
						1	8	171	1819.70	1814.27	**1.003**		
		zono	off	direct_pgd	0.5	1	4	172	384.34	455.04	0.845	***1.007***	*0.04*
						1	6	172	1816.47	1814.82	**1.001**		
						1	7	172	309.44	262.45	**1.179**		
						1	8	172	1819.70	1814.06	**1.003**		
		zono	off	direct_pgd	0.0	1	4	0	384.34	376.73	**1.020**	*0.977*	*0.09*
						1	6	0	1816.47	1816.24	**1.000**		
						1	7	0	309.44	350.17	0.884		
						1	8	0	1819.70	1815.69	**1.002**		

Table 58: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 13–18 of 18 by mean `sp`); continuation of Table 36 (6 combos, 22 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`3,1`	zono	off	direct_pgd	0.5	0	3	190	458.41	225.16	**2.036**	***1.738***	*0.02*
						0	4	190	270.41	211.84	**1.276**		
						0	5	190	285.88	115.96	**2.465**		
						0	6	190	258.24	239.06	**1.080**		
						0	7	190	323.42	317.45	**1.019**		
						0	8	190	296.48	116.28	**2.550**		
		zono	off	direct_pgd	0.0	0	3	0	458.41	229.76	**1.995**	***1.689***	*0.06*
						0	4	0	270.41	278.69	0.970		
						0	5	0	285.88	118.80	**2.406**		
						0	6	0	258.24	118.90	**2.172**		
						0	8	0	296.48	328.32	0.903		
		interval	off	direct_pgd	0.0	0	3	0	458.41	226.43	**2.025**	***1.555***	*0.06*
						0	4	0	270.41	262.12	**1.032**		
						0	5	0	285.88	121.21	**2.359**		
						0	6	0	258.24	118.51	**2.179**		
						0	7	0	323.42	297.66	**1.087**		
						0	8	0	296.48	454.33	0.653		
		interval	off	direct	0.5	0	3	181	458.41	329.18	**1.393**	***1.371***	*0.02*
						0	4	181	270.41	309.18	0.875		
						0	5	181	285.88	174.14	**1.642**		
						0	6	181	258.24	105.96	**2.437**		
						0	7	181	323.42	359.77	0.899		
						0	8	181	296.48	302.96	0.979		
		interval	off	direct_pgd	0.5	0	3	181	458.41	249.76	**1.835**	***1.197***	*0.02*
						0	4	181	270.41	298.42	0.906		
						0	5	181	285.88	266.16	**1.074**		
						0	6	181	258.24	301.82	0.856		
						0	7	181	323.42	298.44	**1.084**		
						0	8	181	296.48	208.11	**1.425**		
		zono	off	direct	0.5	0	3	190	458.41	226.78	**2.021**	***1.128***	*0.01*
						0	4	190	270.41	382.27	0.707		
						0	5	190	285.88	250.77	**1.140**		
						0	6	190	258.24	204.66	**1.262**		
						0	7	190	323.42	344.60	0.939		
						0	8	190	296.48	423.52	0.700		
		zono	off	prev_pgd	0.5	0	3	190	458.41	275.19	**1.666**	***1.085***	*0.02*
						0	4	190	270.41	241.81	**1.118**		
						0	5	190	285.88	348.23	0.821		
						0	6	190	258.24	308.46	0.837		
						0	7	190	323.42	529.48	0.611		
						0	8	190	296.48	203.91	**1.454**		

Table 59: Architecture **cnn1** — perturbation **translation**(`3,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–7 of 20 by mean `sp`); continuation of Table 36 (7 combos, 41 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_\text{src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_\text{src}}$
translation	`3,1`	interval	off	prev	0.5	0	3	181	458.41	348.84	**1.314**	***1.011***	*0.02*
						0	4	181	270.41	460.91	0.587		
						0	5	181	285.88	152.19	**1.878**		
						0	6	181	258.24	404.90	0.638		
						0	7	181	323.42	301.75	**1.072**		
						0	8	181	296.48	514.48	0.576		
		interval	off	no	0.0	0	3	0	458.41	279.15	**1.642**	***1.010***	*0.01*
						0	4	0	270.41	488.16	0.554		
						0	5	0	285.88	241.79	**1.182**		
						0	6	0	258.24	304.00	0.849		
						0	7	0	323.42	340.76	0.949		
						0	8	0	296.48	336.22	0.882		
		interval	off	prev_pgd	0.0	0	3	0	458.41	225.74	**2.031**	***1.006***	*0.02*
						0	4	0	270.41	475.66	0.568		
						0	5	0	285.88	367.51	0.778		
						0	6	0	258.24	296.39	0.871		
						0	7	0	323.42	275.71	**1.173**		
						0	8	0	296.48	483.81	0.613		
		interval	off	prev	0.0	0	3	0	458.41	391.18	**1.172**	*0.991*	*0.02*
						0	4	0	270.41	302.55	0.894		
						0	5	0	285.88	228.67	**1.250**		
						0	6	0	258.24	210.92	**1.224**		
						0	7	0	323.42	478.02	0.677		
						0	8	0	296.48	404.98	0.732		
		zono	off	prev	0.0	0	3	0	458.41	389.31	**1.177**	*0.977*	*0.02*
						0	4	0	270.41	296.33	0.913		
						0	5	0	285.88	212.75	**1.344**		
						0	6	0	258.24	243.42	**1.061**		
						0	7	0	323.42	646.14	0.501		
						0	8	0	296.48	342.40	0.866		
		zono	off	prev_pgd	0.0	0	3	0	458.41	249.86	**1.835**	*0.948*	*0.02*
						0	4	0	270.41	465.54	0.581		
						0	5	0	285.88	378.39	0.756		
						0	6	0	258.24	328.08	0.787		
						0	7	0	323.42	294.08	**1.100**		
						0	8	0	296.48	471.74	0.628		
		interval	off	no	0.5	0	3	181	458.41	440.50	**1.041**	*0.868*	*0.07*
						0	4	181	270.41	343.01	0.788		
						0	5	181	285.88	259.82	**1.100**		
						0	6	181	258.24	461.41	0.560		
						0	7	181	323.42	482.64	0.670		
						0	8	181	296.48	282.36	**1.050**		

Table 60: Architecture **cnn1** — perturbation **translation**(`3,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 8–14 of 20 by mean `sp`); continuation of Table 36 (7 combos, 42 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`3,1`	zono	off	direct	0.0	0	3	0	458.41	298.41	**1.536**	*0.841*	*0.42*
						0	4	0	270.41	1811.60	0.149		
						0	5	0	285.88	306.77	0.932		
						0	6	0	258.24	305.43	0.845		
						0	7	0	323.42	352.90	0.916		
						0	8	0	296.48	445.98	0.665		
		zono	off	prev	0.5	0	3	190	458.41	486.35	0.943	*0.778*	*0.01*
						0	4	190	270.41	439.54	0.615		
						0	5	190	285.88	426.34	0.671		
						0	6	190	258.24	308.42	0.837		
						0	7	190	323.42	350.64	0.922		
						0	8	190	296.48	437.08	0.678		
		zono	off	no	0.0	0	3	0	458.41	404.77	**1.133**	*0.774*	*0.01*
						0	4	0	270.41	376.75	0.718		
						0	6	0	258.24	502.61	0.514		
						0	7	0	323.42	479.76	0.674		
						0	8	0	296.48	355.67	0.834		
		interval	off	direct	0.0	0	3	0	458.41	333.69	**1.374**	*0.770*	*0.37*
						0	4	0	270.41	1814.53	0.149		
						0	5	0	285.88	318.91	0.896		
						0	6	0	258.24	273.54	0.944		
						0	7	0	323.42	517.08	0.625		
						0	8	0	296.48	471.66	0.629		
		interval	off	prev_pgd	0.5	0	3	181	458.41	1817.39	0.252	*0.693*	*0.31*
						0	4	181	270.41	472.20	0.573		
						0	5	181	285.88	227.51	**1.257**		
						0	6	181	258.24	532.65	0.485		
						0	7	181	323.42	403.74	0.801		
						0	8	181	296.48	373.95	0.793		
		zono	off	no	0.5	0	3	190	458.41	241.72	**1.896**	*0.655*	*0.72*
						0	4	190	270.41	1813.27	0.149		
						0	5	190	285.88	625.34	0.457		
						0	6	190	258.24	454.30	0.568		
						0	7	190	323.42	463.93	0.697		
						0	8	190	296.48	1813.48	0.163		

Table 61: Architecture **cnn1** — perturbation **translation**(`3,1`), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 15–20 of 20 by mean `sp`); continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	3,1	zono	off	prev_pgd	0.0	1	3	0	285.99	242.06	**1.181**	***1.714***	*0.12*
						1	4	0	111.60	147.10	0.759		
						1	5	0	501.62	121.32	**4.135**		
						1	6	0	249.93	169.21	**1.477**		
						1	7	0	245.31	240.75	**1.019**		
						1	8	0	264.15	153.94	**1.716**		
		interval	off	no	0.5	1	3	181	285.99	124.76	**2.292**	***1.631***	*0.13*
						1	4	181	111.60	148.45	0.752		
						1	5	181	501.62	130.20	**3.853**		
						1	6	181	249.93	241.73	**1.034**		
						1	7	181	245.31	345.19	0.711		
						1	8	181	264.15	230.79	**1.145**		
		interval	off	prev_pgd	0.5	1	3	181	285.99	168.95	**1.693**	***1.484***	*0.14*
						1	4	181	111.60	108.92	**1.025**		
						1	5	181	501.62	202.35	**2.479**		
						1	6	181	249.93	140.68	**1.777**		
						1	7	181	245.31	411.83	0.596		
						1	8	181	264.15	197.64	**1.337**		
		interval	off	direct_pgd	0.5	1	3	181	285.99	327.27	0.874	***1.445***	*0.10*
						1	4	181	111.60	141.92	0.786		
						1	5	181	501.62	157.57	**3.183**		
						1	6	181	249.93	127.73	**1.957**		
						1	7	181	245.31	187.92	**1.305**		
						1	8	181	264.15	469.13	0.563		
		zono	off	direct_pgd	0.5	1	3	190	285.99	361.59	0.791	***1.431***	*0.14*
						1	4	190	111.60	336.76	0.331		
						1	5	190	501.62	184.61	**2.717**		
						1	6	190	249.93	88.95	**2.810**		
						1	7	190	245.31	178.60	**1.374**		
						1	8	190	264.15	471.44	0.560		
		interval	off	prev_pgd	0.0	1	3	0	285.99	648.09	0.441	***1.389***	*0.12*
						1	4	0	111.60	165.79	0.673		
						1	5	0	501.62	133.79	**3.749**		
						1	6	0	249.93	360.44	0.693		
						1	7	0	245.31	183.05	**1.340**		
						1	8	0	264.15	183.51	**1.439**		

Table 62: Architecture **cnn1** — perturbation **translation**(3,1), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean sp); continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`3,1`	zono	off	no	0.0	1	3	0	285.99	225.87	**1.266**	***1.389***	*0.10*
						1	4	0	111.60	266.23	0.419		
						1	5	0	501.62	191.86	**2.615**		
						1	6	0	249.93	105.87	**2.361**		
						1	7	0	245.31	497.23	0.493		
						1	8	0	264.15	223.71	**1.181**		
		interval	off	no	0.0	1	3	0	285.99	194.40	**1.471**	***1.256***	*0.09*
						1	4	0	111.60	218.18	0.512		
						1	5	0	501.62	334.48	**1.500**		
						1	6	0	249.93	327.83	0.762		
						1	7	0	245.31	102.06	**2.404**		
						1	8	0	264.15	297.86	0.887		
		zono	off	prev_pgd	0.5	1	3	190	285.99	404.25	0.707	***1.121***	*0.13*
						1	4	190	111.60	202.48	0.551		
						1	5	190	501.62	333.21	**1.505**		
						1	6	190	249.93	193.54	**1.291**		
						1	7	190	245.31	199.66	**1.229**		
						1	8	190	264.15	182.86	**1.445**		
		interval	off	direct_pgd	0.0	1	3	0	285.99	359.95	0.795	*0.861*	*0.14*
						1	4	0	111.60	586.27	0.190		
						1	5	0	501.62	197.57	**2.539**		
						1	6	0	249.93	451.87	0.553		
						1	7	0	245.31	416.52	0.589		
						1	8	0	264.15	527.36	0.501		
		zono	off	no	0.5	1	3	190	285.99	406.70	0.703	*0.833*	*0.15*
						1	4	190	111.60	298.54	0.374		
						1	5	190	501.62	310.09	**1.618**		
						1	6	190	249.93	258.10	0.968		
						1	7	190	245.31	514.64	0.477		
						1	8	190	264.15	308.21	0.857		
		zono	off	direct_pgd	0.0	1	3	0	285.99	546.00	0.524	*0.559*	*0.12*
						1	4	0	111.60	417.92	0.267		
						1	6	0	249.93	443.52	0.564		
						1	7	0	245.31	292.67	0.838		
						1	8	0	264.15	439.55	0.601		

Table 63: Architecture **cnn1** — perturbation **translation**(`3,1`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean `sp`); continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}}-\delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	3,3	zono	off	prev_pgd	0.5	0	3	190	800.69	350.15	**2.287**	*2.843*	*0.01*
						0	5	190	682.97	141.61	**4.823**		
						0	6	190	765.68	338.72	**2.261**		
						0	7	190	1231.36	391.75	**3.143**		
						0	8	190	375.90	220.73	**1.703**		
		interval	off	direct_pgd	0.5	0	3	180	800.69	228.83	**3.499**	*2.657*	*0.01*
						0	5	181	682.97	235.99	**2.894**		
						0	6	180	765.68	305.42	**2.507**		
						0	7	180	1231.36	415.64	**2.963**		
						0	8	180	375.90	264.02	**1.424**		
		interval	off	prev	0.5	0	3	180	800.69	318.38	**2.515**	*2.535*	*0.00*
						0	5	180	682.97	297.99	**2.292**		
						0	6	180	765.68	547.73	**1.398**		
						0	7	181	1231.36	341.67	**3.604**		
						0	8	180	375.90	131.14	**2.866**		
		zono	off	direct_pgd	0.5	0	3	190	800.69	301.29	**2.658**	*2.317*	*0.03*
						0	4	190	1821.48	1406.06	**1.295**		
						0	5	190	682.97	248.62	**2.747**		
						0	6	190	765.68	297.86	**2.571**		
						0	7	190	1231.36	404.42	**3.045**		
						0	8	190	375.90	236.66	**1.588**		
		interval	off	direct_pgd	0.0	0	3	0	800.69	289.09	**2.770**	*2.280*	*0.04*
						0	5	0	682.97	260.32	**2.624**		
						0	6	0	765.68	514.44	**1.488**		
						0	7	0	1231.36	866.65	**1.421**		
						0	8	0	375.90	121.39	**3.097**		
		zono	off	direct	0.5	0	3	190	800.69	421.60	**1.899**	*2.213*	*0.03*
						0	4	190	1821.48	1073.49	**1.697**		
						0	5	190	682.97	372.72	**1.832**		
						0	6	190	765.68	345.22	**2.218**		
						0	7	190	1231.36	475.25	**2.591**		
						0	8	190	375.90	123.54	**3.043**		
		zono	off	direct_pgd	0.0	0	3	0	800.69	278.16	**2.879**	*2.189*	*0.04*
						0	5	0	682.97	260.72	**2.620**		
						0	6	0	765.68	545.35	**1.404**		
						0	7	0	1231.36	946.91	**1.300**		
						0	8	0	375.90	137.15	**2.741**		

Table 64: Architecture **cnn1** — perturbation **translation**(3,3), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/3, combos ranked 1–7 of 20 by mean sp); continuation of Table 36 (7 combos, 37 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	3,3	zono	off	prev_pgd	0.0	0	3	0	800.69	337.58	**2.372**	***2.130***	*0.00*
						0	5	0	682.97	345.88	**1.975**		
						0	6	0	765.68	363.34	**2.107**		
						0	7	0	1231.36	510.71	**2.411**		
						0	8	0	375.90	210.38	**1.787**		
		interval	off	prev_pgd	0.5	0	3	181	800.69	887.89	0.902	***2.098***	*0.02*
						0	4	180	1821.48	1265.58	**1.439**		
						0	5	180	682.97	148.72	**4.592**		
						0	6	180	765.68	388.40	**1.971**		
						0	7	181	1231.36	446.44	**2.758**		
						0	8	180	375.90	407.43	0.923		
		interval	off	prev_pgd	0.0	0	3	0	800.69	361.59	**2.214**	***2.050***	*0.00*
						0	5	0	682.97	413.35	**1.652**		
						0	6	0	765.68	435.35	**1.759**		
						0	7	0	1231.36	418.63	**2.941**		
						0	8	0	375.90	223.40	**1.683**		
		zono	off	no	0.5	0	3	190	800.69	402.39	**1.990**	***2.010***	*0.03*
						0	4	190	1821.48	1341.89	**1.357**		
						0	5	190	682.97	330.07	**2.069**		
						0	6	190	765.68	341.53	**2.242**		
						0	7	190	1231.36	485.35	**2.537**		
						0	8	190	375.90	201.49	**1.866**		
		zono	off	no	0.0	0	3	0	800.69	350.15	**2.287**	***1.951***	*0.01*
						0	5	0	682.97	264.20	**2.585**		
						0	6	0	765.68	473.32	**1.618**		
						0	7	0	1231.36	625.19	**1.970**		
						0	8	0	375.90	290.05	**1.296**		
		interval	off	direct	0.0	0	3	0	800.69	452.79	**1.768**	***1.937***	*0.03*
						0	5	0	682.97	172.08	**3.969**		
						0	6	0	765.68	580.59	**1.319**		
						0	7	0	1231.36	680.04	**1.811**		
						0	8	0	375.90	458.53	0.820		
		interval	off	no	0.5	0	3	180	800.69	455.08	**1.759**	***1.886***	*0.01*
						0	5	180	682.97	437.85	**1.560**		
						0	6	180	765.68	329.28	**2.325**		
						0	7	180	1231.36	450.77	**2.732**		
						0	8	181	375.90	355.96	**1.056**		

Table 65: Architecture **cnn1** — perturbation **translation**(3,3), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/3, combos ranked 8–14 of 20 by mean sp); continuation of Table 36 (7 combos, 37 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	3,3	interval	off	direct	0.5	0	3	180	800.69	451.31	**1.774**	***1.839***	*0.01*
						0	5	180	682.97	404.03	**1.690**		
						0	6	180	765.68	420.61	**1.820**		
						0	7	180	1231.36	449.51	**2.739**		
						0	8	180	375.90	320.47	**1.173**		
		interval	off	no	0.0	0	3	0	800.69	340.33	**2.353**	***1.777***	*0.01*
						0	5	0	682.97	284.48	**2.401**		
						0	6	0	765.68	543.74	**1.408**		
						0	7	0	1231.36	621.39	**1.982**		
						0	8	0	375.90	505.89	0.743		
		zono	off	prev	0.0	0	3	0	800.69	306.25	**2.614**	***1.740***	*0.00*
						0	5	0	682.97	472.59	**1.445**		
						0	6	0	765.68	732.63	**1.045**		
						0	7	0	1231.36	872.97	**1.411**		
						0	8	0	375.90	172.22	**2.183**		
		interval	off	prev	0.0	0	3	0	800.69	310.78	**2.576**	***1.735***	*0.00*
						0	5	0	682.97	470.68	**1.451**		
						0	6	0	765.68	740.33	**1.034**		
						0	7	0	1231.36	875.66	**1.406**		
						0	8	0	375.90	170.18	**2.209**		
		zono	off	direct	0.0	0	3	0	800.69	478.93	**1.672**	***1.698***	*0.03*
						0	5	0	682.97	272.04	**2.511**		
						0	6	0	765.68	506.42	**1.512**		
						0	7	0	1231.36	645.35	**1.908**		
						0	8	0	375.90	424.32	0.886		
		zono	off	prev	0.5	0	3	190	800.69	546.20	**1.466**	***1.666***	*0.01*
						0	5	190	682.97	472.47	**1.446**		
						0	6	190	765.68	475.41	**1.611**		
						0	7	190	1231.36	419.32	**2.937**		
						0	8	190	375.90	432.50	0.869		

Table 66: Architecture **cnn1** — perturbation **translation**(3,3), $c_{\text{src}} = 0$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 3/3, combos ranked 15–20 of 20 by mean sp); continuation of Table 36 (6 combos, 30 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`3,3`	interval	off	no	0.0	1	3	0	1812.65	1815.72	0.998	***1.068***	*0.10*
						1	4	0	1086.37	1820.20	0.597		
						1	7	0	176.49	105.38	**1.675**		
						1	8	0	1820.58	1814.55	**1.003**		
		zono	off	prev_pgd	0.5	1	3	190	1812.65	1815.95	0.998	***1.065***	*0.07*
						1	4	190	1086.37	1326.67	0.819		
						1	6	190	1815.56	1823.05	0.996		
						1	7	190	176.49	116.99	**1.509**		
						1	8	190	1820.58	1816.18	**1.002**		
		interval	off	no	0.5	1	3	180	1812.65	1815.41	0.998	*0.976*	*0.11*
						1	4	180	1086.37	1816.44	0.598		
						1	6	181	1815.56	1814.60	**1.001**		
						1	7	181	176.49	137.95	**1.279**		
						1	8	181	1820.58	1817.76	**1.002**		
		interval	off	prev_pgd	0.0	1	3	0	1812.65	1815.38	0.998	*0.960*	*0.07*
						1	4	0	1086.37	1814.67	0.599		
						1	6	0	1815.56	1814.10	**1.001**		
						1	7	0	176.49	147.30	**1.198**		
						1	8	0	1820.58	1813.81	**1.004**		
		zono	off	no	0.0	1	4	0	1086.37	1813.42	0.599	*0.959*	*0.09*
						1	6	0	1815.56	1813.14	**1.001**		
						1	7	0	176.49	142.70	**1.237**		
						1	8	0	1820.58	1819.30	**1.001**		
		zono	off	prev_pgd	0.0	1	4	0	1086.37	1813.65	0.599	*0.948*	*0.11*
						1	6	0	1815.56	1816.59	0.999		
						1	7	0	176.49	147.80	**1.194**		
						1	8	0	1820.58	1823.02	0.999		

Table 67: Architecture **cnn1** — perturbation **translation**(`3,3`), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 1/2, combos ranked 1–6 of 12 by mean `sp`); continuation of Table 36 (6 combos, 27 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	3,3	zono	off	no	0.5	1	3	190	1812.65	1817.10	0.998	*0.934*	*0.08*
						1	4	190	1086.37	1817.93	0.598		
						1	6	190	1815.56	1815.52	**1.000**		
						1	7	190	176.49	164.25	**1.075**		
						1	8	190	1820.58	1816.41	**1.002**		
		interval	off	direct_pgd	0.0	1	3	0	1812.65	1813.61	0.999	*0.871*	*0.10*
						1	4	0	1086.37	1813.57	0.599		
						1	6	0	1815.56	1813.43	**1.001**		
						1	7	0	176.49	233.95	0.754		
						1	8	0	1820.58	1814.40	**1.003**		
		interval	off	direct_pgd	0.5	1	3	181	1812.65	1822.16	0.995	*0.862*	*0.08*
						1	4	181	1086.37	1823.43	0.596		
						1	6	181	1815.56	1819.69	0.998		
						1	7	181	176.49	243.61	0.724		
						1	8	181	1820.58	1822.56	0.999		
		zono	off	direct_pgd	0.5	1	4	190	1086.37	1816.03	0.598	*0.823*	*1.77*
						1	6	190	1815.56	1815.10	**1.000**		
						1	7	190	176.49	254.89	0.692		
						1	8	190	1820.58	1820.42	**1.000**		
		interval	off	prev_pgd	0.5	1	3	181	1812.65	1823.00	0.994	*0.816*	*0.06*
						1	4	181	1086.37	1823.35	0.596		
						1	6	181	1815.56	1825.91	0.994		
						1	7	181	176.49	357.46	0.494		
						1	8	181	1820.58	1816.23	**1.002**		
		zono	off	direct_pgd	0.0	1	3	0	1812.65	1816.32	0.998	*0.737*	*0.06*
						1	4	0	1086.37	1817.13	0.598		
						1	7	0	176.49	503.95	0.350		
						1	8	0	1820.58	1819.58	**1.001**		

Table 68: Architecture **cnn1** — perturbation **translation**(3,3), $c_{\text{src}} = 1$, $\tau \in \{0.0, 0.5, 1.0\}$ (part 2/2, combos ranked 7–12 of 12 by mean sp); continuation of Table 36 (6 combos, 28 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	n_timeout	n_advstd_tighter	avg_gap_std	avg_gap_adv
translation	`1,1`	zono	off	direct	0.5	1	1	1	0.8551	**0.8047**
		interval	off	prev	0.5	1	1	1	0.8551	**0.7487**
		zono	off	prev_pgd	0.5	1	1	0	0.8551	0.8928
		zono	off	prev_pgd	0.0	1	1	0	0.8551	1.0141
		zono	off	direct_pgd	0.5	1	1	0	0.8551	1.0302
		interval	off	prev_pgd	0.0	1	1	0	0.8551	0.9293
		interval	off	direct	0.5	1	1	0	0.8551	1.0162
		interval	off	direct_pgd	0.0	1	1	0	0.8551	0.9989
		interval	off	no	0.5	1	1	0	0.8551	0.9452
		interval	off	prev	0.0	1	1	0	0.8551	0.9355
		zono	off	direct_pgd	0.0	1	1	0	0.8551	0.9808
		interval	off	direct_pgd	0.5	1	1	0	0.8551	0.9520
		zono	off	no	0.1	1	1	0	0.8551	1.0911
		interval	off	no	0.1	1	1	0	0.8551	0.8652
		zono	off	no	0.05	1	1	0	0.8551	0.9476
		interval	off	prev_pgd	0.5	1	1	0	0.8551	0.8965
		interval	off	no	0.05	1	1	0	0.8551	0.9576
		interval	off	no	0.0	1	1	0	0.8551	0.8820
		interval	off	direct	0.0	1	1	0	0.8551	1.0925
		zono	off	direct	0.0	1	1	0	0.8551	0.9919
		zono	off	no	0.0	1	1	0	0.8551	0.9702
		interval	off	no	0.01	1	1	0	0.8551	15.1060
		zono	off	prev	0.0	1	1	0	0.8551	1.0129
		zono	off	no	0.01	1	1	0	0.8551	1.0039
translation	`1,3`	zono	off	direct_pgd	0.5	1	2	2	1.2931	**1.1876**
		zono	off	prev_pgd	0.5	1	2	2	1.2931	**1.2121**
		interval	off	prev_pgd	0.5	1	2	2	1.2931	**1.0703**
		interval	off	no	0.05	1	2	2	1.2931	**1.1454**
		zono	off	prev_pgd	0.0	1	2	2	1.2931	**1.0828**
		interval	off	no	0.1	1	2	2	1.2931	**1.2509**
		interval	off	direct_pgd	0.5	1	2	1	1.2931	**1.2831**
		zono	off	no	0.0	1	2	1	1.2931	**1.2896**
		zono	off	direct	0.0	1	1	1	0.9654	**0.8280**
		interval	off	no	0.0	1	2	1	1.2931	**1.2788**
		interval	off	no	0.5	1	1	1	1.6208	**1.2862**
		zono	off	direct	0.5	1	1	1	1.6208	**1.5457**
		zono	off	no	0.5	1	1	1	1.6208	**1.5080**
		interval	off	prev	0.0	1	1	1	1.6208	**1.5602**
		interval	off	prev_pgd	0.0	1	2	1	1.2931	1.2943
		interval	off	direct_pgd	0.0	1	2	0	1.2931	1.3488
		zono	off	direct_pgd	0.0	1	2	0	1.2931	1.3858
		interval	off	direct	0.0	1	2	0	1.2931	1.5078
		interval	off	direct	0.5	1	2	0	1.2931	1.3978
translation	`3,3`	zono	off	prev_pgd	0.5	1	3	2	2.2997	2.3025
		interval	off	prev_pgd	0.5	1	3	2	2.2997	**2.2728**
		interval	off	direct_pgd	0.5	1	3	1	2.2997	**2.2363**
		zono	off	direct_pgd	0.5	1	2	1	2.3049	9.0515
		zono	off	no	0.0	1	2	1	2.3049	2.3950
		interval	off	prev_pgd	0.0	1	3	1	2.2997	2.3569
		zono	off	direct_pgd	0.0	1	2	0	2.0928	2.2357
		interval	off	direct_pgd	0.0	1	3	0	2.2997	2.4615
		zono	off	prev_pgd	0.0	1	2	0	2.3049	2.5059
		interval	off	no	0.0	1	2	0	2.0928	2.2541
		zono	off	no	0.5	1	3	0	2.2997	2.4861
		interval	off	no	0.5	1	3	0	2.2997	2.5268

Table 69: Architecture **cnn1** — gap comparison on cells where both modes hit `TIME_LIMIT`. One row per (combo, c_{src}), aggregated over all tested c_{tgt}. n_advstd_tighter: count where advstd's $u - l$ gap < standard's. avg_gap_*: mean $u - l$ per method. avg_gap_adv<avg_gap_std in **bold**. Includes every combo with at least one such cell (may include combos absent from Table 36). Orange rows: bp=off.

Mid-τ detail tables ($\tau \in \{0.01, 0.05, 0.1\}$)

The following tables break out the same per-architecture advstd combinations as the green tables above, but for the intermediate Technique-4 thresholds $\tau \in \{0.01, 0.05, 0.1\}$. They are grouped at the end of the paper because these mid-range thresholds are denser and more exploratory than the $\tau \in \{0.0, 0.5, 1.0\}$ headline tables. Shaded rows (cyan) satisfy bp=off; column meanings match the green tables above.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
patch	`1,14,14,3`	zono	off	no	0.1	0	3	0	14.75	11.91	**1.238**	***1.098***	*0.01*
						0	4	0	15.24	14.79	**1.030**		
						0	5	0	16.13	13.00	**1.241**		
						0	6	0	13.07	13.30	0.983		
						0	7	0	13.37	14.13	0.946		
						0	8	0	13.05	11.33	**1.152**		
		interval	off	no	0.05	0	3	0	14.75	13.89	**1.062**	***1.065***	*0.01*
						0	4	0	15.24	12.72	**1.198**		
						0	5	0	16.13	15.10	**1.068**		
						0	6	0	13.07	14.00	0.934		
						0	7	0	13.37	12.54	**1.066**		
						0	8	0	13.05	12.29	**1.062**		
		interval	off	no	0.01	0	3	0	14.75	15.73	0.938	***1.009***	*0.01*
						0	4	0	15.24	13.95	**1.092**		
						0	5	0	16.13	14.83	**1.088**		
						0	6	0	13.07	13.73	0.952		
						0	7	0	13.37	13.14	**1.018**		
						0	8	0	13.05	13.46	0.970		
		interval	off	no	0.1	0	3	0	14.75	16.63	0.887	*0.993*	*0.01*
						0	4	0	15.24	14.12	**1.079**		
						0	5	0	16.13	14.53	**1.110**		
						0	6	0	13.07	15.24	0.858		
						0	7	0	13.37	11.76	**1.137**		
						0	8	0	13.05	14.71	0.887		
		zono	off	no	0.01	0	3	0	14.75	14.10	**1.046**	*0.989*	*0.01*
						0	4	0	15.24	13.23	**1.152**		
						0	5	0	16.13	15.02	**1.074**		
						0	6	0	13.07	17.18	0.761		
						0	7	0	13.37	14.50	0.922		
						0	8	0	13.05	13.35	0.978		
		zono	off	no	0.05	0	3	0	14.75	16.53	0.892	*0.950*	*0.01*
						0	4	0	15.24	18.00	0.847		
						0	5	0	16.13	17.19	0.938		
						0	6	0	13.07	14.41	0.907		
						0	7	0	13.37	11.57	**1.156**		
						0	8	0	13.05	13.59	0.960		

Table 70: Architecture **3x10** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
occ	5,5,5	zono	off	no	0.1	0	3	0	21.76	13.19	**1.650**	*1.275*	*0.01*
						0	4	0	15.18	12.67	**1.198**		
						0	5	0	15.59	12.40	**1.257**		
						0	6	0	14.17	12.26	**1.156**		
						0	7	0	17.19	11.62	**1.479**		
						0	8	0	14.13	15.52	0.910		
		interval	off	no	0.05	0	3	0	21.76	13.75	**1.583**	*1.249*	*0.01*
						0	4	0	15.18	12.21	**1.243**		
						0	5	0	15.59	13.11	**1.189**		
						0	7	0	17.19	15.62	**1.101**		
						0	8	0	14.13	12.49	**1.131**		
		interval	off	no	0.1	0	3	0	21.76	13.46	**1.617**	*1.220*	*0.01*
						0	4	0	15.18	13.14	**1.155**		
						0	5	0	15.59	12.67	**1.230**		
						0	6	0	14.17	13.17	**1.076**		
						0	7	0	17.19	14.62	**1.176**		
						0	8	0	14.13	13.27	**1.065**		
		zono	off	no	0.01	0	3	0	21.76	14.80	**1.470**	*1.177*	*0.01*
						0	4	0	15.18	13.46	**1.128**		
						0	5	0	15.59	14.10	**1.106**		
						0	6	0	14.17	13.26	**1.069**		
						0	7	0	17.19	13.49	**1.274**		
						0	8	0	14.13	13.88	**1.018**		
		interval	off	no	0.01	0	3	0	21.76	13.95	**1.560**	*1.174*	*0.01*
						0	4	0	15.18	13.13	**1.156**		
						0	5	0	15.59	14.16	**1.101**		
						0	6	0	14.17	12.56	**1.128**		
						0	7	0	17.19	14.79	**1.162**		
						0	8	0	14.13	15.13	0.934		
		zono	off	no	0.05	0	3	0	21.76	16.85	**1.291**	*1.144*	*0.01*
						0	4	0	15.18	13.66	**1.111**		
						0	5	0	15.59	13.36	**1.167**		
						0	6	0	14.17	13.12	**1.080**		
						0	7	0	17.19	12.80	**1.343**		
						0	8	0	14.13	16.19	0.873		

Table 71: Architecture **3x10** — perturbation **occ**(5,5,5), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{\mathrm{c\_src}}$	$\overline{\lvert\delta_{\mathrm{std}}-\delta_{\mathrm{adv}}\rvert}_{\mathrm{c\_src}}$
translation	`1,1`	interval	off	no	0.1	0	3	0	13.93	15.89	0.877	***1.163***	*0.00*
						0	4	0	15.18	14.26	**1.065**		
						0	5	0	15.40	13.85	**1.112**		
						0	6	0	18.67	13.88	**1.345**		
						0	7	0	20.37	14.12	**1.443**		
						0	8	0	16.12	14.20	**1.135**		
		interval	off	no	0.01	0	3	0	13.93	16.08	0.866	***1.123***	*0.00*
						0	4	0	15.18	14.41	**1.053**		
						0	5	0	15.40	14.98	**1.028**		
						0	6	0	18.67	13.89	**1.344**		
						0	7	0	20.37	15.19	**1.341**		
						0	8	0	16.12	14.61	**1.103**		
		zono	off	no	0.01	0	3	0	13.93	13.67	**1.019**	***1.111***	*0.00*
						0	4	0	15.18	14.92	**1.017**		
						0	5	0	15.40	15.19	**1.014**		
						0	6	0	18.67	16.30	**1.145**		
						0	7	0	20.37	15.63	**1.303**		
						0	8	0	16.12	13.80	**1.168**		
		interval	off	no	0.05	0	3	0	13.93	17.39	0.801	***1.017***	*0.00*
						0	4	0	15.18	24.94	0.609		
						0	5	0	15.40	13.41	**1.148**		
						0	6	0	18.67	14.90	**1.253**		
						0	7	0	20.37	14.18	**1.437**		
						0	8	0	16.12	18.90	0.853		
		zono	off	no	0.05	0	3	0	13.93	17.19	0.810	*0.974*	*0.00*
						0	4	0	15.18	19.20	0.791		
						0	6	0	18.67	15.89	**1.175**		
						0	7	0	20.37	16.09	**1.266**		
						0	8	0	16.12	19.52	0.826		
		zono	off	no	0.1	0	3	0	13.93	16.20	0.860	*0.954*	*0.00*
						0	4	0	15.18	15.51	0.979		
						0	5	0	15.40	18.18	0.847		
						0	6	0	18.67	17.39	**1.074**		
						0	7	0	20.37	21.69	0.939		
						0	8	0	16.12	15.69	**1.027**		

Table 72: Architecture **3x10** — perturbation **translation**(`1,1`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	interval	off	no	0.01	0	3	0	12.26	15.34	0.799	*0.741*	*0.02*
						0	4	0	14.39	33.68	0.427		
						0	5	0	14.10	15.03	0.938		
						0	6	0	11.86	15.98	0.742		
						0	7	0	12.10	30.79	0.393		
						0	8	0	16.83	14.65	**1.149**		
		interval	off	no	0.05	0	3	0	12.26	32.05	0.383	*0.612*	*0.02*
						0	4	0	14.39	16.53	0.871		
						0	5	0	14.10	17.10	0.825		
						0	6	0	11.86	24.58	0.483		
						0	7	0	12.10	32.01	0.378		
						0	8	0	16.83	22.93	0.734		
		interval	off	no	0.1	0	3	0	12.26	17.65	0.695	*0.592*	*0.02*
						0	4	0	14.39	19.91	0.723		
						0	5	0	14.10	30.36	0.464		
						0	6	0	11.86	33.40	0.355		
						0	7	0	12.10	27.46	0.441		
						0	8	0	16.83	19.27	0.873		
		zono	off	no	0.01	0	3	0	12.26	26.68	0.460	*0.467*	*0.02*
						0	4	0	14.39	34.78	0.414		
						0	5	0	14.10	18.56	0.760		
						0	6	0	11.86	35.44	0.335		
						0	7	0	12.10	30.59	0.396		
						0	8	0	16.83	38.40	0.438		
		zono	off	no	0.1	0	3	0	12.26	35.48	0.346	*0.464*	*0.02*
						0	4	0	14.39	19.25	0.748		
						0	5	0	14.10	35.22	0.400		
						0	6	0	11.86	36.56	0.324		
						0	7	0	12.10	26.13	0.463		
						0	8	0	16.83	33.54	0.502		
		zono	off	no	0.05	0	3	0	12.26	35.91	0.341	*0.424*	*0.02*
						0	4	0	14.39	26.08	0.552		
						0	5	0	14.10	35.51	0.397		
						0	6	0	11.86	36.74	0.323		
						0	7	0	12.10	26.32	0.460		
						0	8	0	16.83	35.79	0.470		

Table 73: Architecture **3x10** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}} - \delta_{\mathrm{adv}}\rvert}_{c\_src}$
translation	`3,1`	zono	off	no	0.01	0	3	0	13.40	37.33	0.359	*0.466*	*0.02*
						0	4	0	14.38	13.31	**1.080**		
						0	5	0	13.93	35.98	0.387		
						0	6	0	11.80	36.43	0.324		
						0	7	0	11.82	36.11	0.327		
						0	8	0	12.12	38.06	0.318		
		zono	off	no	0.1	0	3	0	13.40	36.71	0.365	*0.439*	*0.02*
						0	4	0	14.38	38.45	0.374		
						0	5	0	13.93	36.30	0.384		
						0	6	0	11.80	36.56	0.323		
						0	7	0	11.82	14.53	0.813		
						0	8	0	12.12	32.27	0.376		
		interval	off	no	0.01	0	3	0	13.40	38.22	0.351	*0.433*	*0.02*
						0	4	0	14.38	38.12	0.377		
						0	5	0	13.93	36.96	0.377		
						0	6	0	11.80	36.43	0.324		
						0	7	0	11.82	37.41	0.316		
						0	8	0	12.12	14.24	0.851		
		interval	off	no	0.05	0	3	0	13.40	37.48	0.358	*0.431*	*0.02*
						0	4	0	14.38	37.64	0.382		
						0	5	0	13.93	36.15	0.385		
						0	6	0	11.80	37.15	0.318		
						0	7	0	11.82	38.27	0.309		
						0	8	0	12.12	14.48	0.837		
		interval	off	no	0.1	0	3	0	13.40	35.31	0.379	*0.405*	*0.02*
						0	4	0	14.38	36.97	0.389		
						0	5	0	13.93	35.96	0.387		
						0	6	0	11.80	18.58	0.635		
						0	7	0	11.82	35.91	0.329		
						0	8	0	12.12	39.04	0.310		
		zono	off	no	0.05	0	3	0	13.40	35.31	0.379	*0.342*	*0.02*
						0	5	0	13.93	36.52	0.381		
						0	6	0	11.80	38.38	0.307		
						0	7	0	11.82	36.53	0.324		
						0	8	0	12.12	37.92	0.320		

Table 74: Architecture **3x10** — perturbation **translation**(`3,1`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 6 (6 combos, 35 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{c\_src}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{c\_src}$
translation	`3,3`	zono	off	no	0.01	0	3	0	12.78	10.92	**1.170**	***1.136***	*0.03*
						0	4	0	14.57	11.49	**1.268**		
						0	5	0	14.10	11.05	**1.276**		
						0	6	0	10.86	11.55	0.940		
						0	7	0	11.99	11.59	**1.035**		
						0	8	0	12.37	10.99	**1.126**		
		zono	off	no	0.05	0	3	0	12.78	13.22	0.967	***1.019***	*0.03*
						0	4	0	14.57	12.17	**1.197**		
						0	5	0	14.10	11.57	**1.219**		
						0	6	0	10.86	13.65	0.796		
						0	7	0	11.99	14.91	0.804		
						0	8	0	12.37	10.95	**1.130**		
		interval	off	no	0.1	0	3	0	12.78	14.12	0.905	***1.008***	*0.03*
						0	4	0	14.57	13.32	**1.094**		
						0	5	0	14.10	12.20	**1.156**		
						0	6	0	10.86	12.99	0.836		
						0	7	0	11.99	11.70	**1.025**		
						0	8	0	12.37	11.98	**1.033**		
		zono	off	no	0.1	0	3	0	12.78	14.22	0.899	*0.999*	*0.03*
						0	4	0	14.57	14.92	0.977		
						0	5	0	14.10	11.45	**1.231**		
						0	6	0	10.86	11.23	0.967		
						0	7	0	11.99	12.17	0.985		
						0	8	0	12.37	13.26	0.933		
		interval	off	no	0.05	0	3	0	12.78	13.16	0.971	*0.953*	*0.03*
						0	4	0	14.57	15.04	0.969		
						0	5	0	14.10	13.19	**1.069**		
						0	6	0	10.86	12.50	0.869		
						0	7	0	11.99	13.67	0.877		
						0	8	0	12.37	12.87	0.961		
		interval	off	no	0.01	0	3	0	12.78	15.65	0.817	*0.850*	*0.03*
						0	4	0	14.57	16.21	0.899		
						0	5	0	14.10	12.73	**1.108**		
						0	6	0	10.86	13.10	0.829		
						0	7	0	11.99	13.61	0.881		
						0	8	0	12.37	21.93	0.564		

Table 75: Architecture **3x10** — perturbation **translation**(`3,3`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 6 (6 combos, 36 cell rows). Column meanings: see Table 6.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
patch	`1,14,14,3`	zono	off	no	0.1	0	3	2	96.17	63.02	**1.526**	*1.425*	*0.03*
						0	4	2	328.21	140.28	**2.340**		
						0	5	2	62.33	45.44	**1.372**		
						0	6	2	34.38	35.26	0.975		
						0	7	2	71.72	127.33	0.563		
						0	8	2	55.30	31.22	**1.771**		
		zono	off	no	0.05	0	3	2	96.17	69.80	**1.378**	*1.379*	*0.03*
						0	4	2	328.21	154.01	**2.131**		
						0	5	2	62.33	45.35	**1.374**		
						0	6	2	34.38	30.55	**1.125**		
						0	7	2	71.72	144.75	0.495		
						0	8	2	55.30	31.29	**1.767**		
		interval	off	no	0.1	0	3	2	96.17	71.74	**1.341**	*1.284*	*0.03*
						0	4	2	328.21	166.42	**1.972**		
						0	5	2	62.33	47.86	**1.302**		
						0	6	2	34.38	31.35	**1.097**		
						0	7	2	71.72	127.37	0.563		
						0	8	2	55.30	38.66	**1.430**		
		interval	off	no	0.05	0	3	2	96.17	74.63	**1.289**	*1.253*	*0.03*
						0	4	2	328.21	174.54	**1.880**		
						0	5	2	62.33	49.29	**1.265**		
						0	6	2	34.38	30.98	**1.110**		
						0	7	2	71.72	133.13	0.539		
						0	8	2	55.30	38.44	**1.439**		
		interval	off	no	0.01	0	3	0	96.17	51.64	**1.862**	*0.939*	*0.02*
						0	4	0	328.21	468.98	0.700		
						0	5	0	62.33	598.51	0.104		
						0	6	0	34.38	57.09	0.602		
						0	7	0	71.72	45.58	**1.573**		
						0	8	0	55.30	69.66	0.794		
		zono	off	no	0.01	0	3	0	96.17	57.25	**1.680**	*0.814*	*0.02*
						0	4	0	328.21	545.86	0.601		
						0	5	0	62.33	652.89	0.095		
						0	6	0	34.38	65.59	0.524		
						0	7	0	71.72	55.96	**1.282**		
						0	8	0	55.30	78.50	0.704		

Table 76: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}} - \delta_{\mathrm{adv}}\rvert}_{c\_src}$
patch	`1,14,14,3`	interval	off	no	0.1	1	3	2	94.60	33.11	**2.857**	***1.950***	*0.01*
						1	4	2	113.19	83.83	**1.350**		
						1	5	2	138.25	76.61	**1.805**		
						1	6	2	206.61	126.97	**1.627**		
						1	7	2	83.24	38.47	**2.164**		
						1	8	2	67.32	35.53	**1.895**		
		zono	off	no	0.01	1	3	0	94.60	41.99	**2.253**	***1.883***	*0.01*
						1	4	0	113.19	65.59	**1.726**		
						1	5	0	138.25	41.85	**3.303**		
						1	6	0	206.61	138.90	**1.487**		
						1	7	0	83.24	76.31	**1.091**		
						1	8	0	67.32	46.82	**1.438**		
		zono	off	no	0.1	1	3	2	94.60	46.55	**2.032**	***1.546***	*0.01*
						1	4	2	113.19	73.82	**1.533**		
						1	5	2	138.25	79.21	**1.745**		
						1	6	2	206.61	124.57	**1.659**		
						1	7	2	83.24	66.33	**1.255**		
						1	8	2	67.32	64.04	**1.051**		
		interval	off	no	0.05	1	3	2	94.60	52.20	**1.812**	***1.455***	*0.06*
						1	4	2	113.19	52.42	**2.159**		
						1	5	2	138.25	122.41	**1.129**		
						1	6	2	206.61	154.74	**1.335**		
						1	7	2	83.24	36.84	**2.260**		
						1	8	2	67.32	1815.66	0.037		
		zono	off	no	0.05	1	3	2	94.60	46.15	**2.050**	***1.453***	*0.01*
						1	4	2	113.19	127.16	0.890		
						1	5	2	138.25	163.32	0.846		
						1	6	2	206.61	131.70	**1.569**		
						1	7	2	83.24	36.05	**2.309**		
						1	8	2	67.32	63.85	**1.054**		
		interval	off	no	0.01	1	3	0	94.60	39.81	**2.376**	***1.347***	*0.01*
						1	4	0	113.19	178.66	0.634		
						1	5	0	138.25	163.03	0.848		
						1	6	0	206.61	109.53	**1.886**		
						1	7	0	83.24	64.07	**1.299**		
						1	8	0	67.32	64.81	**1.039**		

Table 77: Architecture **cnn1** — perturbation **patch**(`1,14,14,3`), $c_{\mathrm{src}} = 1$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}} - \delta_{\mathrm{adv}}\rvert}_{c\_src}$
occ	`5,5,5`	interval	off	no	0.01	0	3	0	42.18	81.38	0.518	***1.071***	*0.02*
						0	4	0	536.01	341.64	**1.569**		
						0	5	0	110.91	60.70	**1.827**		
						0	6	0	33.37	64.98	0.514		
						0	7	0	56.20	61.39	0.915		
						0	8	0	57.73	53.32	**1.083**		
		zono	off	no	0.05	0	3	3	42.18	135.90	0.310	*0.927*	*0.02*
						0	4	3	536.01	321.09	**1.669**		
						0	5	3	110.91	74.19	**1.495**		
						0	6	3	33.37	41.54	0.803		
						0	7	3	56.20	168.64	0.333		
						0	8	3	57.73	60.66	0.952		
		interval	off	no	0.1	0	3	3	42.18	89.09	0.473	*0.922*	*0.02*
						0	4	3	536.01	331.69	**1.616**		
						0	5	3	110.91	86.97	**1.275**		
						0	6	3	33.37	40.92	0.815		
						0	7	3	56.20	205.69	0.273		
						0	8	3	57.73	53.47	**1.080**		
		zono	off	no	0.01	0	3	0	42.18	95.77	0.440	*0.921*	*0.02*
						0	4	0	536.01	529.98	**1.011**		
						0	5	0	110.91	56.54	**1.962**		
						0	6	0	33.37	50.15	0.665		
						0	7	0	56.20	68.52	0.820		
						0	8	0	57.73	92.15	0.626		
		zono	off	no	0.1	0	3	3	42.18	97.84	0.431	*0.897*	*0.01*
						0	4	3	536.01	350.02	**1.531**		
						0	5	3	110.91	80.51	**1.378**		
						0	6	3	33.37	45.81	0.728		
						0	7	3	56.20	190.17	0.296		
						0	8	3	57.73	56.84	**1.016**		
		interval	off	no	0.05	0	3	3	42.18	118.64	0.356	*0.766*	*0.02*
						0	4	3	536.01	464.45	**1.154**		
						0	5	3	110.91	68.00	**1.631**		
						0	6	3	33.37	67.29	0.496		
						0	7	3	56.20	218.07	0.258		
						0	8	3	57.73	82.56	0.699		

Table 78: Architecture **cnn1** — perturbation **occ**(`5,5,5`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 36 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\mathrm{sp}}_{c\_src}$	$\overline{\lvert\delta_{\mathrm{std}}-\delta_{\mathrm{adv}}\rvert}_{c\_src}$
translation	`1,1`	zono	off	no	0.05	0	3	18	344.88	377.32	0.914	***1.427***	*0.03*
						0	4	18	442.26	307.06	**1.440**		
						0	5	18	393.83	283.71	**1.388**		
						0	6	18	526.31	438.50	**1.200**		
						0	7	18	568.70	272.56	**2.087**		
						0	8	18	1392.84	908.78	**1.533**		
		zono	off	no	0.1	0	3	27	344.88	328.94	**1.048**	***1.388***	*0.02*
						0	4	27	442.26	322.87	**1.370**		
						0	5	27	393.83	336.59	**1.170**		
						0	6	27	526.31	347.48	**1.515**		
						0	7	27	568.70	292.15	**1.947**		
						0	8	27	1392.84	1086.99	**1.281**		
		interval	off	no	0.1	0	3	27	344.88	337.33	**1.022**	***1.356***	*0.02*
						0	4	27	442.26	329.69	**1.341**		
						0	5	27	393.83	303.11	**1.299**		
						0	6	27	526.31	346.59	**1.519**		
						0	7	27	568.70	302.30	**1.881**		
						0	8	27	1392.84	1295.56	**1.075**		
		zono	off	no	0.01	0	3	8	344.88	382.66	0.901	***1.342***	*0.02*
						0	5	8	393.83	304.68	**1.293**		
						0	6	8	526.31	342.38	**1.537**		
						0	7	8	568.70	285.52	**1.992**		
						0	8	8	1392.84	1409.59	0.988		
		interval	off	no	0.05	0	3	18	344.88	411.08	0.839	***1.303***	*0.03*
						0	4	18	442.26	341.09	**1.297**		
						0	5	18	393.83	295.32	**1.334**		
						0	6	18	526.31	438.36	**1.201**		
						0	7	18	568.70	293.53	**1.937**		
						0	8	18	1392.84	1150.00	**1.211**		
		interval	off	no	0.01	0	3	8	344.88	397.29	0.868	***1.250***	*0.02*
						0	4	8	442.26	428.04	**1.033**		
						0	5	8	393.83	315.89	**1.247**		
						0	6	8	526.31	368.46	**1.428**		
						0	7	8	568.70	298.87	**1.903**		
						0	8	8	1392.84	1368.32	**1.018**		

Table 79: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\mathrm{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 35 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,1`	interval	off	no	0.01	1	4	8	932.29	1265.54	0.737	***1.131***	*1.46*
						1	5	8	226.98	243.45	0.932		
						1	6	8	336.06	241.75	**1.390**		
						1	7	8	776.98	487.72	**1.593**		
						1	8	8	1820.97	1814.38	**1.004**		
		zono	off	no	0.01	1	3	8	298.04	306.20	0.973	***1.097***	*0.04*
						1	4	8	932.29	628.43	**1.484**		
						1	5	8	226.98	293.35	0.774		
						1	6	8	336.06	321.03	**1.047**		
						1	7	8	776.98	594.32	**1.307**		
						1	8	8	1820.97	1830.39	0.995		
		zono	off	no	0.05	1	3	18	298.04	168.40	**1.770**	***1.062***	*0.02*
						1	4	18	932.29	946.31	0.985		
						1	5	18	226.98	192.35	**1.180**		
						1	6	18	336.06	472.33	0.711		
						1	7	18	776.98	1076.43	0.722		
						1	8	18	1820.97	1812.91	**1.004**		
		zono	off	no	0.1	1	3	27	298.04	235.02	**1.268**	***1.012***	*0.04*
						1	4	27	932.29	1086.29	0.858		
						1	5	27	226.98	150.21	**1.511**		
						1	6	27	336.06	386.79	0.869		
						1	7	27	776.98	1357.40	0.572		
						1	8	27	1820.97	1835.40	0.992		
		interval	off	no	0.1	1	3	27	298.04	521.27	0.572	***1.005***	*0.04*
						1	4	27	932.29	949.49	0.982		
						1	5	27	226.98	261.02	0.870		
						1	6	27	336.06	225.40	**1.491**		
						1	7	27	776.98	691.40	**1.124**		
						1	8	27	1820.97	1831.25	0.994		
		interval	off	no	0.05	1	3	18	298.04	325.90	0.915	*0.936*	*0.02*
						1	4	18	932.29	1328.07	0.702		
						1	5	18	226.98	280.39	0.810		
						1	7	18	776.98	616.05	**1.261**		
						1	8	18	1820.97	1830.87	0.995		

Table 80: Architecture **cnn1** — perturbation **translation**(`1,1`), $c_{\text{src}} = 1$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 34 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	interval	off	no	0.01	0	3	8	251.33	295.58	0.850	*0.934*	*0.05*
						0	5	8	270.72	244.67	**1.106**		
						0	6	8	999.68	1107.91	0.902		
						0	7	8	246.58	212.41	**1.161**		
						0	8	8	276.94	427.62	0.648		
		zono	off	no	0.01	0	3	8	251.33	360.00	0.698	*0.794*	*0.06*
						0	5	8	270.72	322.18	0.840		
						0	6	8	999.68	947.17	**1.055**		
						0	7	8	246.58	515.82	0.478		
						0	8	8	276.94	308.05	0.899		
		interval	off	no	0.05	0	3	18	251.33	265.52	0.947	*0.767*	*0.48*
						0	5	18	270.72	310.55	0.872		
						0	6	18	999.68	972.80	**1.028**		
						0	7	18	246.58	1813.38	0.136		
						0	8	18	276.94	323.70	0.856		
		zono	off	no	0.05	0	3	18	251.33	299.27	0.840	*0.664*	*0.48*
						0	5	18	270.72	325.27	0.832		
						0	6	18	999.68	1205.12	0.830		
						0	7	18	246.58	1818.06	0.136		
						0	8	18	276.94	405.20	0.683		
		zono	off	no	0.1	0	3	18	251.33	290.20	0.866	*0.662*	*0.48*
						0	5	18	270.72	298.87	0.906		
						0	6	18	999.68	1191.95	0.839		
						0	7	18	246.58	1812.03	0.136		
						0	8	18	276.94	491.46	0.564		
		interval	off	no	0.1	0	3	18	251.33	399.41	0.629	*0.648*	*0.48*
						0	5	18	270.72	308.58	0.877		
						0	6	18	999.68	1092.79	0.915		
						0	7	18	246.58	1812.44	0.136		
						0	8	18	276.94	405.33	0.683		

Table 81: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 30 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`1,3`	interval	off	no	0.1	1	3	18	1814.42	1833.53	0.990	***1.349***	*0.02*
						1	6	18	1816.47	1812.04	**1.002**		
						1	7	18	309.44	128.91	**2.400**		
						1	8	18	1819.70	1812.30	**1.004**		
		interval	off	no	0.05	1	4	18	384.34	413.30	0.930	***1.347***	*0.06*
						1	6	18	1816.47	1814.64	**1.001**		
						1	7	18	309.44	126.17	**2.453**		
						1	8	18	1819.70	1813.20	**1.004**		

Table 82: Architecture **cnn1** — perturbation **translation**(`1,3`), $c_{\text{src}} = 1$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (2 combos, 8 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`3,1`	interval	off	no	0.1	0	3	28	458.41	280.45	**1.635**	*1.044*	*0.02*
						0	4	28	270.41	262.27	**1.031**		
						0	5	28	285.88	394.12	0.725		
						0	6	28	258.24	430.82	0.599		
						0	8	28	296.48	241.08	**1.230**		
		interval	off	no	0.01	0	3	0	458.41	381.05	**1.203**	*0.980*	*0.00*
						0	4	0	270.41	304.85	0.887		
						0	5	0	285.88	244.84	**1.168**		
						0	6	0	258.24	295.52	0.874		
						0	7	0	323.42	348.51	0.928		
						0	8	0	296.48	360.69	0.822		
		zono	off	no	0.05	0	3	20	458.41	324.44	**1.413**	*0.923*	*0.05*
						0	5	20	285.88	399.48	0.716		
						0	6	20	258.24	334.59	0.772		
						0	7	20	323.42	381.87	0.847		
						0	8	20	296.48	341.31	0.869		
		interval	off	no	0.05	0	3	20	458.41	274.22	**1.672**	*0.891*	*0.41*
						0	4	20	270.41	1811.15	0.149		
						0	6	20	258.24	310.48	0.832		
						0	7	20	323.42	454.67	0.711		
						0	8	20	296.48	272.18	**1.089**		
		zono	off	no	0.1	0	3	28	458.41	350.38	**1.308**	*0.841*	*0.02*
						0	4	28	270.41	581.72	0.465		
						0	5	28	285.88	478.13	0.598		
						0	6	28	258.24	384.41	0.672		
						0	7	28	323.42	333.04	0.971		
						0	8	28	296.48	287.17	**1.032**		
		zono	off	no	0.01	0	3	0	458.41	405.39	**1.131**	*0.762*	*0.00*
						0	4	0	270.41	385.65	0.701		
						0	5	0	285.88	381.31	0.750		
						0	6	0	258.24	654.47	0.395		
						0	7	0	323.42	441.97	0.732		
						0	8	0	296.48	342.02	0.867		

Table 83: Architecture **cnn1** — perturbation **translation**(`3,1`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 33 cell rows). Column meanings: see Table 36.

p_type	p_size	bound_tight	bp	varHint	τ	c_src	c_tgt	n_relax	t_std	t_adv	sp	$\overline{\text{sp}}_{\text{c\_src}}$	$\overline{\lvert\delta_{\text{std}} - \delta_{\text{adv}}\rvert}_{\text{c\_src}}$
translation	`3,3`	interval	off	no	0.01	0	3	0	800.69	271.91	**2.945**	***2.026***	*0.01*
						0	5	0	682.97	277.00	**2.466**		
						0	6	0	765.68	460.65	**1.662**		
						0	7	0	1231.36	667.26	**1.845**		
						0	8	0	375.90	309.98	**1.213**		
		interval	off	no	0.1	0	3	20	800.69	285.27	**2.807**	***1.979***	*0.02*
						0	5	20	682.97	326.08	**2.094**		
						0	6	20	765.68	402.96	**1.900**		
						0	7	20	1231.36	609.89	**2.019**		
						0	8	20	375.90	349.21	**1.076**		
		zono	off	no	0.01	0	3	0	800.69	279.03	**2.870**	***1.875***	*0.01*
						0	5	0	682.97	324.47	**2.105**		
						0	6	0	765.68	597.87	**1.281**		
						0	7	0	1231.36	690.41	**1.784**		
						0	8	0	375.90	280.79	**1.339**		
		interval	off	no	0.05	0	3	19	800.69	417.32	**1.919**	***1.715***	*0.02*
						0	4	19	1821.48	1749.19	**1.041**		
						0	5	19	682.97	481.51	**1.418**		
						0	6	19	765.68	471.99	**1.622**		
						0	7	19	1231.36	380.61	**3.235**		
						0	8	19	375.90	357.52	**1.051**		
		zono	off	no	0.1	0	3	20	800.69	322.64	**2.482**	***1.645***	*0.02*
						0	5	20	682.97	449.25	**1.520**		
						0	6	20	765.68	460.43	**1.663**		
						0	7	20	1231.36	727.96	**1.692**		
						0	8	20	375.90	433.39	0.867		
		zono	off	no	0.05	0	3	19	800.69	528.94	**1.514**	***1.553***	*0.02*
						0	4	19	1821.48	1575.09	**1.156**		
						0	5	19	682.97	495.78	**1.378**		
						0	6	19	765.68	436.60	**1.754**		
						0	7	19	1231.36	472.87	**2.604**		
						0	8	19	375.90	412.28	0.912		

Table 84: Architecture **cnn1** — perturbation **translation**(`3,3`), $c_{\text{src}} = 0$, $\tau \in \{0.01, 0.05, 0.1\}$; continuation of Table 36 (6 combos, 32 cell rows). Column meanings: see Table 36.